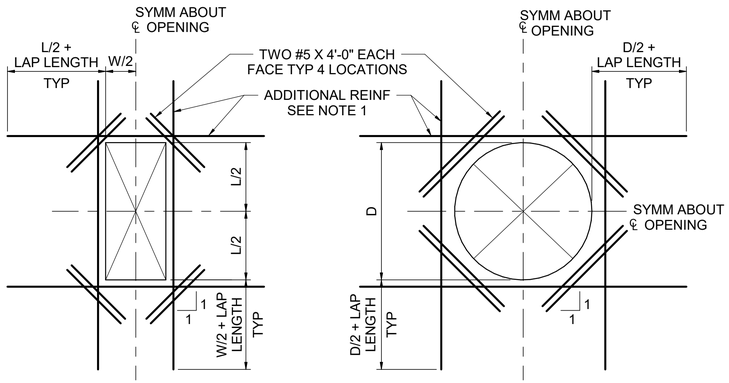


RECTANGULAR OPENING DETAIL

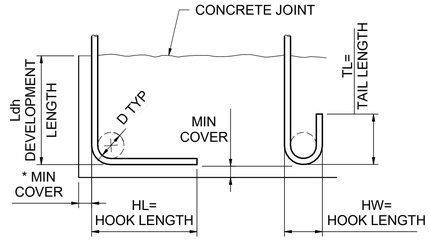
CIRCULAR OPENING DETAIL

NOTES:

1. PROVIDE ADDITIONAL REINFORCING THE SAME SIZE AS DISCONTINUOUS REINFORCEMENT AT OPENING. QUANTITY OF REINFORCING IN EACH DIRECTION SHALL BE EQUAL TO OR ONE GREATER THAN THE NUMBER OF DISCONTINUOUS BARS. PLACE 1/2 OF ADDITIONAL REINFORCING BARS EACH SIDE OF OPENING, PLACE ADDITIONAL REINFORCEMENT AT 3" OC (TYPICAL BOTH DIRECTIONS AND ALL LAYERS OF REINFORCEMENT). START FIRST BAR 2" CLEAR TO OPENING.
2. EXTEND ADDITIONAL REINFORCING BEYOND EDGE OF OPENING AS SHOWN. ADDITIONAL BARS MAY TERMINATE AT THE END OF THE WALL WITH A STANDARD HOOK WHERE THE LENGTH OF THE WALL WILL NOT PERMIT BARS TO EXTEND AS SHOWN ABOVE.
3. TYPICAL WALL OR SLAB REINFORCING NOT SHOWN FOR CLARITY. TERMINATE TYPICAL REINFORCING 2" CLEAR TO OPENING.
4. DETAIL APPLIES TO ALL INSTANCES WHERE THE OPENING SIZE EQUALS OR EXCEEDS THE SPACING OF WALL OR SLAB REINFORCING.
5. UNLESS SHOWN OTHERWISE ON DRAWINGS, PROVIDE EXTRA REINFORCING AROUND OPENINGS AS SHOWN AND INDICATED ABOVE.
6. PROVIDE ADDITIONAL DOWELS PER NOTE 1 ABOVE FOR ALL OPENINGS NEAR THE FLOOR SLAB, BASE SLAB, OR CORNERS.

1 003S504 EXTRA REINFORCING AROUND OPENINGS

NOT TO SCALE



90 DEG STD HOOK **180 DEG STD HOOK**

BAR SIZE GRADE 60	HL	HW	TL	D	f'c=4.5 OR 5.0 KSI
					Ldh *
#3	6"	3"	3"	2 1/4"	6"
#4	8"	4"	4 1/2"	3"	7"
#5	10"	5"	5"	3 3/4"	9"
#6	1'-0"	6"	6"	4 1/2"	10"
#7	1'-2"	7"	7"	5 1/4"	12"
#8	1'-4"	8"	8"	6"	14"
#9	1'-7"	11 3/4"	10 1/2"	9 1/2"	15"
#10	1'-10"	1'-1 1/4"	11 1/2"	10 3/4"	17"
#11	2'-0"	1'-2 3/4"	1'-1"	12"	19"

* COMPLYING WITH MINIMUM COVER REQUIREMENTS OF ACI 318. OTHERWISE Ldh MUST BE RE-CALCULATED.

2 003S504 REINFORCING HOOK SCHEDULE

NOT TO SCALE

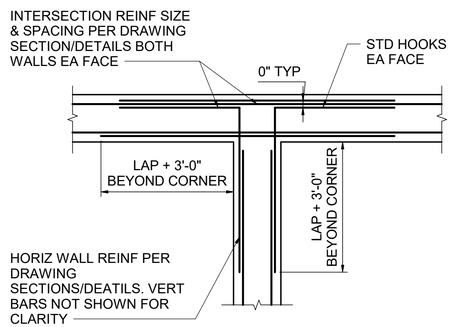
LAP SPLICE AND EMDEDMENT LENGTHS f'c =4.5 ksi fy = 60 ksi f'c =5.0 ksi		
BAR	BARS SPACED GREATER THAN 4"	BARS SPACED LESS THAN OR EQUAL TO 4"
#3	14"	20"
#4	19"	32"
#5	29"	46"
#6	39"	62"
#7	55"	87"
#8	69"	107"
#9	76"	116"
#10	97"	140"
#11	120"	146"

NOTES:

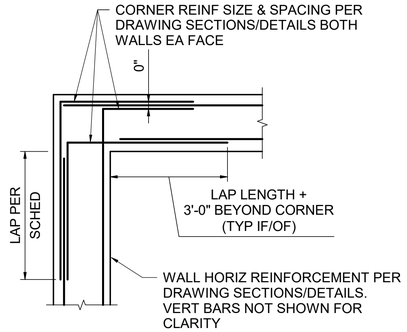
1. PROVIDE MINIMUM LAP SPLICE LENGTHS AND EMBEDMENTS PER TABLE UNLESS NOTED OTHERWISE. EMBEDMENT LENGTH EQUALS THE LAP SPLICE LENGTH UNLESS OTHERWISE NOTED.
2. BAR SPACING AT LAP SPLICE IS THE MINIMUM CLEAR DISTANCE BETWEEN LAPPED BARS PLUS ONE BAR DIAMETER.
3. ALL SPLICES TO BE CONTACT SPLICES AND WIRED TOGETHER UNLESS OTHERWISE APPROVED BY THE ENGINEER.

3 003S504 CONCRETE REINFORCING LAP AND EMBEDMENT SCHEDULE

NOT TO SCALE



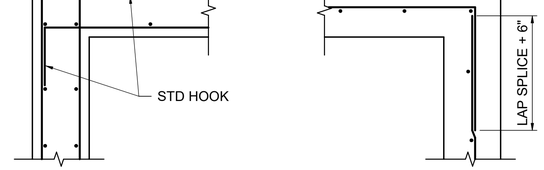
AT INTERSECTING CONDITIONS

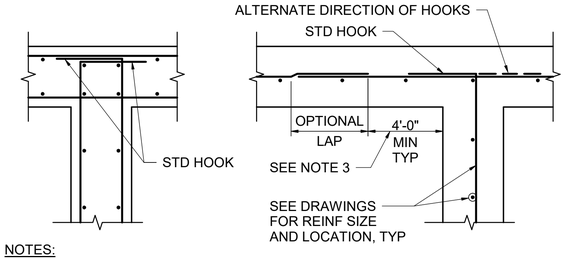


AT CORNER CONDITIONS

4 003S504 WALL REINF AT CORNERS & INTERSECTIONS WHERE BAR SIZE DIFFERS AT CORNER FROM WALL HORIZ

NOT TO SCALE





NOTES:

1. ALL HOOKS SHALL BE STD 90 DEGREE HOOKS.
2. SEE DRAWINGS FOR ADDITIONAL HORIZONTAL BARS. STAGGER BETWEEN TYPICAL REINF SPACING, EXTEND TO 1/5 OF DISTANCE TO NEAREST ADJACENT WALL IN EACH DIRECTION, UNO.
3. OPTIONAL LAP LOCATION. APPLIES TO BOTH DOUBLE AND SINGLE LAYER CONDITIONS TYP.

5 003S504 WALL REINFORCEMENT AT CORNERS AND INTERSECTIONS

NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 05/20/26 ARIZONA, U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 85509 JOSE M. CORONADO Date Signed 05/20/26 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. CORONADO	003 S504	GENERAL STRUCTURAL DETAILS - REINFORCING
	DRAWN: R. NELSON		
	TECH. REVIEW: M. HERRERA		NORTH RIM - GRAND CANYON NATIONAL PARK
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	38 OF 99

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_000_GENERAL.rvt
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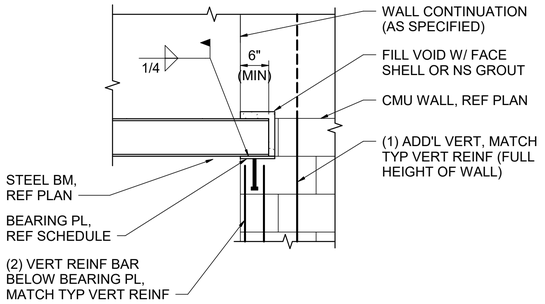
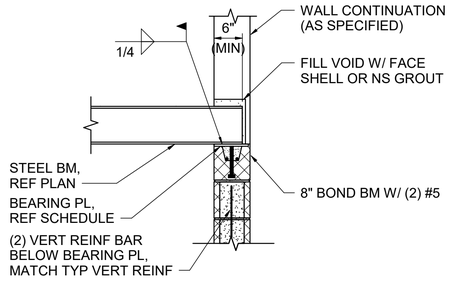


BEAM BEARING PARALLEL TO WALL

BEARING PLATE SCHEDULE				
MARK	THICKNESS	WIDTH	LENGTH	# OF HEADED STUDS
CASE 1	1/2"	7"	7"	(2) 5/8"Ø
CASE 2	1/2"	11"	11"	(4) 5/8"Ø
CASE 3	3/4"	7"	12"	(4) 5/8"Ø
CASE 4	3/4"	11"	16"	(4) 5/8"Ø



BEAM BEARING PERPENDICULAR TO WALL

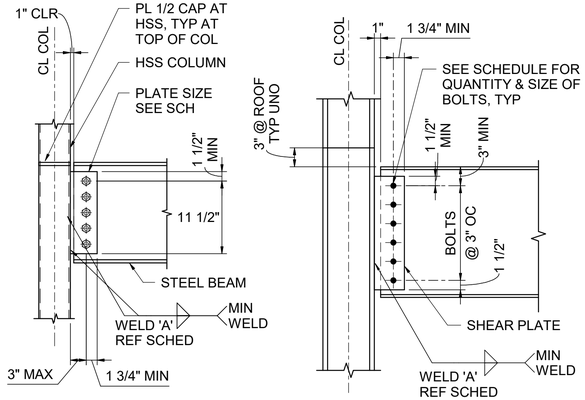
CASE 1 = FLOOR OR ROOF BEAM NOT SUPPORTING OTHER BEAMS, BEARING ON 8" WALLS

CASE 2 = FLOOR OR ROOF BEAM NOT SUPPORTING OTHER BEAMS, BEARING ON 12" WALLS

CASE 3 = GIRDER BEAM SUPPORTING OTHER BEAMS OR JOISTS, BEARING ON 8" WALLS

CASE 4 = GIRDER BEAM SUPPORTING OTHER BEAMS OR JOISTS, BEARING ON 12" WALLS

1 MASONRY WITH STEEL BEAM BEARING
003S505
NOT TO SCALE



TYPICAL CONDITION AT HSS COLUMNS

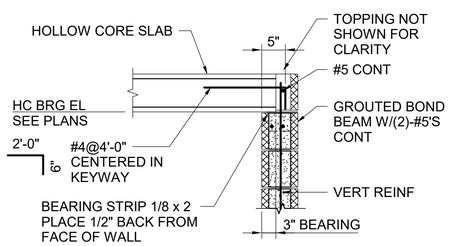
TYPICAL CONDITION AT WF COLUMNS

3/4" DIAMETER F1554 BOLTS SINGLE PLATE SHEAR CONNECTION SCHEDULE			
BEAM SIZE	SHEAR PLATE SIZE	FILLET WELD "A"	MIN # OF BOLTS
W8, W10	5/16"	3/16"	2
W12	5/16"	3/16"	3
W14	3/8"	1/4"	3
W16	3/8"	1/4"	4
W18	7/16"	5/16"	5
W21, 24	7/16"	5/16"	6
W27	7/16"	5/16	7

NOTES:

1. ALL BOLTS SHALL BE F1852 AND FULLY TIGHTENED TWIST-OFF TYPE TENSION CONTROL BOLTS.
2. USE AISC STANDARD HOLES OR HORIZONTAL SHORT SLOTTED HOLES IN SHEAR PLATE. HARDENED WASHERS ARE REQUIRED WITH SLOTTED HOLES.

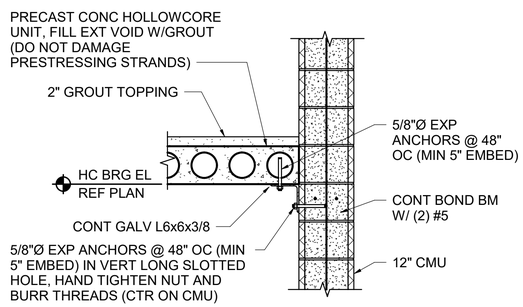
3 TYPICAL STEEL GRAVITY BEAM/GIRDER TO COLUMN CONNECTION
003S505
NOT TO SCALE



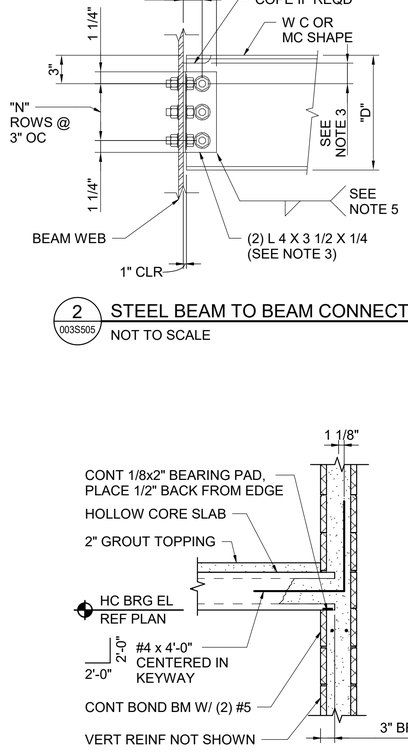
NOTE:

1. FILL HOLLOW CORE KEYWAY COMPLETELY WITH SAND/CEMENT GROUT.

4 TYPICAL HOLLOW CORE BEARING AT CMU
003S505
NOT TO SCALE



5 TYPICAL HOLLOW CORE CONNECTION AT CMU
003S505
NOT TO SCALE



STANDARD BOLTED CONNECTION SCHEDULE		
NOMINAL BEAM SIZE "D"	NUMBER OF BOLT ROWS "N"	WELD SIZE
W8	2	3/16
W10	2	3/16
W12	3	3/16
W14	3	3/16
W16	3	1/4
W18	4	1/4
W21	4	1/4
W24	4	1/4
W27	5	1/4
W30	5	5/16
W33	6	5/16
W36	6	5/16

NOTES:

1. ALL BOLTS SHALL BE 3/4"ø F1554 UNLESS NOTED OTHERWISE.
2. PROVIDE MINIMUM NUMBER OF BOLT ROWS "N" SHOWN AS THE TYPICAL CONN. INCREASE NUMBER OF ROWS AND / OR BOLT DIA. IF INDICATED ON PLANS.
3. MINIMUM DISTANCE FROM CL OF TOP BOLT TO A COPE SHALL BE 1 1/2". WHERE DEEP COPES ARE REQD, INCREASE DISTANCE FROM TOP OF BEAM TO CL OF TOP OF BOLT.
4. USE STANDARD OR SHORT HORIZONTAL SLOTTED HOLES AS REQUIRED.
5. WELD DOUBLE ANGLES TO BEAM WEB IN LIEU OF BOLTING AT CONTRACTOR'S OPTION.
6. IN THE CASE OF ALUMINUM CONSTRUCTION, CONNECTIONS SHALL BE ALL-BOLTED AND BOLTS SHALL BE ALUMINUM OR STAINLESS STEEL; SAME NUMBER AND DIAMETER AS INDICATED.

2 STEEL BEAM TO BEAM CONNECTION
003S505
NOT TO SCALE

NOTES:

1. FILL HOLLOW CORE KEY WAY COMPLETELY W/ SAND/CEMENT GROUT.

6 TYPICAL HOLLOW CORE BEARING EXTERIOR
003S505
NOT TO SCALE

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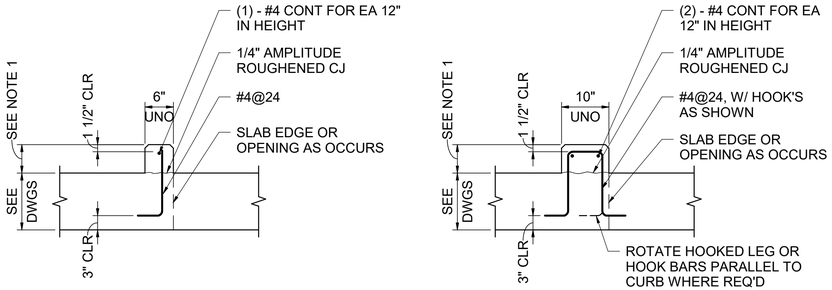
Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
Date Signed 05/01/2026
ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER (Civil)
CERTIFICATE NO.
85509
JOEL
CORONADO
Date Signed 05/01/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. CORONADO	003 S505	GENERAL STRUCTURAL DETAILS - CONNECTIONS
	CADD R. NELSON		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: M. HERRERA		
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	39 OF 99



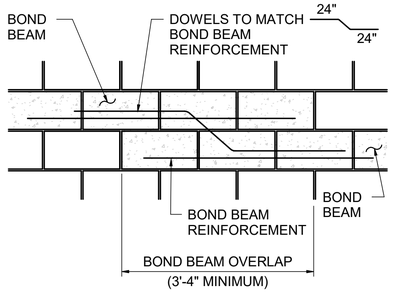
CURBS LESS THAN 10" WIDE

CURBS 10" WIDE AND WIDER

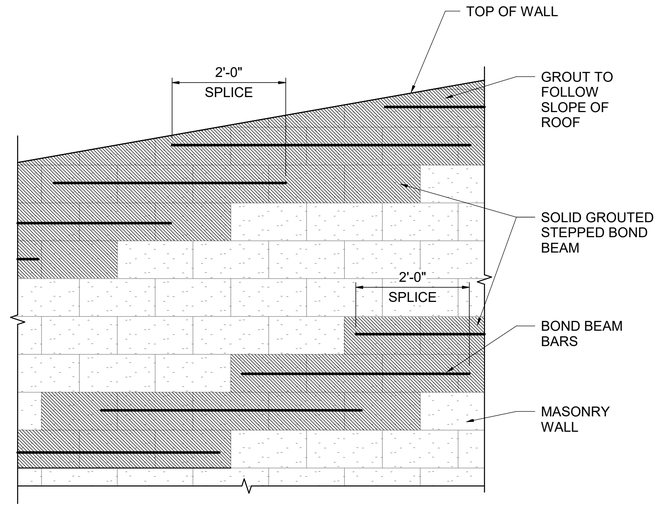
NOTES:

1. CURB SHALL BE 6" MINIMUM AT FLOORS AND 12" MINIMUM AT ROOFS UNLESS NOTED OTHERWISE IN THE DRAWINGS.
2. PLACE STRIP-TYPE WATERSTOP AT CJ WHERE CURB IS REQUIRED FOR CONTAINMENT. CENTER WATERSTOP IN CURB.
3. COORDINATE ROOF CURB WIDTHS WITH HATCHES AND EQUIPMENT.

1 003S506 CONCRETE CURB
NOT TO SCALE



2 003S506 TYPICAL BOND BEAM STEP
NOT TO SCALE



3 003S506 TYPICAL SLOPED BOND BEAM ELEVATION
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 4/27/2026 ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JOSE CORONADO Date Signed 4/27/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. CORONADO DRAWN: R. NELSON TECH. REVIEW: M. HERRERA DATE: MAY 2026	SUB SHEET NO. 003 S506	TITLE OF SHEET GENERAL STRUCTURAL DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 99

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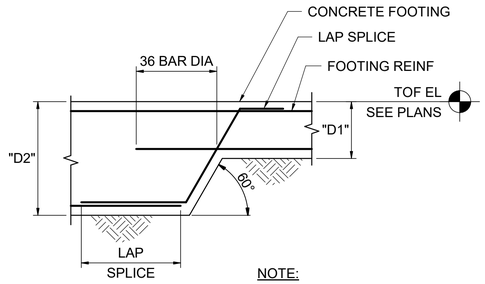
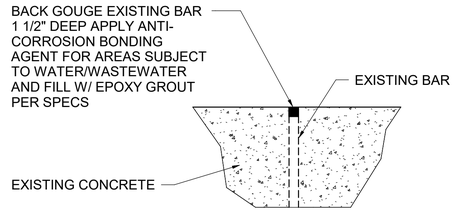


NOTE:

1. FOR FTG EL, DEPTH AND REINF, SEE PLANS AND TYP FTG DET.

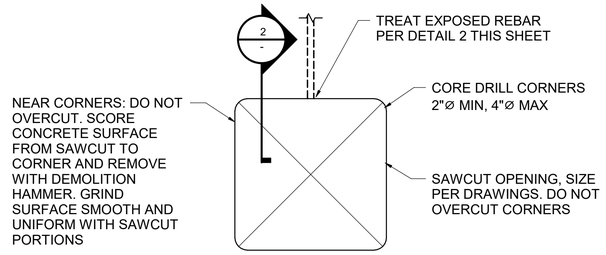
1 003S507 TYPICAL BOTTOM STEP FOOTING

NOT TO SCALE



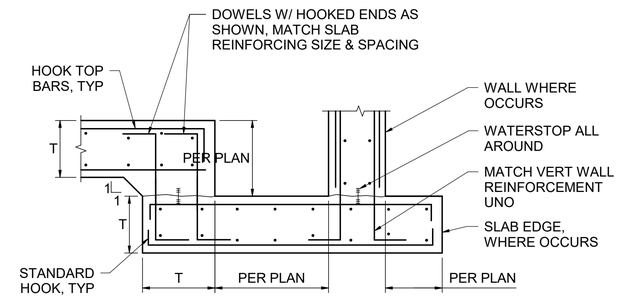
2 003S507 TYPICAL AT ALL SAWCUT WALLS AND EXPOSED REBAR

NOT TO SCALE



3 003S507 TYP SAWCUT OPENING IN EXISTING CONCRETE

NOT TO SCALE

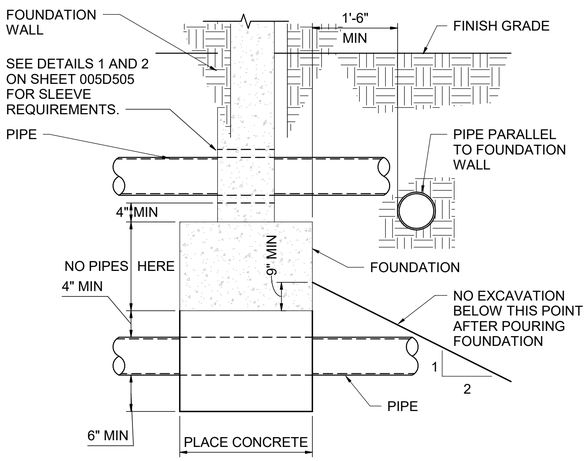


NOTE:

1. REINFORCING SIZE AND SPACING OF ALL SUMP BARS TO MATCH SLAB HORIZONTAL BAR SIZE AND SPACING.
2. GRATING NOT SHOWN FOR CLARITY. SEE PLAN FOR WHERE GRATING IS REQUIRED.

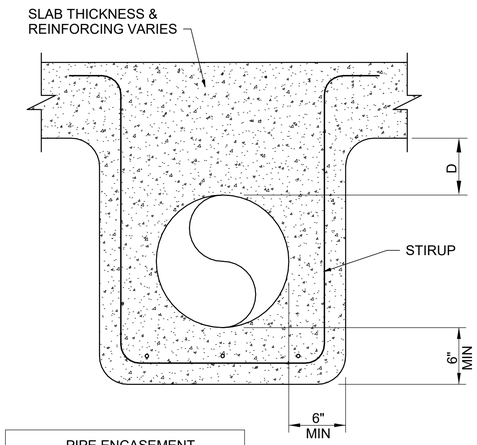
4 003S507 TYPICAL SUMP GREATER THAN 12-INCH DEEP

NOT TO SCALE



6 003S507 PIPE THROUGH FOUNDATION STEM WALLS AND FOOTINGS

NOT TO SCALE



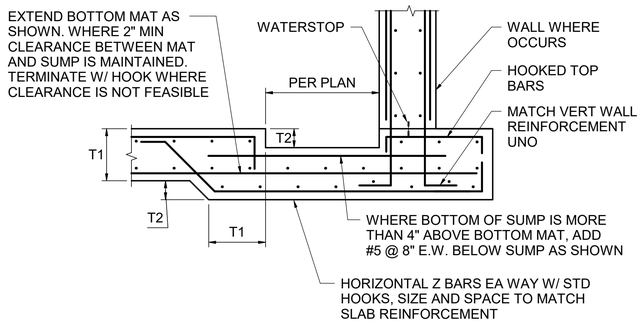
PIPE ENCASEMENT REINFORCING TABLE		
PIPE SIZE	STIRUPS	BOTTOM BARS
2"-14"	#3@12	3#4
16"-20"	#3@12	4#4
24"-30"	#4@12	4#4
36"-48"	#4@8	5#4

NOTES:

1. THIS DETAIL IS FOR ALL UNDERSLAB PIPE ENCASEMENT WHEN D < 4'-0".
2. USE CONCRETE ENCASEMENT DETAIL 3 THIS PAGE FOR D > 4'-0".

7 003S507 UNDERSLAB PIPE ENCASEMENT

NOT TO SCALE



5 003S507 TYPICAL SUMP 12" DEEP OR LESS

NOT TO SCALE

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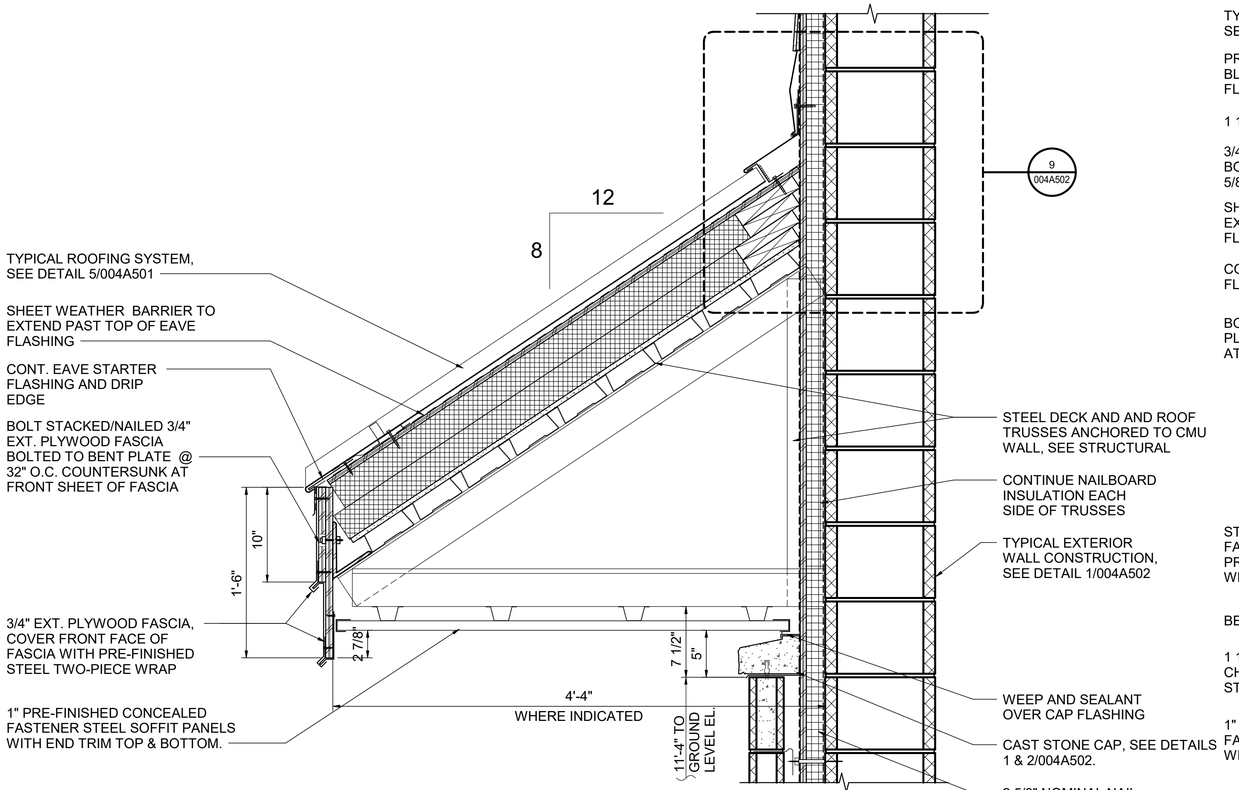
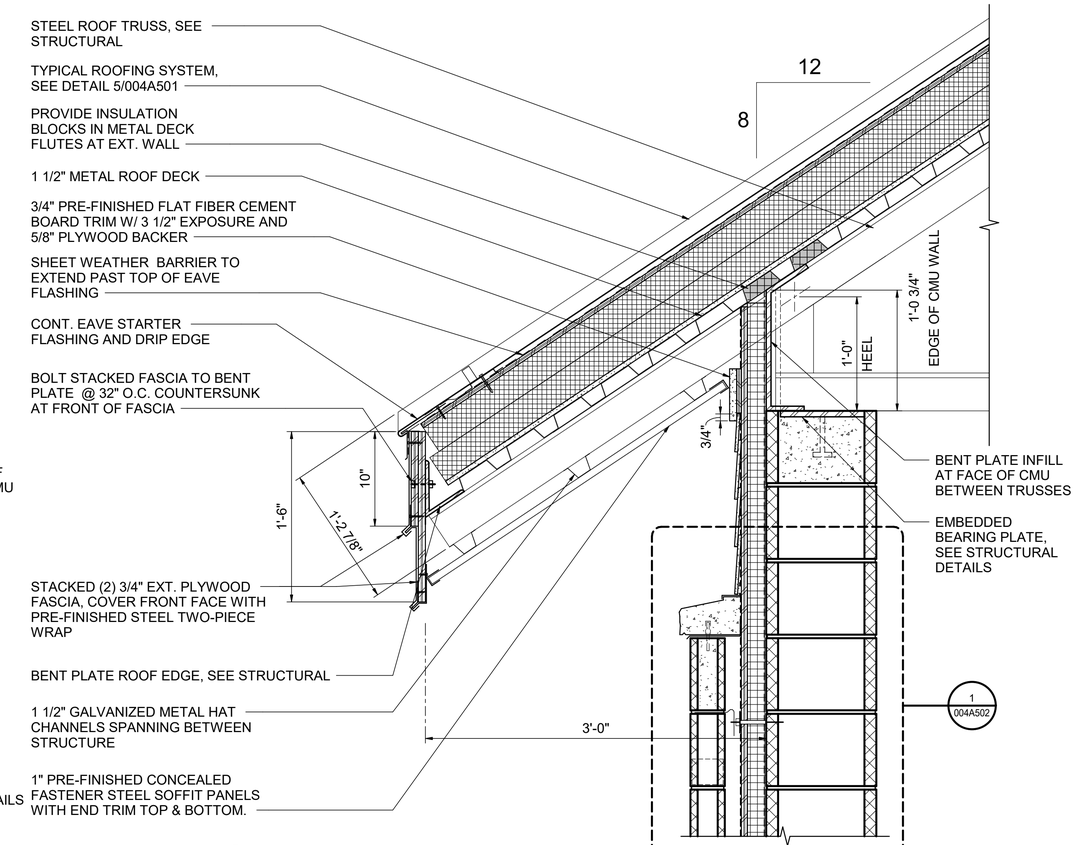
Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 05/11/2026 ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JUAN CORONADO Date Signed 05/11/2026 ARIZONA U.S.A.

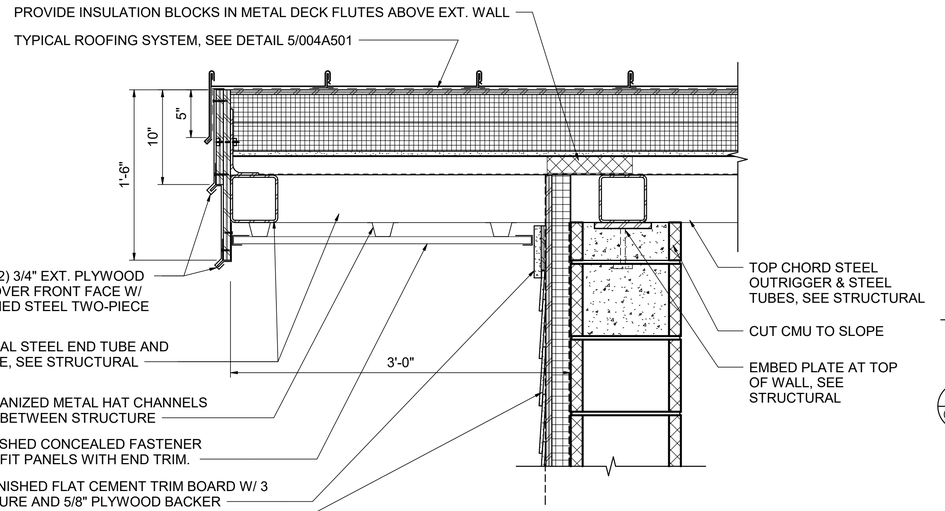
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. CORONADO	003 S507	GENERAL STRUCTURAL DETAILS - GENERAL CONTINUED	113 182752
	CADD R. NELSON			PMIS/PKG NO. 318719
	TECH. REVIEW: M. HERRERA			SHEET 41 OF 99
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	



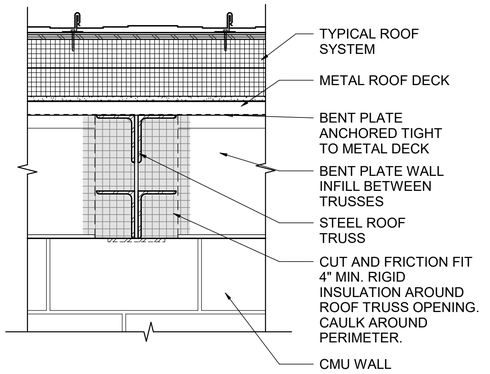
1 004A501 LOWER ROOF EDGE @ 062 BUILDING MECHINCAL UNITS
SCALE A



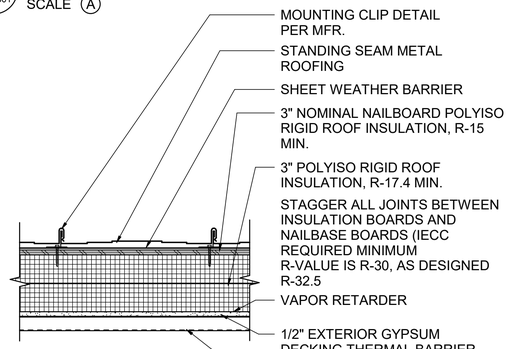
2 004A501 ROOF EDGE @ 062, 072 & 073 BUILDINGS
SCALE A



3 004A501 RAKE EDGE
SCALE A



4 004A501 ROOF TRUSS BEARING POCKET INFILL
SCALE A



5 004A501 TYPICAL ROOF SYSTEM
SCALE A

GENERAL ROOF NOTES:

A. SEE SERIES 004 SHEETS FOR STANDARD DETAILS.

B. PROVIDE ICE AND WATER SHIELD PROTECTION AT ALL EAVES, VALLEYS, PENETRATIONS, AND ROOF EDGES TO 36" MIN AT EA SIDE.

C. COORDINATE ALL ROOF PENETRATIONS THAT REQUIRE FLASHING; SEE MECHANICAL. SEE ALSO DETAILS 4 & 5/004A503.

D. PROVIDE ALL FLASHING, JOINTS, DRIPS, TRIM, CAPS, AND COPINGS REQUIRED AS A COMPONENT OF THE ROOF ASSEMBLY.

E. REFER TO ENGINEERING DISCIPLINE DOCUMENTS FOR ADDITIONAL EQUIPMENT, PIPE, CONDUIT, VALVES, ETC.

F. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

G. INSTALL ALL ROOFING ASSEMBLIES IN ACCORDANCE WITH PROJECT SPECIFICATIONS, DETAILS, AND MANUFACTURER'S STANDARD INSTALLATION REQUIREMENTS.

SCALE A
12 6 0 12
SCALE OF FEET
1-1/2" = 1'-0"

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY STATE REGISTERED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A501	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 42 OF 99

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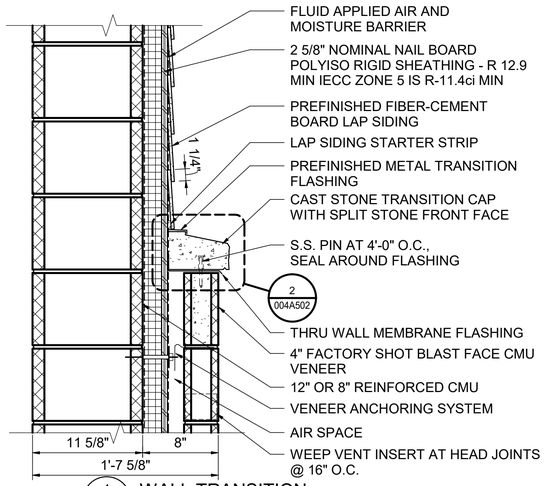
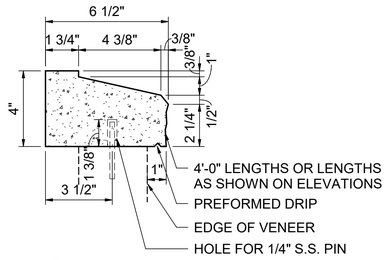
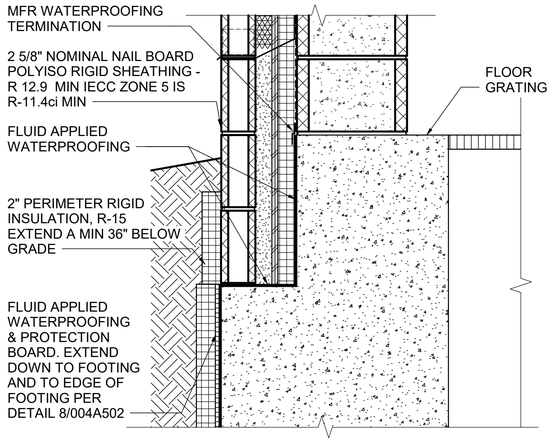


1 004A502 WALL TRANSITION
SCALE A

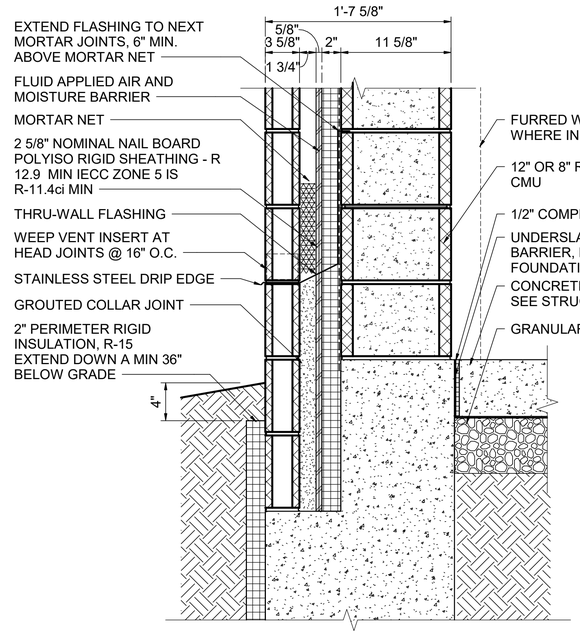


2 004A502 CAST-STONE CAP
SCALE B

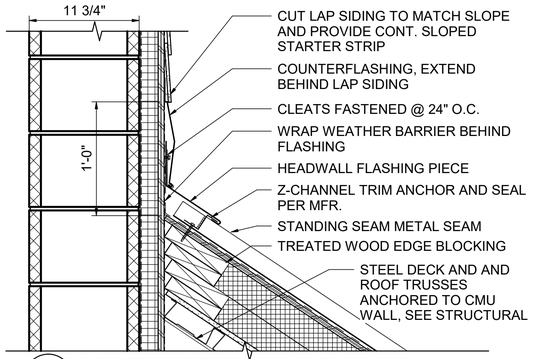


5 004A502 FOUNDATION WATERPROOFING
SCALE A

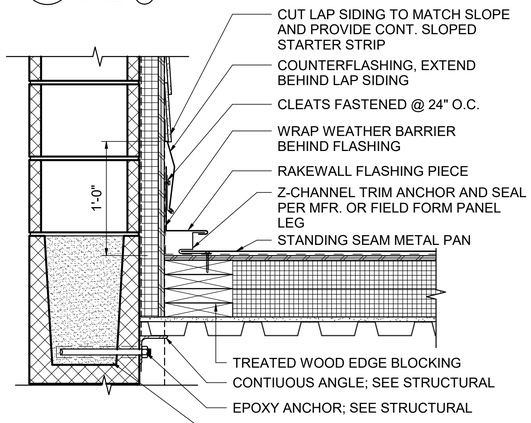
3 004A502 NOT USED
SCALE A



6 004A502 WALL BASE
SCALE A

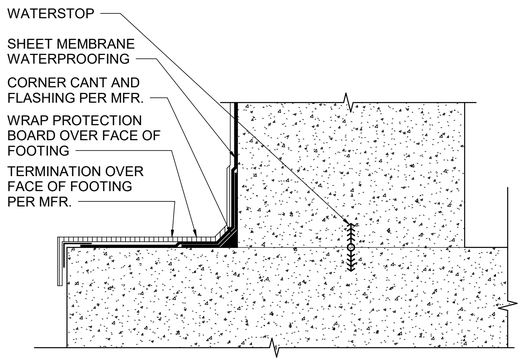


9 004A502 LOWER ROOF HEADWALL FLASHING
SCALE A

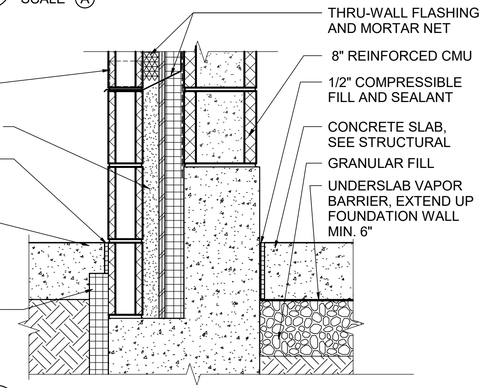


10 004A502 LOWER ROOF RAKE FLASHING
SCALE A

4 004A502 NOT USED
SCALE A



8 004A502 WATERPROOFING AT FOOTING
SCALE A



11 004A502 WALL BASE @ 072 BUILDING
SCALE A

SCALE B
6 3 0 6
3" = 1'-0"
SCALE OF FEET

SCALE A
12 6 0 12
1-1/2" = 1'-0"
SCALE OF FEET

7 004A502 NOT USED
SCALE A

LICENSED ARCHITECT
CERTIFICATE NO.
85785
RICHARD MCKINLEY
DATE SIGNED 2026.04.01
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A502	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 43 OF 99

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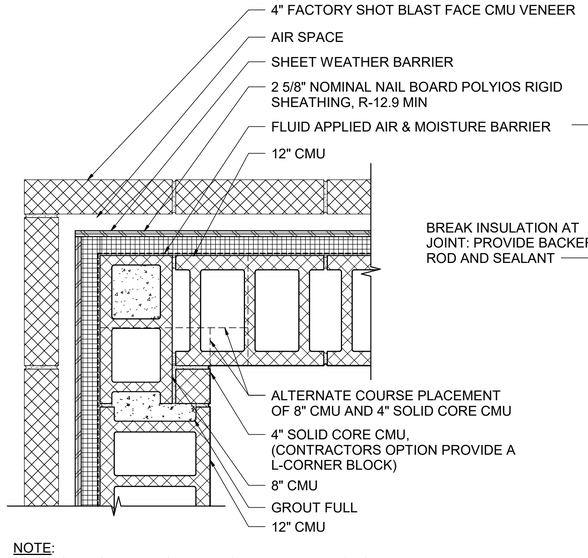
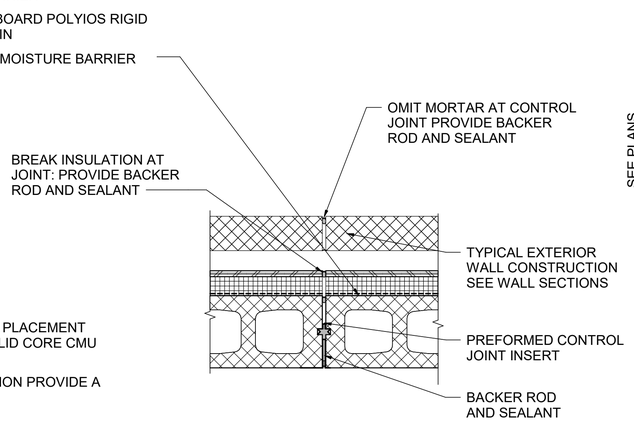


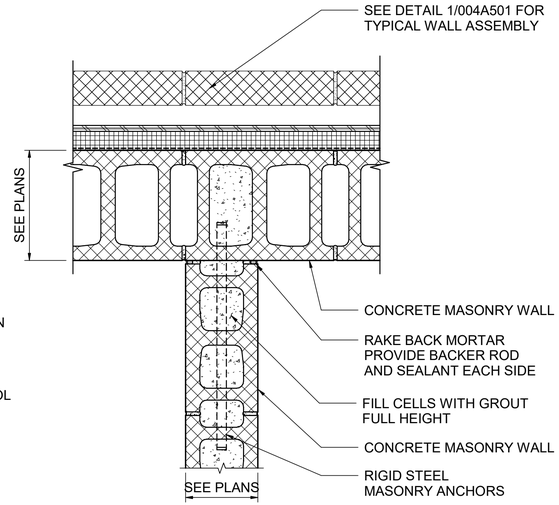
NOTE:
1. SEE STRUCTURAL FOR VERTICAL WALL REINFORCEMENT.

1 004A503 MASONRY CORNER @ 12" CMU
SCALE A



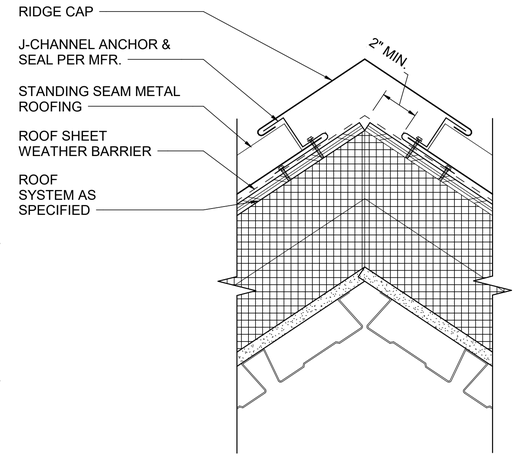
NOTE:
1. SEE STRUCTURAL FOR VERTICAL WALL REINFORCEMENT.

2 004A503 MASONRY CONTROL JT.
SCALE A

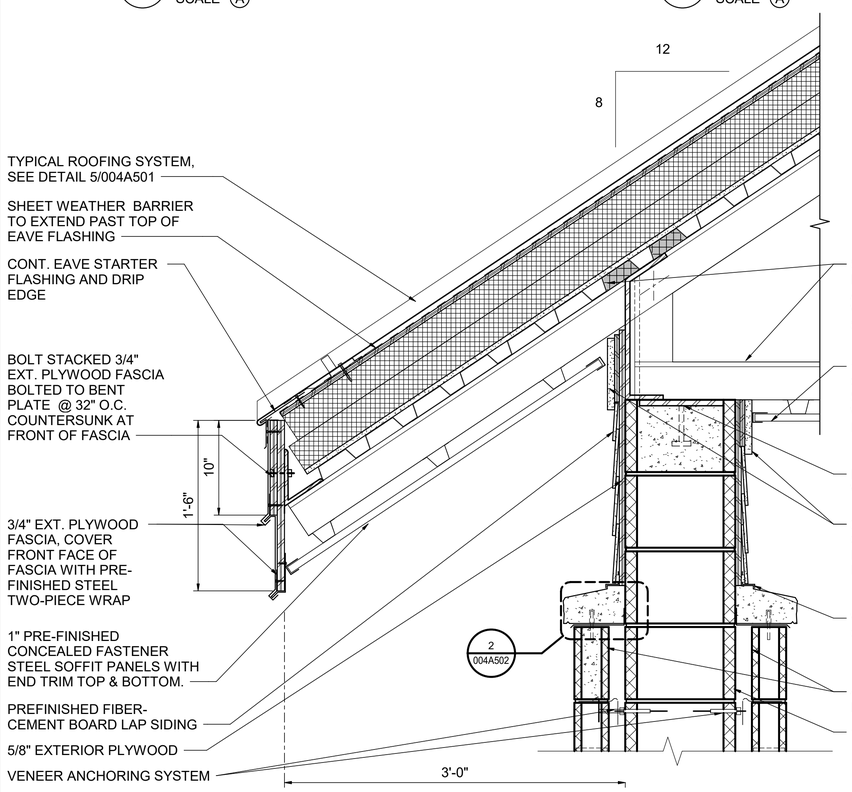


NOTES:
1. SEE STRUCTURAL FOR VERTICAL WALL REINFORCING.

3 004A503 MASONRY INTERIOR CONTROL JT.
SCALE A



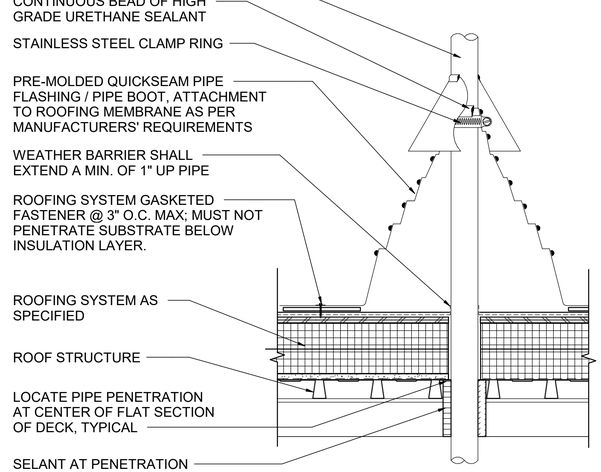
4 004A503 METAL ROOF RIDGE
SCALE A



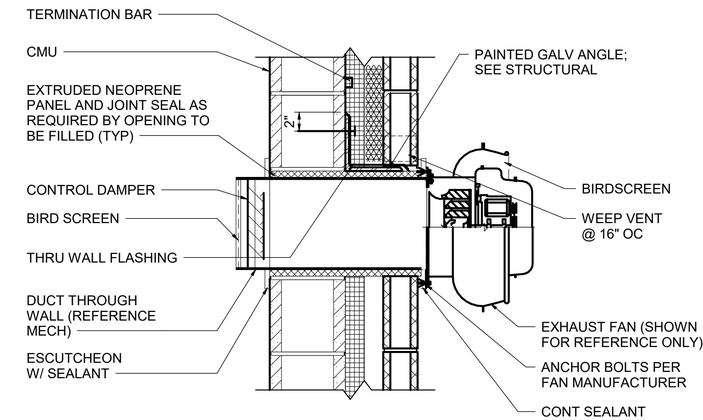
8 004A503 LOWER ROOF EDGE AT 062 BUILDING ENTRY
SCALE A

DETAIL MOVED
SEE WALL PENETRATION DETAIL
3/005D505

6 004A503 WALL PENETRATION
SCALE A



5 ROOF PIPE PENETRATION FLASHING
SCALE A



7 004A503 EXHAUST VENT
SCALE A

SCALE B: 6 3 0 6 — SCALE OF FEET — 3" = 1'-0"
SCALE A: 12 6 0 12 — SCALE OF FEET — 1-1/2" = 1'-0"

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 05/08/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A503	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 44 OF 99

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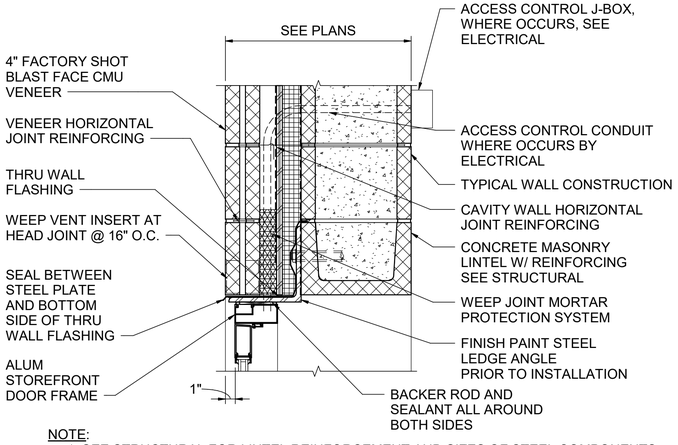
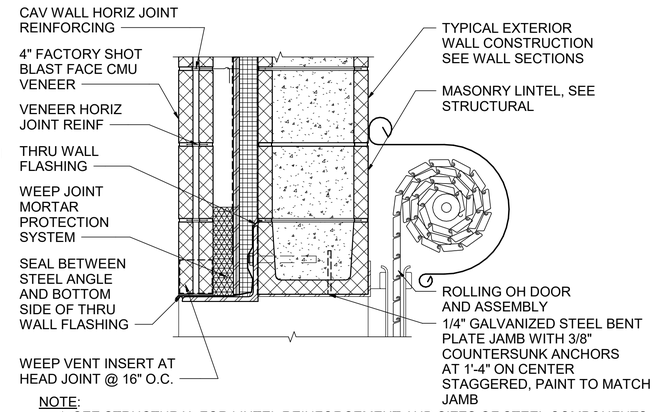
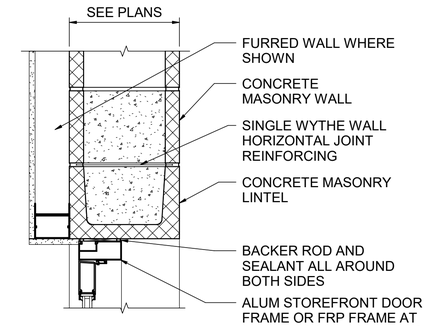


NOTE:
1. SEE STRUCTURAL FOR LINTEL REINFORCEMENT AND SIZES OF STEEL COMPONENTS.

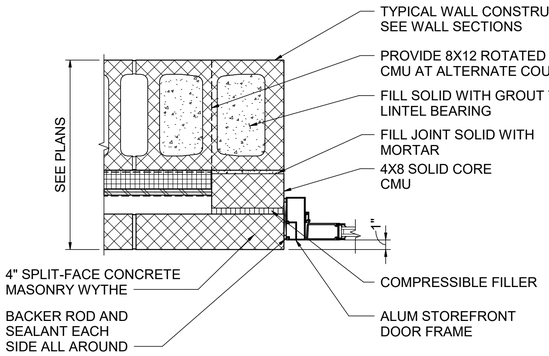


NOTE:
1. SEE STRUCTURAL FOR LINTEL REINFORCEMENT AND SIZES OF STEEL COMPONENTS.



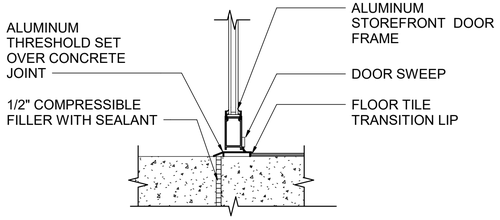
NOTES:
1. SEE STRUCTURAL FOR LINTEL SIZE AND VERTICAL WALL REINFORCING.

1 004A504 ALUMINUM DOOR HEAD IN CMU CAVITY WALL
SCALE (A)



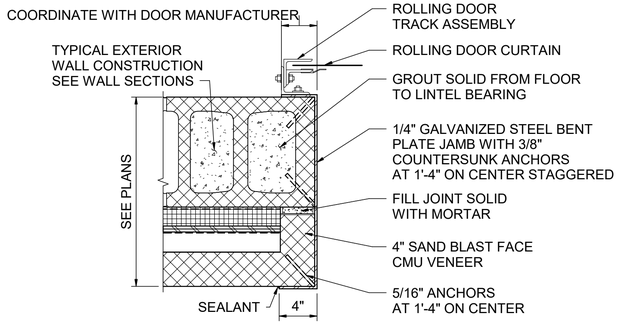
NOTE:
1. SEE STRUCTURAL FOR REINFORCEMENT.

2 004A504 ALUMINUM DOOR JAMB IN CMU CAVITY WALL
SCALE (A)



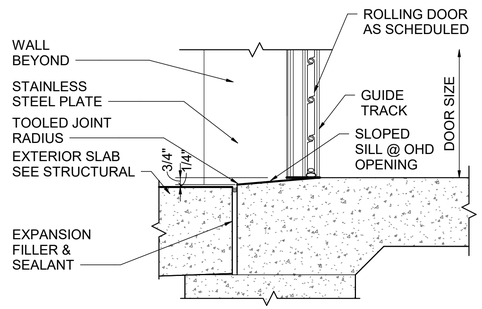
3 004A504 ALUM SILL AT STOOP
SCALE (A)

4 004A504 COILING DOOR HEAD
SCALE (A)



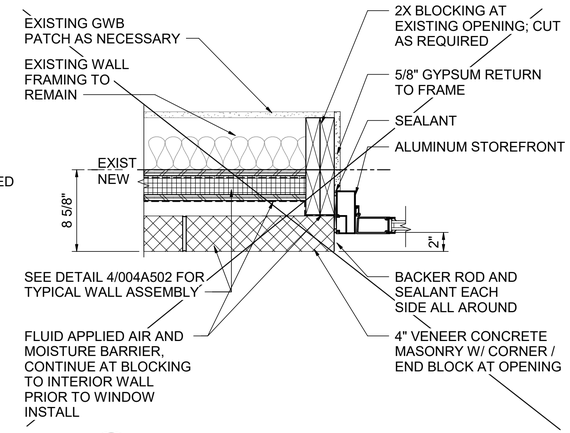
NOTES:
1. FINISH PAINT ENTIRE BENT PLATE JAMB PRIOR TO INSTALLATION.
2. SEE STRUCTURAL FOR VERTICAL REINFORCEMENT.
3. HORIZONTAL REINFORCEMENT NOT SHOWN.
4. SEE 004A516 FOR BOLLARDS

5 004A504 COILING DOOR JAMB
SCALE (A)

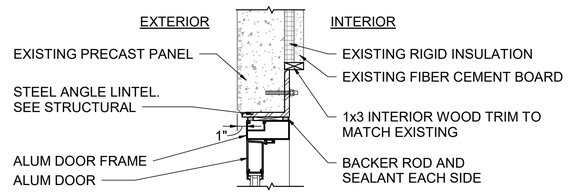


6 004A504 COILING DOOR SILL
SCALE (A)

7 004A504 NOT USED
SCALE (A)

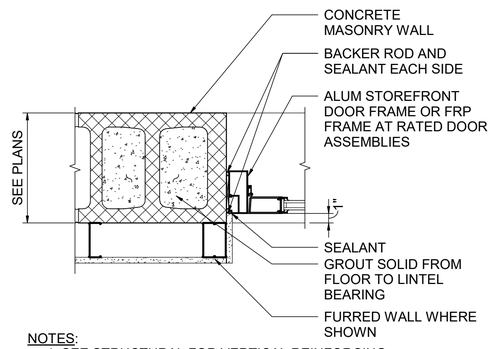


8 004A504 NOT USED
SCALE (A)



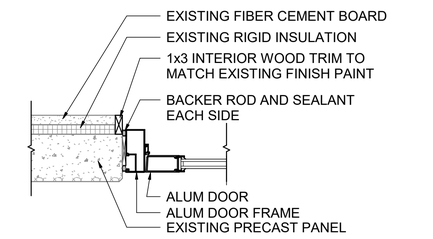
11 004A504 ALUM DOOR HEAD IN EXISTING WALL
SCALE (A)

9 004A504 ALUMINUM DOOR HEAD IN CMU WALL
SCALE (A)



NOTES:
1. SEE STRUCTURAL FOR VERTICAL REINFORCING.
2. HORIZONTAL JOINT REINFORCING NOT SHOWN.

10 004A504 ALUMINUM DOOR JAMB IN CMU WALL
SCALE (A)



12 004A504 ALUM DOOR JAMB IN EXISTING WALL
SCALE (A)

SCALE (A) 12 6 0 12 1-1/2" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85788 RICHARD MCKINLEY DATE SIGNED 4/19/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A504	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - DOOR FRAMES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 45 OF 99

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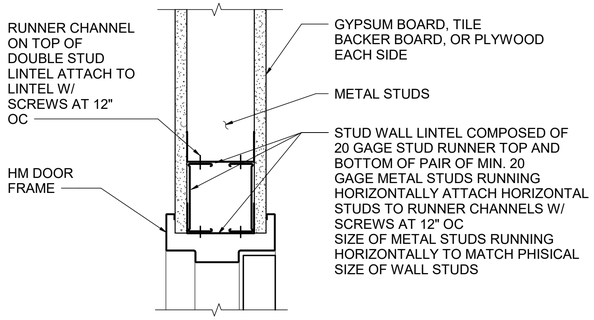
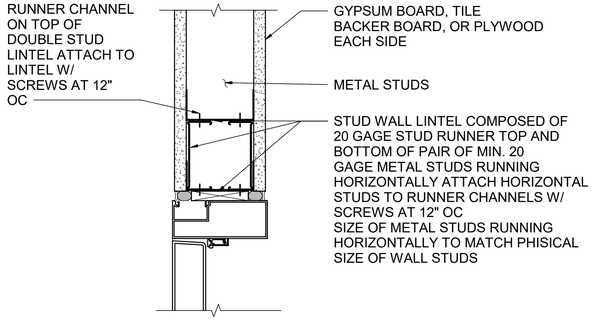


NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

1 / 004A505 HM DOOR IN METAL STUD WALL
SCALE (B)

4 / 004A505 NOT USED
SCALE (B)

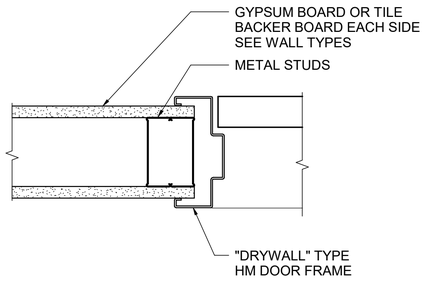


NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

6 / 004A505 ALUMINUM INTERIOR DOOR HEAD
SCALE (B)

8 / 004A505 NOT USED
SCALE (B)

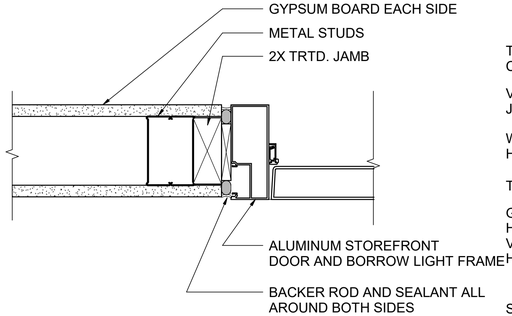
9 / 004A505 NOT USED
SCALE (B)



NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

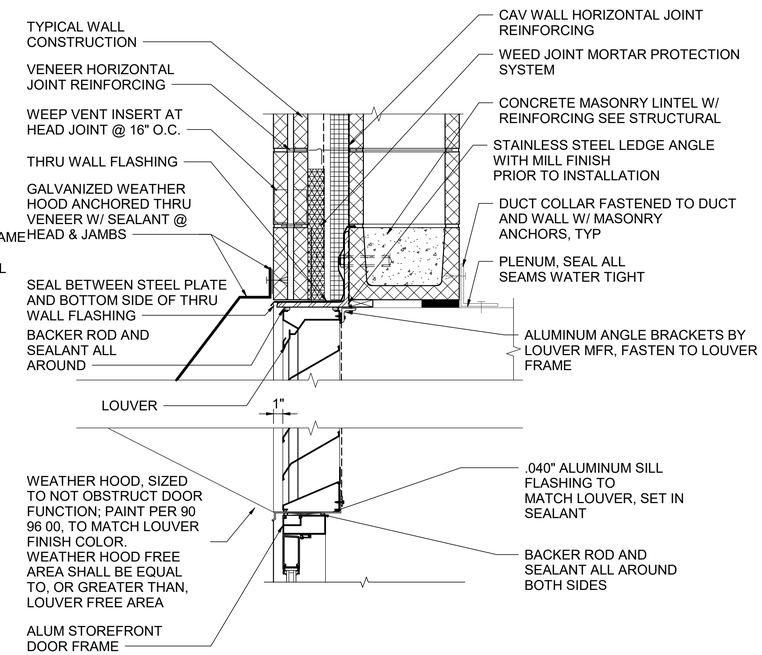
2 / 004A505 HM DOOR JAMB IN METAL STUD WALL
SCALE (B)

5 / 004A505 NOT USED
SCALE (B)



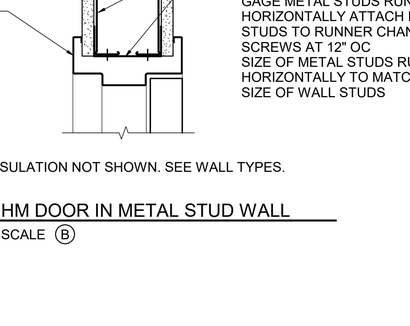
NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

7 / 004A505 ALUMINUM INTERIOR DOOR JAMB
SCALE (B)



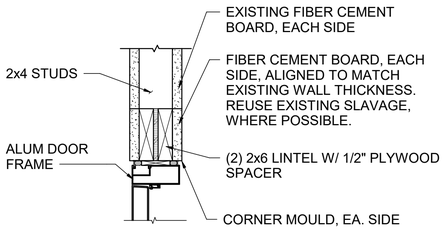
NOTE:
1. SEE STRUCTURAL FOR LINTEL REINFORCEMENT AND SIZES OF STEEL COMPONENTS.

12 / 004A505 LOUVER HEAD, SILL, AND WEATHER HOOD AT DOOR
SCALE (A)

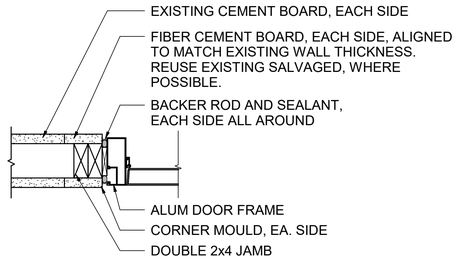


NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

3 / 004A505 HM DOOR JAMB IN CORNER INTERSECTION
SCALE (B)



10 / 004A505 ALUM DOOR HEAD IN EXISTING WALL
SCALE (A)



11 / 004A505 ALUM DOOR JAMB IN EXISTING WALL
SCALE (A)

SCALE (B) 6 3 0 6 — 3" = 1'-0"
SCALE OF FEET

SCALE (A) 12 6 0 12 — 1-1/2" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 2026.04.16 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. HEANEY	004 A505	GENERAL ARCHITECTURAL DETAILS - DOOR FRAMES CONTINUED	113 182752
	DRAWN: R. MILBRATH			PMIS/PKG NO. 318719
	TECH. REVIEW: R. McKINLEY			SHEET 46 OF 99
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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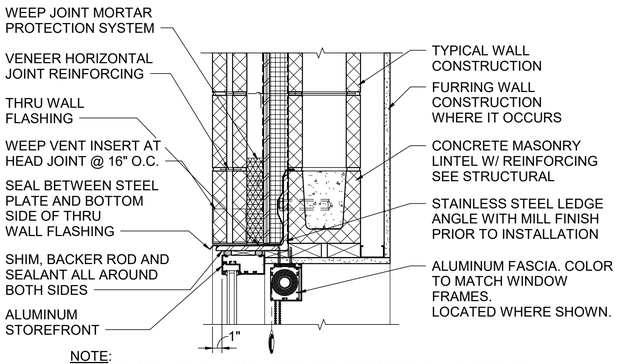
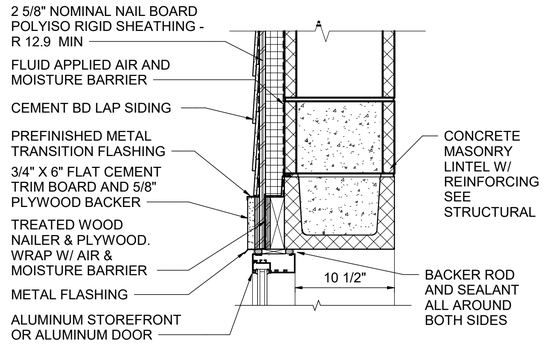


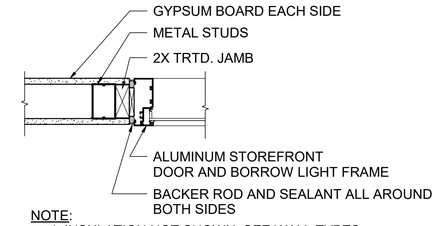
NOTE:
1. SEE STRUCTURAL FOR LINTEL REINFORCEMENT AND SIZES OF STEEL LINTEL COMPONENTS.

1 004A506 ALUMINUM STOREFRONT HEAD AT VENEER
SCALE (A)



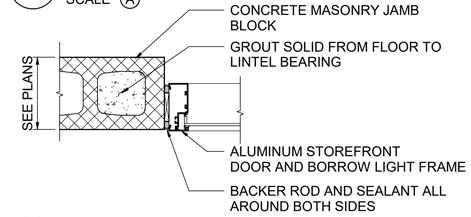
4 004A506 ALUMINUM STOREFRONT HEAD AT SIDING
SCALE (A)

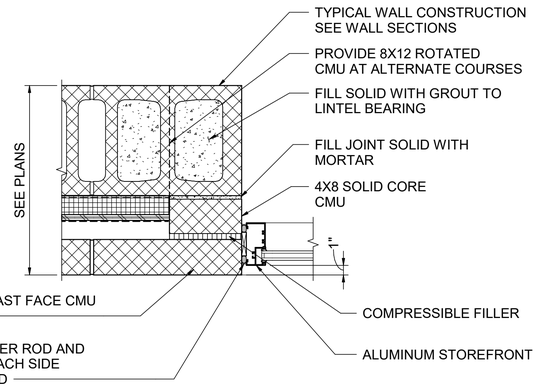
6 004A506 NOT USED
SCALE (A)



NOTE:
1. INSULATION NOT SHOWN. SEE WALL TYPES.

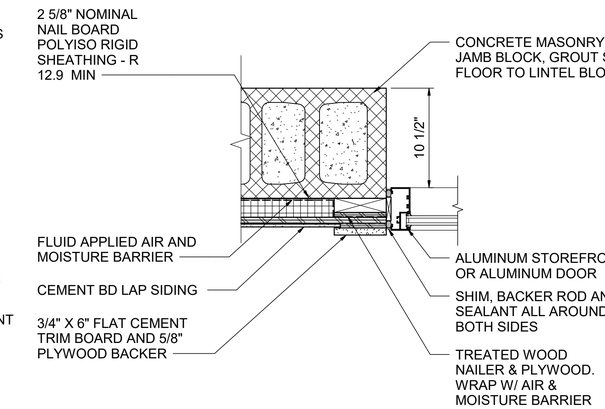
9 004A506 ALUMINUM INTERIOR WINDOW JAMB
SCALE (A)





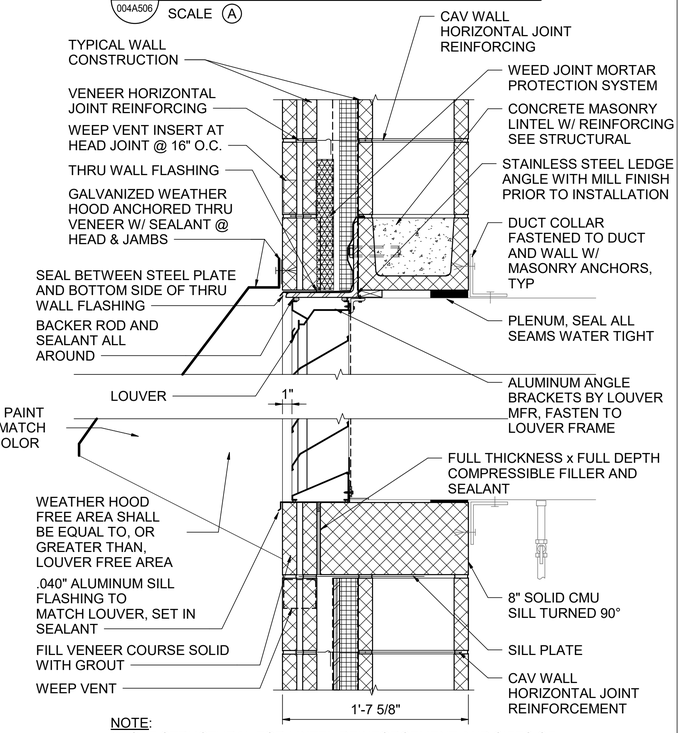
NOTE:
1. SEE STRUCTURAL FOR REINFORCEMENT.

2 004A506 ALUMINUM STOREFRONT JAMB AT VENEER
SCALE (A)



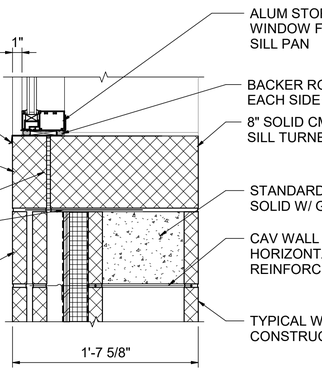
5 004A506 ALUMINUM STOREFRONT JAMB AT SIDING
SCALE (A)

7 004A506 NOT USED
SCALE (A)



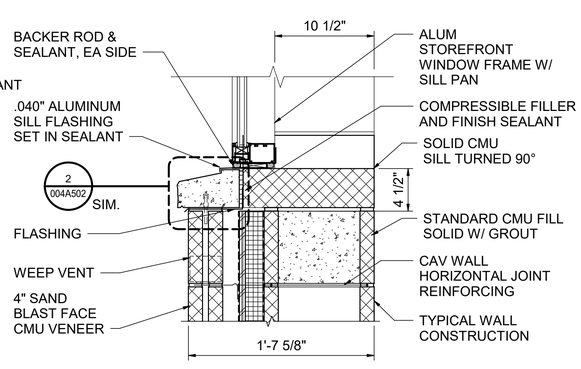
NOTE:
1. SEE STRUCTURAL FOR LINTEL REINFORCEMENT AND SIZES OF STEEL COMPONENTS.

10 004A506 ALUMINUM INTERIOR WINDOW CMU
SCALE (A)



NOTE:
1. SEE STRUCTURAL FOR VERTICAL REINFORCEMENT.

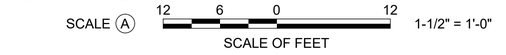
3A 004A506 ALUMINUM STOREFRONT SILL AT VENEER
SCALE (A)



NOTE:
1. SEE STRUCTURAL FOR VERTICAL REINFORCEMENT.

3B 004A506 ALUMINUM STOREFRONT SILL AT SIDING
SCALE (A)

8 004A506 NOT USED
SCALE (A)



11 004A506 LOUVER HEAD, SILL AND WEATHER HOOD
SCALE (A)

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A506	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - ALUMINUM WINDOW FRAMES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 47 OF 99

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_000_GENERAL.rvt
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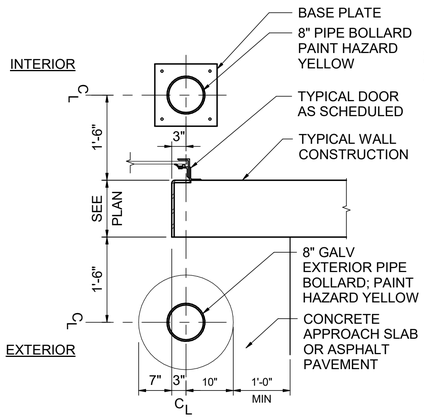
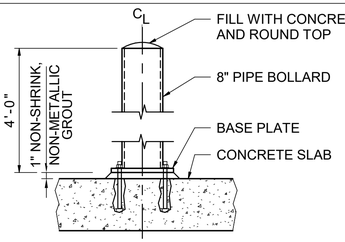
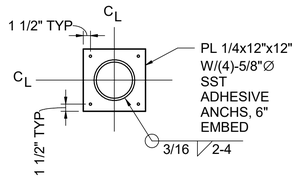


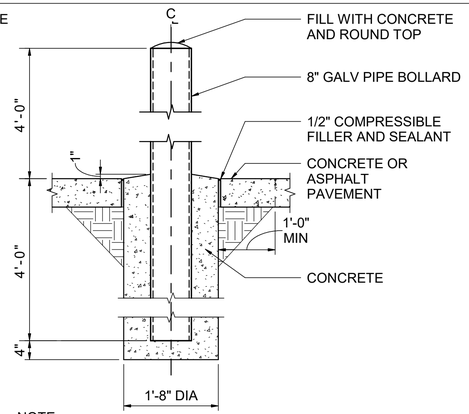
PLAN AT OVERHEAD DOOR



INTERIOR BOLLARD SECTION

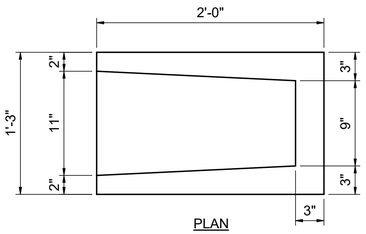


INTERIOR BASE PLATE PLAN

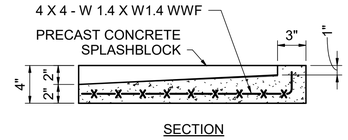


NOTE:
1. REINFORCING FOR CONCRETE SLAB NOT SHOWN

EXTERIOR BOLLARD SECTION

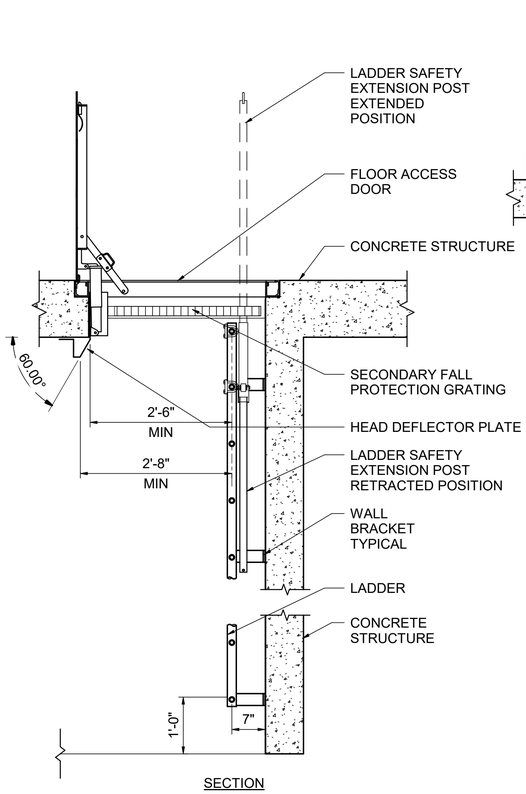


PLAN

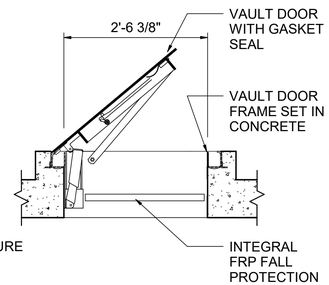


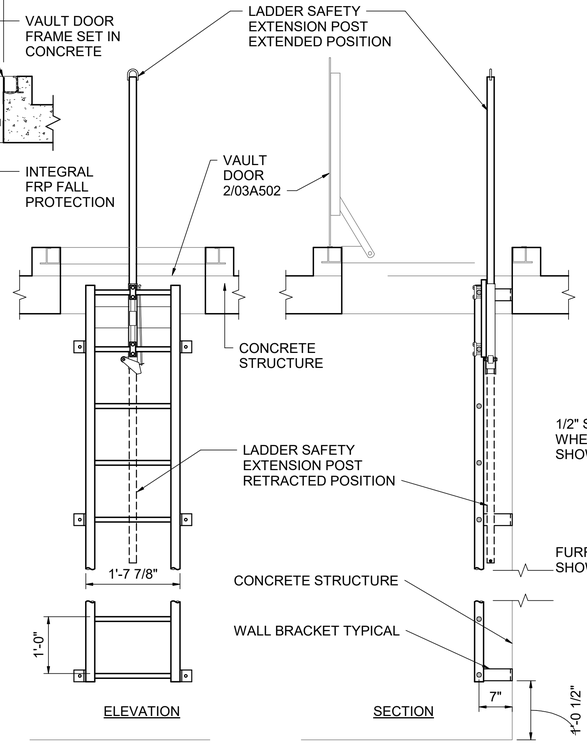
SECTION

1 004A507 BOLLARD
SCALE (A)



SECTION





ELEVATION

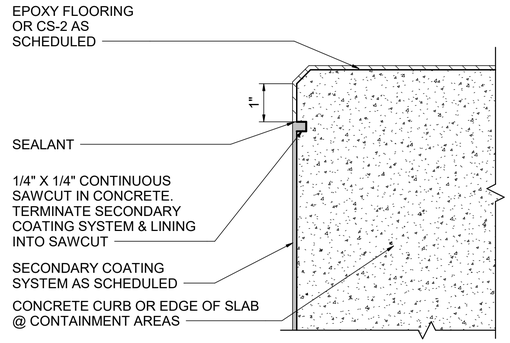
SECTION

NOTE:
1. MAINTAIN MINIMUM 7" FROM CENTERLINE OF RUNG TO ANY OBSTRUCTION.
2. LADDER, SAFETY EXTENSION POST, AND HEAD DEFLECTOR PLATE OMMITTED AT 01-FAD-01, 01-FAD-02, 01-FAD-06, AND 01-FAD-07

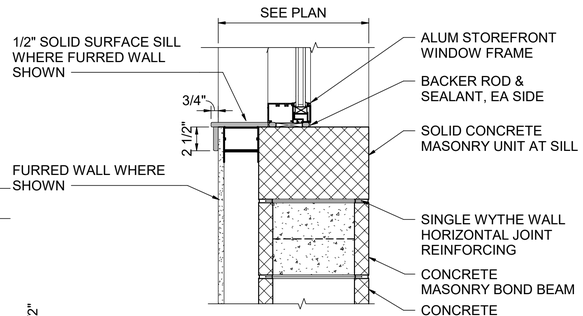
NOTE:
1. PROVIDE HEAD DEFLECTOR PLATE WHEN CLEARANCE ON CLIMBING SIDE OF LADDER IS LESS THAN 2'-6"

3 004A507 LADDER AT FLOOR ACCESS DOOR
SCALE (A)

2 004A507 SPLASHBLOCK
SCALE (C)

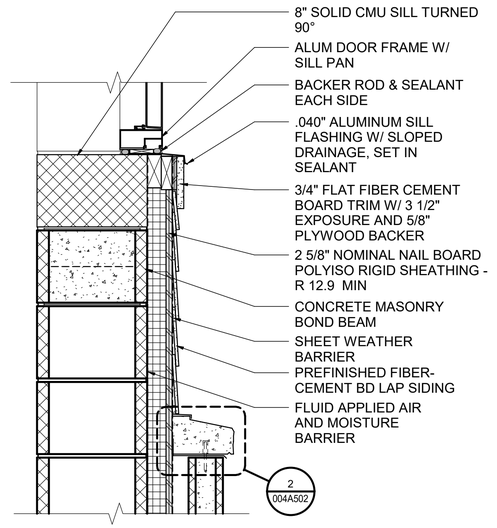


5 004A507 SECONDARY CONTAINMENT CURB TERMINATIION DETAIL



6 004A601 ALUMINUM INTERIOR WINDOW SILL
SCALE (A)

4 004A507 NOT USED
SCALE (B)



7 004A507 DOOR FRAME SILL AT SIDING UPPER WALL
SCALE (A)

SCALE (C) 12 6 0 12 — SCALE OF FEET — 1-1/2" = 1'-0"

SCALE (B) 1 0 1 2 — SCALE OF FEET — 1" = 1'-0"

SCALE (A) 1 0 1 2 3 — SCALE OF FEET — 3/4" = 1'-0"

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.01 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A507	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - HATCHES AND MISCELLANOUS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 48 OF 99

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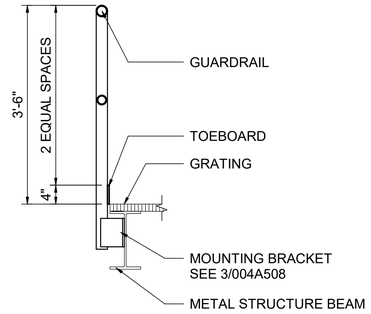
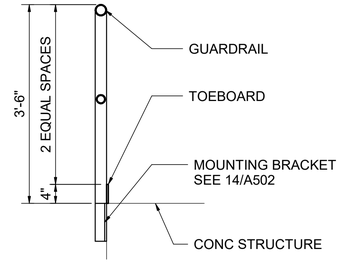
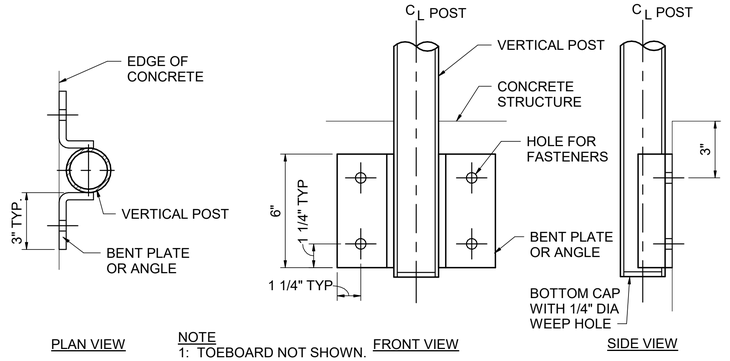


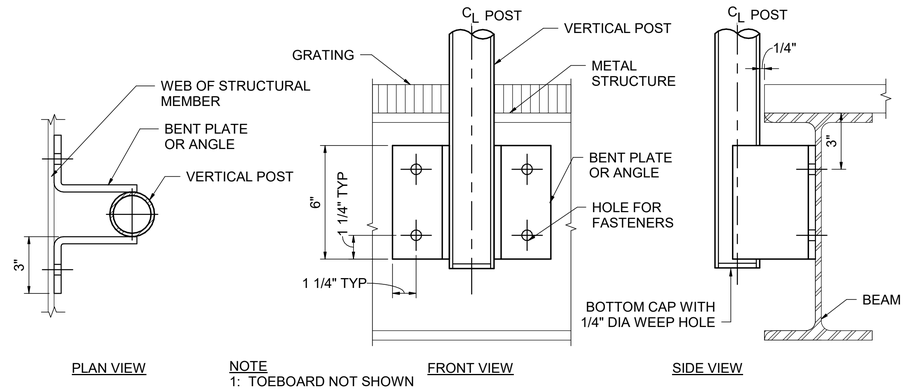
1 004A508 GUARDRAIL SIDE-MOUNT ON METAL CHANNEL
SCALE (A)



2 004A508 GUARDRAIL SIDE-MOUNT ON CONCRETE STRUCTURE
SCALE (A)



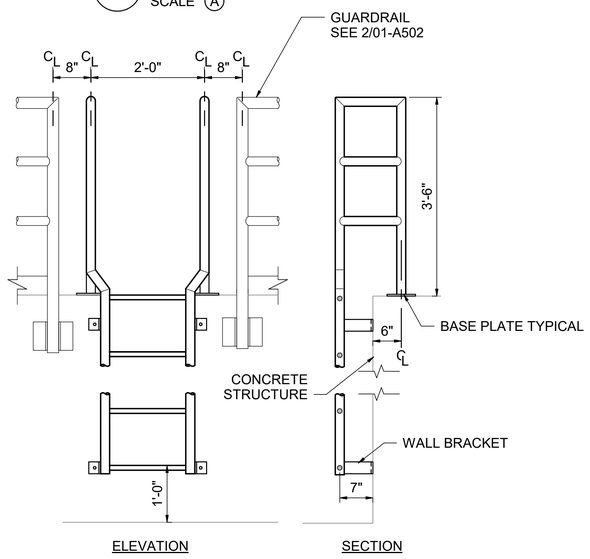
3 004A508 GUARDRAIL MOUNTING BRACKET DETAIL @ CONCRETE
SCALE (B)

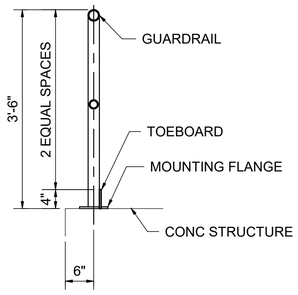


6 004A508 GUARDRAIL MOUNTING BRACKET DETAIL @ METAL BEAM
SCALE (B)

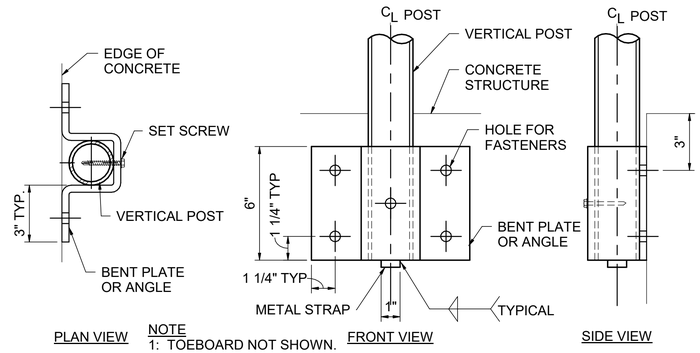
4 004A508 NOT USED
SCALE (A)

5 004A508 NOT USED
SCALE (A)





9 004A508 GUARDRAIL ON CONCRETE STRUCTURE
SCALE (A)



10 004A508 REMOVABLE GUARDRAIL POST MOUNTING BRACKET
SCALE (B)

7 004A508 NOT USED
SCALE (A)

8 004A508 LADDER
SCALE (A)

SCALE (B) 6 3 0 6 3" = 1'-0"
SCALE OF FEET

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.01 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A508	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - RAILINGS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 49 OF 99

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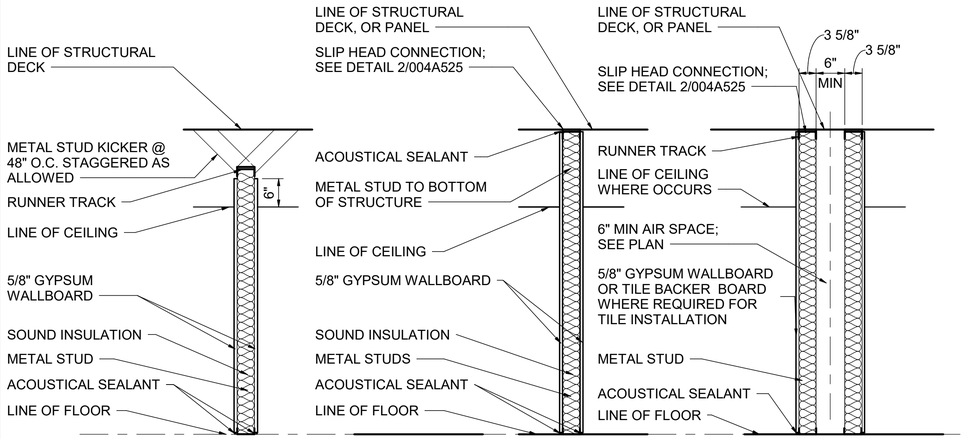
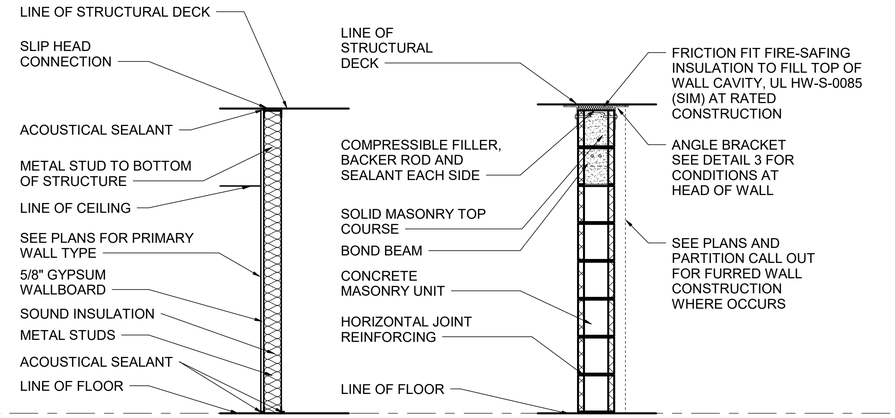




S METAL STUD WALL

- S1.1 3 5/8" METAL STUDS W/ GWB TO 6" ABOVE CEILING
- S1.2 6" METAL STUDS - GWB TO 6" ABOVE CEILING
- S1.3 3 5/8" METAL STUDS W/ MOISTURE RESISTANT BACKER TO 6" ABOVE CEILING
- S1.4 6" METAL STUDS - MOISTURE RESISTANT BACKER TO 6" ABOVE CEILING

S METAL STUD WALL

- S2.1 3 5/8" METAL STUDS - GYPSUM WALL BOARD TO CONTINUE TO BOTTOM OF STRUCTURE
- S2.2 6" METAL STUDS - GYPSUM WALL BOARD TO CONTINUE TO BOTTOM OF STRUCTURE
- S5.2 6" METAL STUDS - GYPSUM WALL BOARD TO CONTINUE TO BOTTOM OF STRUCTURE

S METAL STUD WALL

- S3.1 (2) 3 5/8" METAL STUD PLUMBING WALL W/ BACKER BOARD AT TILE WHERE OCCURS

S METAL STUD WALL

- S4.1 NOT USED

SF METAL STUD FURRING WALL

- SF1.1 2 1/2" METAL STUDS
- SF1.2 3 5/8" METAL STUDS

M CONCRETE MASONRY UNIT WALL

- M1.1 8" CMU - NOT RATED
- M1.2 8" CMU - 1 HR RATED - UL U905
- M2.1 12" CMU - NOT RATED
- M2.2 12" CMU - 1 HR RATED - UL U905

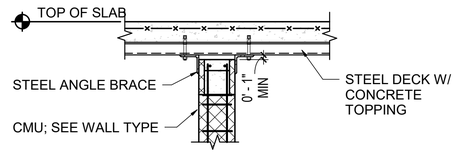
M CONCRETE MASONRY UNIT WALL

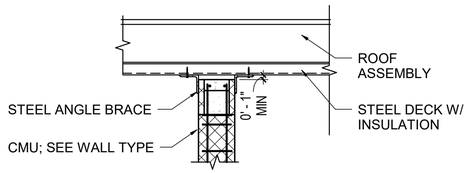
- M3.1 NOT USED

1 004A509 PARTITION TYPES

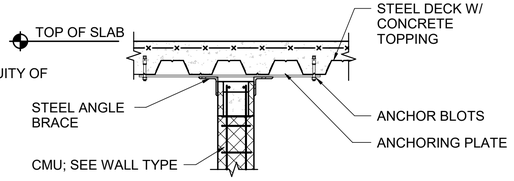
SCALE (A)

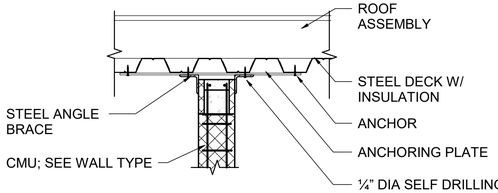
IN SOME INSTANCES, THERMAL & SOUND BATT INSULATION HAS BEEN OMITTED FROM THE DETAILS FOR CLARITY. SEE SPECIFICATIONS, WALL TYPES & REFLECTED CEILING PLANS FOR LOCATIONS REQUIRING INSULATION.

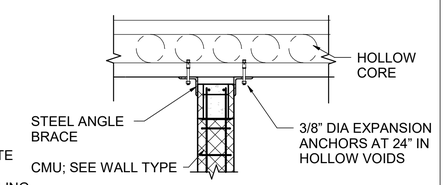




NOTE: SEE STRUCTURAL DETAILS SHEET 003S505 FOR ADDITIONAL INFORMATION REFER ALSO TO DETAIL 4/004A501 AT SIMILAR CONDITION WHERE ROOF JOISTS RUN PERPENDICULAR TO WALL AND PENETRATE MASONRY.

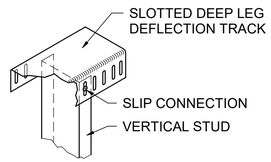






3 TOP OF WALL DETAILS

SCALE (A)



ISOMETRIC VIEW OF SLOTTED TOP TRACK

2 SLIP HEAD DETAIL

SCALE (A)

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"

SCALE OF FEET

GENERAL NOTES

A. CONDITIONS INDICATED SHOW DESIGN INTENT, ESPECIALLY IN REGARD TO ACCOMMODATION OF STRUCTURAL DEFLECTION AND CONTINUITY OF INTEGRITY OF SOUND, SMOKE AND FIRE WALLS.

B. DESIGN INTENT DETAILS ARE NOT INTENDED TO BE PROPRIETARY. CONTRACTOR MAY EMPLOY SIMILAR HEAD-OF-WALL DESIGNS AND COMPONENTS AS APPROPRIATE FOR CONDITIONS.

C. DESIGN INTENT DETAILS MAY NOT SHOW ALL CONDITIONS TO BE ENCOUNTERED ON A PROJECT. COMPLY WITH MANUFACTURER AND UL APPROVED METHODS FOR ATYPICAL CONDITIONS.

D. SECURE TOP OF INTERIOR PARTITIONS TO SUPERSTRUCTURE IN A WAY THAT PROVIDES LATERAL STABILITY (PERPENDICULAR-TO AND IN-PLANE WITH WALL) YET ALLOWING FOR A MINIMUM OF 3/4" OF VERTICAL DEFLECTION OF THE SUPERSTRUCTURE.

E. RIGIDLY SECURE SLOTTED TOP TRACK TO BUILDING SUPERSTRUCTURE IN AN APPROVED MANNER. EMPLOY Z-BARS, COLD-ROLLED CHANNELS OR SIMILAR SPACER TO ACCOMMODATE THICKNESS OF SECTION 07 81 16 FIREPROOFING.

F. SLOTTED TOP TRACK, INDICATED ON THESE DETAILS, IS THE BASIS FOR DESIGN AND REFERS TO DEEP-LEG TRACKS WITH VERTICALLY SLOTTED HOLES. OPTIONAL MANUFACTURERS MAY PROVIDE SIMILAR, PROPRIETARY METHODS IN ACCORDANCE WITH THE PROVISIONS OF SPECIFICATION SECTION 09 22 16.

G. SECURE VERTICAL STUDS TO SLOTTED TOP TRACK AT MID-HEIGHT OF VERTICAL SLOTS IN TRACK. COMPONENTS INTENDED TO SLIDE VERTICALLY AS SUPERSTRUCTURE DEFLECTS.

H. DO NOT SCREW GYPSUM WALLBOARD TO TOP TRACK OR SUPERSTRUCTURE. GWB SCREWS INTO THE STUDS MUST BE AT LEAST 1" BELOW THE BOTTOM OF THE TOP TRACK.

I. REFER TO PARTITION STANDARDS FOR SPECIFIC WALL TYPES.

J. AT FIRE -RATED WALLS REFER TO SPECIFICATION SECTION 07 84 00 FOR ADDITIONAL REQUIREMENTS REGARDING HEAD-OF-WALL CONDITIONS.

K. MAINTAIN ACOUSTIC RATING WHERE SOUND-CONTROL WALLS ARE INDICATED.

L. WHERE A WALL IS DESIGNATED AS BOTH A SOUND-CONTROL WALL AND A FIRE-RATED WALL, REFER TO FIRE-RATED HEAD-OF-WALL CONDITIONS.

M. WHERE A WALL IS DESIGNATED AS A SOUND-CONTROL WALL, FILL ALL VOIDS WITH SOUND ATTENTUATION BATTS (SAB).

N. SMOKE PARTITIONS REQUIRE THAT AT LEAST ONE LAYER OF GWB, AND SOUND-CONTROL WALLS REQUIRE GWB ON BOTH SIDES, BE CUT TO FOLLOW UNDULATING SURFACES OF THE SUPERSTRUCTURE INCLUDING, BUT NOT LIMITED TO, FLUTES IN METAL DECKING. PROVIDE A CONTINUOUS BEAD OF SEALANT (AS SPECIFIED) TO SUPERSTRUCTURE.

O. ALL METAL STUD PARTITIONS THAT EXTEND TO THE CEILING BUT DO NOT EXTEND TO STRUCTURE SHALL BE BRACED WITH ANGLED KNEE BRACING AT 4'-0" ON CENTER MAXIMUM.

LICENSED ARCHITECT CERTIFICATE NO. 85788 RICHARD MCKINLEY DATE SIGNED 05/26/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A509	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - WALL TYPES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 50 OF 99

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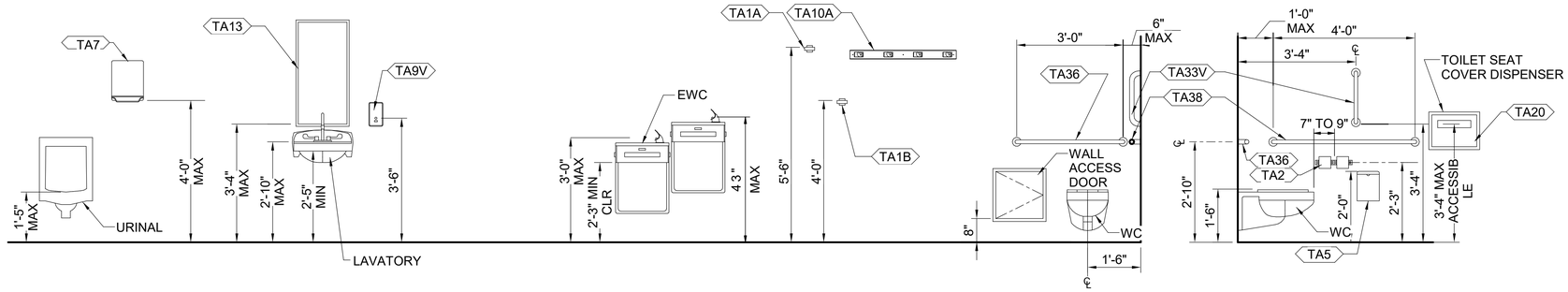

1 004A510 TOILET ACCESSORY MOUNTING HEIGHTS

SCALE (A)

TOILET ACCESSORY LEGEND	
TA1A	SINGLE ROBE HOOK
TA1B	SINGLE ROBE HOOK - ACCESSIBLE
TA2	DOUBLE ROLL TISSUE DISPENSER
TA5	SANITARY NAPKIN DISPOSAL
TA7	PAPER TOWEL DISPENSER
TA9V	SOAP DISPENSER
TA10A	MOP BROOM RACK WITH SHELF
TA13	MIRROR
TA20	TOILET SEAT COVER DISPENSER
TA33V	18 IN VERTICAL GRAB BAR
TA36	36 IN HORIZONTAL GRAB BAR
TA38	48 IN HORIZONTAL GRAB BAR

SCALE (A) 2 0 2 4 1/2" = 1'-0"

SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.05.01 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. HEANEY	004 A510	GENERAL ARCHITECTURAL DETAILS - MOUNTING HEIGHTS	113 182752
	R. MILBRATH			PMIS/PKG NO. 318719
	TECH. REVIEW: R. McKINLEY			SHEET 51 OF 99
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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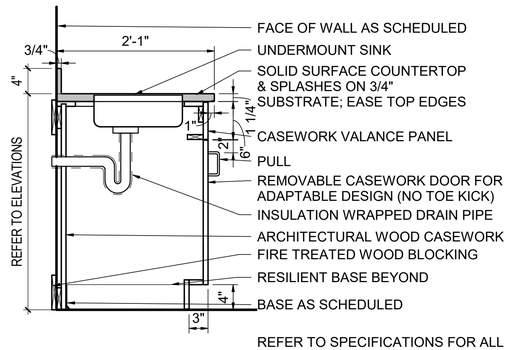
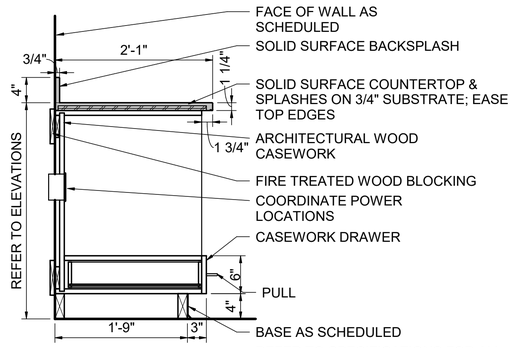


1 004A511 NOT USED
SCALE (A)

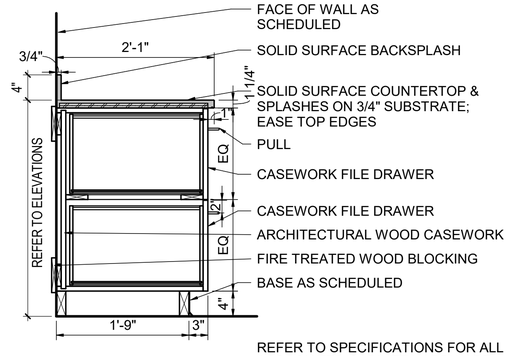
5 004A511 SECTION DETAIL @ OPEN BASE CABINET: 110
SCALE (A)

2 004A511 SECTION DETAIL @ ADA SINK BASE W/DOUBLE DOORS
SCALE (A)

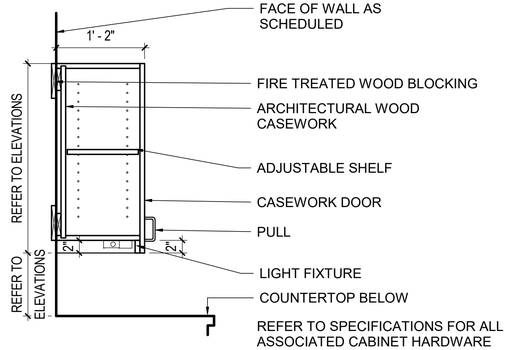
6 004A511 NOT USED
SCALE (A)



3 004A511 SECTION DETAIL @ MICROWAVE BASE CABINET: 218
SCALE (A)



7 SECTION DETAIL @ DOUBLE FILE BASE CABINET: 223
SCALE (A)



4 004A511 SECTION DTL @ UPPER CABINET W/VALANCE SOLID DOOR: 302V
SCALE (A)

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2/19/26 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	004 A511	GENERAL ARCHITECTURAL DETAILS - INTERIORS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 52 OF 99

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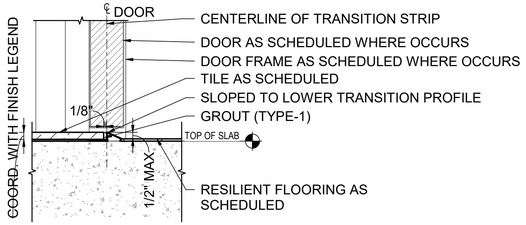
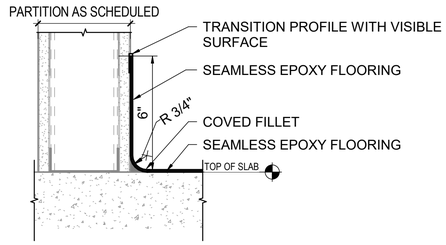
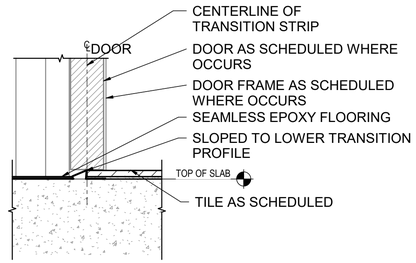


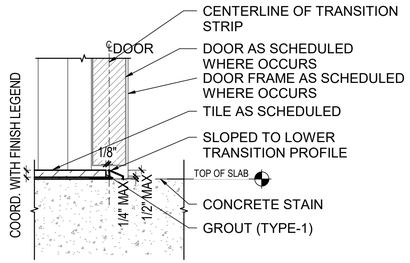
1 DIFFERENTIAL HEIGHT TRANSITION DETAIL @ TILE TO RESILIENT
SCALE (A)



2 SEAMLESS EPOXY INTEGRAL COVE BASE DETAIL
SCALE (A)



5 DIFFERENTIAL HEIGHT TRANSITION DETAIL @ FLUID APPLIED TO TILE
SCALE (A)



6 DIFFERENTIAL HEIGHT TRANSITION DETAIL @ TILE TO CONCRETE
SCALE (A)

3 NOT USED
SCALE (B)

4 NOT USED
004A512
SCALE (B)

7 NOT USED
SCALE (A)

8 NOT USED
SCALE (A)

9 NOT USED
SCALE (B)

SCALE (A) 6 3 0 6 3" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.05.01 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A512	TITLE OF SHEET GENERAL ARCHITECTURAL DETAILS - INTERIORS CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 53 OF 99

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DOOR SCHEDULE

DOOR NUMBER	PAIR	WIDTH	HEIGHT	DOOR TYPE	DOOR MATERIAL	DOOR FINISH	FRAME TYPE	FRAME MATERIAL	FRAME FINISH	GLASS	RATING	HARDWARE SET	DETAILS HEAD	DETAILS JAMB	DETAILS SILL	*REMARKS
052 - RSPH																
52-101 A		3' - 0"	7' - 0"	F	EX	EX	EX	ST	EX	--		6.0	--	--	--	
52-101 B		8' - 0"	8' - 0"	RO	EX	EX	EX	EX	EX	--		37.0	--	--	--	
52-101 C		3' - 0"	7' - 0"	F	AL	FAP/FRP	1	AL	FAP	--		10.0	11/004A504	12/004A504	3/004A504	
52-102 A		3' - 0"	7' - 0"	F	AL	FRP	1	AL	FAP	--		25.0	10/004A505	11/004A505	--	
52-102 B		3' - 0"	7' - 0"	F	EX	EX	EX	EX	EX	--		15.0	--	--	--	
062 - WATER TREATMENT																
62-100 A		3' - 0"	7' - 0"	F	HM	PT	1	HM	PT	-	-	24.0	1/004A505	2/004A505	--	
62-100 B	PR	3' - 0"	7' - 2"	F	AL	FAP/FRP	1	AL	FAP	-	-	14.0	1/004A504	2/004A504	3/004A504	
62-102 A		3' - 0"	7' - 2"	G	AL	FAP/FRP	1	AL	FAP	3	-	17.0	1/004A504	2/004A504	3/004A504	
62-103 A		3' - 0"	7' - 0"	F	WD	ST	1	HM	PT	-	-	36.0	1/004A505	2/004A505	--	E
62-105 A	PR	3' - 0"	7' - 2"	FG	SF	FAP	3	AL	FAP	3	-	1.0	1/004A504	2/004A504	3/004A504	
62-105 B	PR	3' - 0"	7' - 2"	FG	SF	FAP	3	AL	FAP	1	-	22.0	6/004A505	7/004A505	--	
62-106 A		3' - 0"	7' - 0"	G	WD	ST	1	HM	PT	1	-	31.0	1/004A505	2/004A505 &3/004A505 SIM	--	
62-107 A		3' - 0"	7' - 0"	GL	WD	ST	1	HM	PT	1	-	31.0	1/004A505	2/004A505 &3/004A505 SIM	--	E
62-108 A		3' - 0"	7' - 0"	FG	WD	ST	1	AL	FAP	1	-	23.0	6/004A505	7/004A505	--	
62-108 B		3' - 0"	7' - 0"	FG	WD	ST	1	AL	FAP	1		26.0	6/004A505	7/004A505	--	E
62-109 A		3' - 0"	7' - 2"	G	AL	FRP	1	AL	FAP	1	-	33.0	9/004A504	10/004A504	--	
62-109 B		12' - 0"	12' - 0"	RO	AL	FAP	-	AL	-	-	-	37.0	4/004A504	5/004A504	6/004A504	A
62-109 C		3' - 0"	7' - 2"	F	AL	FAP/FRP	1	AL	FAP	-	-	12.0	1/004A504	2/004A504	3/004A504	
62-109 D		3' - 0"	7' - 2"	F	AL	FAP/FRP	1	AL	FAP	-	-	13.0	1/004A504	2/004A504	3/004A504	
62-110 A		3' - 0"	7' - 2"	F	AL	FRP	1	AL	FAP	-	-	20.0	9/004A504	10/004A504	--	
62-110 B	PR	3' - 0"	7' - 0"	F	AL	FAP/FRP	4	AL	FAP	-	-	3.0	1/004A504	2/004A504	3/004A504	D
62-111 A	PR	3' - 0"	7' - 2"	F	AL	FRP	1	AL	FAP	-	-	19.0	9/004A504	10/004A504	--	
62-111 B	PR	3' - 0"	7' - 0"	F	AL	FAP/FRP	4	AL	FAP	-	-	7.0	1/004A504	2/004A504	3/004A504	D
62-112 A		3' - 0"	7' - 2"	F	HM	PT	1	HM	PT	-	1 HR	21.0	9/004A504	10/004A504	--	
62-112 B	PR	3' - 0"	7' - 10"	F	AL	FAP/FRP	4	AL	FAP	-	-	9.0	1/004A504	2/004A504	3/004A504	D
62-113 A		3' - 0"	7' - 2"	F	HM	PT	1	HM	PT	-	1 HR	21.0	9/004A504	10/004A504	--	
62-113 B	PR	3' - 0"	7' - 0"	F	AL	FAP/FRP	4	AL	FAP	-	-	9.0	1/004A504	2/004A504	3/004A504	D

MATERIAL AND FINISH LEGEND

MATERIAL		FINISH	
AL	ALUMINUM/FIBERGLASS REINFORCED POLYMER HYBRID	FAP	FACTORY APPLIED PAINT
HM	HOLLOW METAL	FRP	FIBERGLASS REINFORCED POLYMER
RO	COILING OVERHEAD	PT	FIELD PAINT - SEE COLOR SCHEDULE 09 06 00
SF	STOREFRONT	ST	STAIN
STL	STEEL	EX	EXISTING
WD	WOOD		
EX	EXISTING		

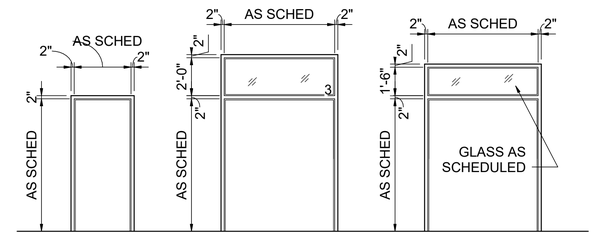
NOTES:
1. SEE DOOR AND FRAME TYPE DETAILS ON THIS SHEET FOR DOOR AND FRAME ELEVATIONS.
2. SEE SECTION 09 06 00 FOR COLOR SCHEDULE.

REMARKS:
A. MOTOR OPERATED ROLLING OVERHEAD DOOR.
B. not used
C. not used
D. TRANSOM TO BE REMOVABLE.
E. DOOR TO BE UNDERCUT.

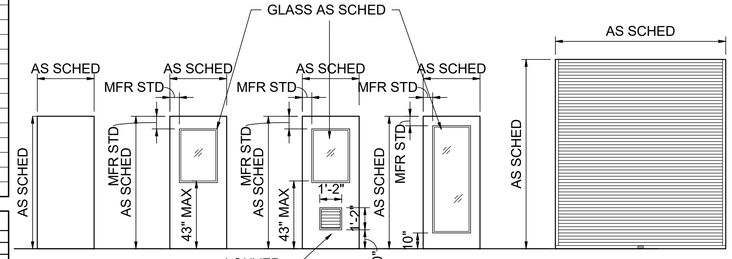
VAULT DOOR SCHEDULE

DOOR NUMBER	DRAWING NO.	PAIR	WIDTH	LENGTH	TYPE	*REMARKS
062 - WATER TREATMENT						
62-VAD01	062A401		3' - 6"	3' - 6"	J-5AL	1-3

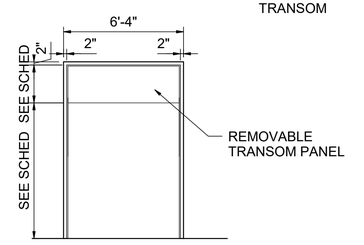
NOTES:
1. PROVIDE SECONDARY FALL PROTECTION SYSTEM
2. H-20 LOAD RATED
3. ALL DIMENSIONS ARE MINIMUM CLEAR OPENING, INCLUDING FALL PROTECTION AND ACCESSORIES
4.



1 TYPICAL FRAME
2 WITH GLASS TRANSOM
3 WITH GLASS TRANSOM



F FLUSH DOOR
G HALF GLASS
GL HALF GLASS
FG FULL GLASS
RO ROLLING OVERHEAD



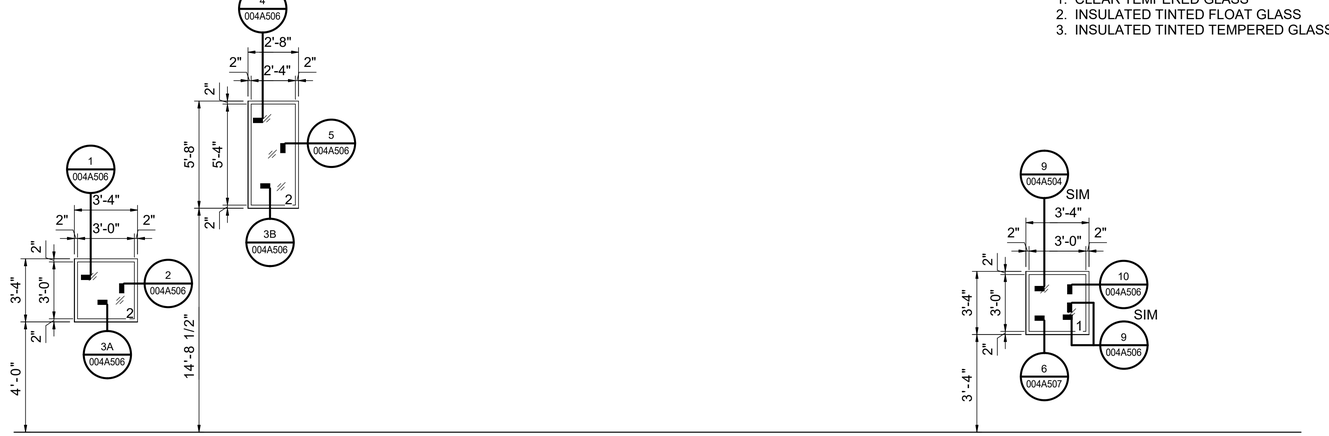
4 WITH TRANSOM

1 004A601 DOOR TYPES
SCALE A

2 FRAME TYPES
SCALE A

GLASS TYPES

1. CLEAR TEMPERED GLASS
2. INSULATED TINTED FLOAT GLASS
3. INSULATED TINTED TEMPERED GLASS



W1 ALUMINUM
W2 ALUMINUM
W7 ALUMINUM (CMU & STUD WALLS)

3 WINDOW ELEVATIONS
SCALE A

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.03.31 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	004 A601	GENERAL ARCHITECTURAL DOOR SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 54 OF 99

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ROOM FINISH SCHEDULE

wt	ROOM NAME	FLOOR	BASE	WALLS NORTH	WALLS EAST	WALLS SOUTH	WALLS WEST	CEILING HEIGHT	CEILING FINISH	*REMARKS
052 - RSPH										
101	PUMP ROOM	CS-2	N	CWFAP	CWFAP	CWFAP	CWAP		ES	
102	OPERATIONS ROOM	CS-2	N	CWFAP	CWFAP	CWFAP	CWAP		ES	
103	ELECTRICAL AREA	CS-2	N	CWFAP	CWFAP	CWFAP	CWAP		ES	
062 - WATER TREATMENT										
101	MECHANICAL ROOM	CS-1	RB-1	P-1	AP-1	AP-1	AP-1		GWB / AP-1	2
102	HALLWAY	CT-1	RB-1	P-1	AP-1	AP-1	AP-1		ACT-1	
103	RESTROOM	CT-1	N	CT-2 / CT-3	CT-2 / CT-3	CT-2 / CT-3	CT-2 / CT-3		ACT-1	
104	BREAK / KITCHEN	CT-1	RB-1	P-1	P-1	P-1	P-1		ACT-1	
105	VESTIBULE	CT-1	RB-1	P-1	P-1	P-1	P-1		ACT-1	
106	CONTROL ROOM	CT-1	RB-1	P-1	P-1	P-1	P-1		ACT-1	
107	LAB	EF-1	EF-1	P-1	P-1	P-1	P-1		ACT-1	
108	SCADA / IT CLOSET	RFT	RB-1	P-1	P-1	P-1	P-1		ACT-1	
109	PROCESS AREA	CS-1	N	P-1	P-1	P-1	P-1			2
110	ELECTRICAL ROOM	CS-1	N	P-1	P-1	P-1	P-1		GWB / AP-1	
111	PROCESS MATERIALS SHOP	CS-1	N	CMU	CMU	CMU	CMU			2
112	BASES DRUM ROOM	CS-1	N	CMU	CMU	CMU	CMU			
113	ACIDS DRUM ROOM	CS-1	N	CMU	CMU	CMU	CMU			

INTERIOR SIGNAGE SCHEDULE

ROOM NUMBER	SIGN COPY
052 - RSPH	
101	PUMP ROOM
102	OPERATIONS ROOM
103	ELECTRICAL AREA
062 - WATER TREATMENT	
101	MECHANICAL ROOM
103	RESTROOM
104	BREAK / KITCHEN
106	CONTROL ROOM
107	LAB
108	SCADA / IT CLOSET
109	PROCESS AREA
110	ELECTRICAL ROOM
111	PROCESS MATERIALS SHOP
112	BASES DRUM ROOM
113	ACIDS DRUM ROOM

MATERIAL AND FINISH LEGEND

FLOOR		BASE	
CPT-#	CARPET TILE	CT-#	CERAMIC TILE
CS-1	CHEMICAL FLOOR SEALER	N	NONE
CS-2	CHEMICAL FLOOR SEALER, HARDENER DENSIFIER	RB-#	RESILIENT BASE
CS-3	CHEMICAL FLOOR SEALER (WATER REPELLENT)	EF-#	TROWELED EPOXY
CT-#	CERAMIC TILE		
EF-#	EPOXY FLOORING		
RFT	RUBBER STATIC CONTROL FLOOR TILE		
EM-1	ENTRANCE MAT		
GT	GRATING		

WALLS		CEILING	
CF-#	CONCRETE FINISH NO. #	ACT-#	ACOUSTICAL CEILING TILE
CMU	CONCRETE MASONRY UNITS - SEALED	C	CONCRETE - NO PAINT
COR	CORTEN STEEL PANEL	ES	EXPOSED STRUCTURE - NO PAINT
CT-#	CERAMIC TILE	HPIC-#	PAINTED STRUCTURE WITH HPIC NO. #
CWFAP	CEMENTITIOUS WOOD FIBER ACOUSTIC PANEL	P-#	ARCHITECTURAL PAINT NO. #
GFMU	GROUND FACE CONCRETE MASONRY UNITS		
HPIC-#	HIGH PERFORMANCE INDUSTRIAL COATING NO. #		
P-#	ARCHITECTURAL PAINT NO. #		
SF	STOREFRONT		
STN	STONE VENEER		

NOTES:
1. SEE DRAWINGS FOR WALL TYPES.
2. PROVIDE CONCRETE FINISH NO. 5 WHERE CONCRETE WALLS ARE INDICATED TO BE PAINTED; SEE SPECIFICATION SECTION 03348.

REMARKS:
1. SEE DETAILS FOR LOWER FRAMED GWB PERIMETER SOFFIT (PAINT PT-1), REMAINDER OF VOLUME IS OPEN TO PAINTED STRUCTURE ABOVE.
2. PAINTED STEEL STRUCTURE. SEE DRAWINGS FOR HEIGHT.

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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A602	TITLE OF SHEET GENERAL ARCHITECTURAL FINISH SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 55 OF 99

INTERIOR FINISH LEGEND

CODE	MANUFACTURER	SERIES/STYLE	COLOR	SIZE	COMMENTS
ACOUSTICAL CEILINGS					
ACT-1	CERTAINTEED	SYMPHONY M HIGH CAC 1222-80-1	WHITE	24" X 24" X 15/16" GRID	PROVIDE HOLD DOWN CLIPS
ACT-2	CERTAINTEED	ECOPHON SOLO CLOUDS RECTANGLE 3544 2077	WHITE	FINAL PANEL SIZE: 48" X 12"	PROVIDE SOLO CLOUDS HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-3	CERTAINTEED	ECOPHON SOLO CLOUDS RECTANGLE 3544 2077	WHITE	FINAL PANEL SIZE: 60" X 12"	PROVIDE SOLO CLOUDS HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-4	CERTAINTEED	ECOPHON SOLO BAFFLES RECTANGLE 3544 2619	WHITE	48" X 8"	PROVIDE SOLO BAFFLES HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-5	CERTAINTEED	ECOPHON SOLO BAFFLES RECTANGLE 3544 2619	WHITE	36" X 8"	PROVIDE SOLO BAFFLES HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-6	CERTAINTEED	ECOPHON SOLO BAFFLES RECTANGLE 3544 7850	WHITE	72" X 24"	PROVIDE SOLO BAFFLES HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-7	CERTAINTEED	ECOPHON SOLO BAFFLES RECTANGLE 3544 7850	WHITE	FINAL PANEL SIZE: 60" X 24"	PROVIDE SOLO BAFFLES HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
ACT-8	CERTAINTEED	ECOPHON SOLO BAFFLES RECTANGLE 3544 7850	WHITE	FINAL PANEL SIZE: 48" X 24"	PROVIDE SOLO BAFFLES HARDWARE (CONNECT ADJUSTABLE WIRE HANGER). PAINT RAW EDGES, COLOR TO MATCH PANEL.
CONCRETE FLOOR SEALER					
CFS	SEE SPECIFICATION	--	--	--	--
CONCRETE FLOOR SEALER AND HARDENER					
CFSH	SEE SPECIFICATION	--	--	--	--
FOLDING PANEL PARTITIONS					
FP	MODERN FOLD	SEE SPECIFICATION FOR REQUIREMENTS	PANEL: MAHARAM TEK-WALL TRUSS - HATCH / FRAME: SHERWIN WILLIAMS NATURAL CHOICE	--	--
INTERIOR PAINTING					
P-1	SHERWIN WILLIAMS	LATEX / EGGSHELL, FLAT FOR CEILINGS	SW 9582 WHITE SAND	--	FIELD COLOR
P-2	SHERWIN WILLIAMS	LATEX / EGGSHELL	SW 9560 NIGHT OUT	--	ACCENT COLOR
P-3	SHERWIN WILLIAMS	LATEX / EGGSHELL	SW 6903 CHEERFUL	--	ACCENT COLOR
P-4	SHERWIN WILLIAMS	LATEX / EGGSHELL	SW 6778 AVIARY BLUE	--	ACCENT COLOR
P-5	SHERWIN WILLIAMS	LATEX / EGGSHELL	SW 6741 DERBYSHIRE	--	ACCENT COLOR
P-D1	SHERWIN WILLIAMS	DRYFALL PAINT	BLACK	--	EXPOSED CEILINGS WHERE NOTED IN RCP
P-M1	SHERWIN WILLIAMS	PAINT TO METAL	SW 9560 NIGHT OUT	--	HM DOOR AND WINDOW FRAMES
PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS					
PL-1	WILSONART	HIGH PRESSURE LAMINATE	PHANTOM ECRU 8212	--	VERTICAL SURFACES
RESILIENT TILE FLOORING					
RB-1	ROPPE	PINNACLE / STANDARD COVE TOE	123 CHARCOAL	4"	--
RESINOUS FLOORING					
EF-1	STONHARD	STONCLAD GS	COOL SHALE	--	SEAMLESS INTEGRAL COVE BASE
ROLLER WINDOW SHADES					
WS-1	MECHO	MECHO 5 MANUAL SHADE SYSTEM / SHADECLOTH: ECOVEIL 1350 SERIES	1369 SILVER BIRCH	5% OPEN	--
SOLID SURFACE FABRICATIONS					
SS-1	WILSONART	--	PEACE GREY 9232SS	1/2" THICK	MATTE / HORIZONTAL SURFACES / WINDOW SILLS
STATIC DISSIPATIVE RUBBER SHEET FLOORING					
RFT	ROPPE	ESD RUBBER STATIC CONTROL TILE	F331 CREMA	24" X 24"	--
TILING					
CT-1	FLORIDA TILE	TIME 2.0	GREY NATURAL	12" X 12"	FLOOR TILE PUBLIC AREAS
CT-2	FLORIDA TILE	WESTMONT	CLAY	12" X 12"	WALL TILE - STACKING BOND
CT-3	FLORIDA TILE	CANAL STREET	CLAY DECO	12" X 24"	WALL TILE ACCENT
TT-1	SCHLUTER	FINEC	ALUMINUM SATIN ANODIZED	MATCH WALL TILE THICKNESS	METAL TRIM - OUTSIDE CORNERS
WALL AND DOOR PROTECTION					
CG-1	INPRO CORPORATION	STAINLESS STEEL CORNER GUARD - SURFACE MOUNTED	304 STAINLESS STEEL / 18-GAUGE	2" LEGS X 10' TO TOP	BOTTOM TO ALIGN TO TOP OF WALL BASE / ALIGN TOP WITH TOP OF WP-2
SSB-1	INPRO CORPORATION	STAINLESS STEEL WALL BASE - PRISM COVE	304 STAINLESS STEEL / 20 GAUGE	6" H	WALL BASE AT WP-1 / WP-2
WP-1	INPRO CORPORATION	STAINLESS STEEL WALL PROTECTION	304 STAINLESS STEEL / 18-GAUGE	48"H AFF PANELS	PROVIDE WITH TOP TRIM CAP / DIVIDER TRIM / INSIDE CORNER TRIM
WP-2	INPRO CORPORATION	STAINLESS STEEL WALL PROTECTION - FULL HEIGHT	304 STAINLESS STEEL / 18-GAUGE	10'H AFF PANELS	PROVIDE WITH INSIDE CORNER TRIM
WP-3	INPRO CORPORATION	STAINLESS STEEL WALL PROTECTION - BACKSPLASH PANEL	304 STAINLESS STEEL / 18-GAUGE	1'-2" PANELS	FIELD VERIFY SIZES BEFORE CUTTING PANEL / BOTTOM OF PANEL TO ALIGN TO TOP OF COUNTERTOP / TOP OF PANEL TO ALIGN TO BOTTOM OF UPPER CASEWORK. IF THERE IS NO UPPER CABINET, MAX HEIGHT OF 24".

NOTE: IN ACCORDANCE WITH OPEN SOURCING REQUIREMENTS, THE FOLLOWING SCHEDULE LISTS THE BASIS OF DESIGN FOR INTERIOR FINISHES. REQUESTS FOR ALTERNATE SOURCING IN COMPLIANCE WITH BUY AMERICA ACT AND ALL OTHER FEDERAL REGULATIONS MAY BE SUBMITTED FOR REVIEW AND CONSIDERATION.

GENERAL FINISH NOTES

A. SEE SERIES 004A SERIES SHEETS FOR STANDARD DETAILS REFERENCED HEREIN.

B. IF FINISH CODE IS NOT SPECIFIED (--), NO FIELD FINISH IS APPLIED. REFER TO CORRESPONDING SPECIFICATION SECTION. IF MANUFACTURER FINISH IS SPECIFIED (MFR), REFER TO CORRESPONDING SPECIFICATION SECTION.

C. MATERIAL AND/OR COLOR SUBSTITUTE SHALL BE REQUESTED PER THE PROJECT POLICIES IDENTIFIED IN THE PROJECT SPECIFICATIONS

D. WHERE ITEMS OR SURFACES ARE NOT SPECIFICALLY MENTIONED, PAINT THE SAME AS SIMILAR ADJACENT MATERIALS OR AREA. IF COLOR OF FINISH IS NOT DESIGNATED, ARCHITECT WILL SELECT FROM STANDARD COLORS OR FINISHES AVAILABLE.

E. GRILLES, DIFFUSERS, ELECTRICAL PANELS, ACCESS PANELS, ETC... WHICH ARE EXPOSED IN FINISHED SPACES SHALL BE PAINTED TO MATCH THE SURFACE ON WHICH THEY OCCUR.

F. PAINT INTERIOR SURFACES OF DUCTS LOW VOC BLACK WHERE SURFACES ARE VISIBLE THROUGH GRILLES AND DIFFUSERS.

G. ALL HOLLOW METAL DOORS AND DOOR FRAMES SHALL BE PAINTED P-M1.

H. ALL WINDOW SILLS SHALL BE SOLID SURFACE UNLESS DETAILED OTHERWISE.

I. REFER TO RCP FOR ROLLER SHADE LOCATIONS.

J. ALL FLOORING TYPE TRANSITIONS AT DOORS SHALL OCCUR ON THE CENTERLINE OF THE DOOR LEAF. TRANSITION TO BE SMOOTH AND EVEN. MAXIMUM VERTICAL CHANGE IN ELEVATION SHALL BE 1/4 IN. REFER TO SPECIFICATIONS FOR REQUIRED FLOOR TRANSITION TRIMS.

K. EXTEND FLOORING UNDER LAV/SINK BASE CABINETS AND OPEN KNEE SPACE.

L. CASEWORK TOE-KICK HEIGHT TO MATCH BASE HEIGHT IN ROOM UNLESS OTHERWISE NOTED (U.N.O.).

M. WALL BASE DOES NOT EXTEND OVER STOREFONT OR OTHER SPECIALTY WALL FINISH UNLESS SHOWN OTHERWISE.

N. ALL WOOD-LOOK PLASTIC LAMINATE ON CASEWORK SHALL RUN VERTICALLY.

O. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.16 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. HEANEY DRAWN: R. MILBRATH TECH. REVIEW: R. McKINLEY DATE: MAY 2026	SUB SHEET NO. 004 A603	TITLE OF SHEET GENERAL ARCHITECTURAL FINISH LEGEND & NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 56 OF 99

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PIPING SERVICE CATEGORIES

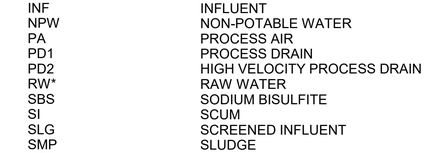
PIPE SERVICE	DISCRIPTION
AIR	COMPRESSED AIR
CA	CAUSTIC SODA
CARB	CARBON
CIP	CLEAN-IN-PLACE
CIT	CITRIC ACID
COAG	COAGULANT
DEC	DECANT
DR	BACKWASH WASTE/CIP DRAIN (VOL 6) POTABLE/FINISHED WATER DRAIN (VOL 5)
DR7	DRAINAGE FROM CHEMICAL ROOMS
EFF	EFFLUENT
FD	FLOOR DRAINS
FLT	FILTRATE
HCL	HYDROCHLORIC ACID
HYPO	SODIUM HYPOCHLORITE
INF	INFLUENT
NPW	NON-POTABLE WATER
PA	PROCESS AIR
PD1	PROCESS DRAIN
PD2	HIGH VELOCITY PROCESS DRAIN
RW*	RAW WATER
SBS	SODIUM BISULFITE
SI	SCUM
SLG	SCREENED INFLUENT
SMP	SLUDGE
UFF	SAMPLE
W*	ULTRAFILTRATION FEED WATER

MODIFIERS	DISCRIPTION
D	DRAIN
DL	DILUTE
FED	FEED
FLL	FILL
OVF	OVERFLOW
REC	RECIRCULATION
RMT	REMOVE
VT	VENT

*RW AND W ARE NUMBERED IN SOME INSTANCES TO CORRESPOND WITH THE PIPE MATERIALS USED IN EACH VOLUME OF THE PROJECT.

INDENTIFICATION SYMBOLOGY

PIPING



EQUIPMENT IDENTIFICATION

EQUIPMENT AND VALVE ABBREVIATIONS

ARV	AIR RELEASE VALVE
AVRV	AIR/VACUUM COMBO RELEASE VALVE
ALV	ALTITUDE VALVE
AE	ANALYTICAL ELEMENT
AIT	ANALYTICAL INDICATING TRANSMITTER
AFV	AUTO-FLUSHING FILTER/STRAINER
FV	AUTOMATED OPEN/CLOSE VALVE
CKV	CHECK VALVE
CMP	COMPRESSOR
END	ENERGY DISSIPATOR SLEEVE
FN	FAN
FL	FLOCCULATORS
FE	FLOW ELEMENT
FIT	COMBINED FLOW ELEMENT / TRANSMITTER
FIT	FLOW INDICATING TRANSMITTER
FT	FLOW TRANSMITTER
HTR	HEATER
H	HOIST
IFT	INTAKE FILTER
KGV	KNIFEGATE VALVE
LE	LEVEL ELEMENT
LIT	LEVEL INDICATING TRANSMITTER
LI	LEVEL INDICATION
LSH	LEVEL SWITCH HIGH
LSL	LEVEL SWITCH LOW
LCP	LOCAL CONTROL PANEL
MEM	MEMBRANE
MX	MIXER
FCV	MODULATING FLOW CONTROL VALVE
PE	PRESSURE ELEMENT
PIT	PRESSURE INDICATING TRANSMITTER
PI	PRESSURE INDICATION
PRV	PRESSURE REGULATING / RELIEF VALVE
PSH	PRESSURE SWITCH HIGH
PSL	PRESSURE SWITCH LOW
PMP	PUMP
PMP	AUX. PUMP (i.e. CHEM INJECTION)
RM	RAPID MIX
SCN	SCREEN
SG	SLIDE GATE
SV	SOLENOID VALVE
STR	STRAINER (MANUAL)
TK	TANK
TIT	TEMPERATURE INDICATING TRANSMITTER
UF	ULTRAFILTRATION
UV	UV UNIT
V	VALVE

PROCESS AREAS

VOLUME 5 - ROARING SPRINGS PUMPHOUSE

1000	GENERAL
1100	PLC
1200	PLANT CONTROL
1300	EWS
1400	DOOR POSITION / INTRUSION / SECURITY
1500	FIRE PUMP / FIRE PANEL
1600	MECHANICAL
1700	HVAC
2000	RAW WATER
2100	SCREENING
2200	PUMPING
2300	FLOW MANAGEMENT
3000	FINISHED WATER AND DISINFECTION
3100	FLOW MANAGEMENT
3200	WATER STORAGE
3300	DILUTE CHLORINE SYSTEMS
3400	STOCK CHLORINE SYSTEMS

PROCESS AREAS

VOLUME 6 - WATER TREATMENT PLANT

1000	GENERAL
1100	PLC
1200	PLANT CONTROL
1300	EWS
1400	DOOR POSITION / INTRUSION / SECURITY
1500	FIRE PANEL
1600	MECHANICAL
1700	HVAC
1800	GAS MONITORING SYSTEM
2000	RAW WATER TANKS & PUMPS
3000	PRETREATMENT
4000	UF FILTRATION & CIP
6000	DISINFECTION
7000	CHEMICAL FEED SYSTEMS
7100	COAGULANT
7200	SODIUM HYPOCHLORITE
7300	CAUSTIC
7400	HYDROCHLORIC ACID
7500	CORROSION CONTROL
7600	CIP
8000	FW TANKS AND PUMPS
8100	FW PUMPING
8200	HIGH DEMAND PUMPING
8300	UPPER ZONE BOOSTER PUMPING
8400	FW STORAGE
8500	FW TEMPORARY VALVES
9000	WASTE RESIDUALS
9100	PROCESS WASTE TANK

PIPE CONNECTIONS

EXPOSED PROCESS PIPING CONNECTION SYMBOLOGY

FLANGE

GROUND COUPLING

WELDED (METAL) OR SOLVENT WELDED (PLASTIC)

THREADED (NPT)

GENERAL NOTES

A. REFER TO CIVIL DRAWINGS FOR CONTINUATION OF YARD PIPING OUTSIDE STRUCTURES.

B. SEE SECTION 40 05 07 FOR DIVISION OF RESPONSIBLITY.

C. BASE ELBOW AND TEE TYPE PIPE SUPPORTS PER DETAIL5/005D501 SHALL BE PROVIDED FOR ALL ELBOW AND TEE TRANSITIONS FROM HORIZONTAL TO VERTICAL PIPING LESS THAN 3'-0" FROM FLOOR.

D. WHERE SHOWN ON PLANS, PENETRATIONS SHALL BE THE TYPE INDICATED, OTHERWISE PROVIDE PENETRATIONS PER DETAILS ON SHEET 005D505.

E. FOR SAMPLE LINES, CHEMICAL PIPING, AND OTHER SMALL DIAMETER PIPING NOT SHOWN ON DRAWINGS, REFER TO P&ID DRAWINGS FOR REQUIRED VALVES.

F. EQUIPMENT AND INSTRUMENTATION TAGS ARE NOT SHOWN ON ALL VIEWS THAT SHOW THE EQUIPMENT AND ARE OFTEN INCLUDED ONLY IN THE MOST REPRESENTATIVE VIEW. REFER TO P&ID DRAWINGS FOR REQUIRED INSTRUMENTATION.

G. CONTRACTOR SHALL COORDINATE THE FOLLOWING PRIOR TO INSTALLATION FOR THE ACTUAL EQUIPMENT BEING SUPPLIED.
   a. EQUIPMENT DIMENSIONS AND LAYOUT
   b. CONNECTING PIPE SIZES, LAYOUTS, AND DIMENSIONS TO CENTERLINE
   c. EQUIPMENT PAD SIZING, THICKNESS FOR PIPE ALIGNMENT AND LOCATION
   d. REQUIRED PIPE PENETRATION SIZE AND LOCATION.

H. REFERENCE SPECIFICATIONS AND P&ID'S FOR TYPES OF VALVES AND VALVE OPERATORS UNLESS SPECIFICALLY NOTED ON DRAWINGS.

I. FOR INSTRUMENTATION WITH LOCAL/REMOTE INDICATOR, COORDINATE MOUNTING LOCATION WITH COR.

CONTROL TYPES

0 - CONTROL PANEL
1 - PRIMARY ELEMENT
2 - AUXILIARY ELEMENT
3 - VALVE
4 - MECHANICAL ELEMENT
5 - LEVEL INSTRUMENT
6 - PRESSURE INSTRUMENT
7 - FLOW INSTRUMENT
8 - ANALYTICAL INSTRUMENT
9 - POSITION / OTHER INSTRUMENT

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN CADD N. LYNCH TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 005 D001	TITLE OF SHEET GENERAL GENERAL NOTES, ABBREVIATIONS, AND LEGEND PROCESS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 57 OF 99

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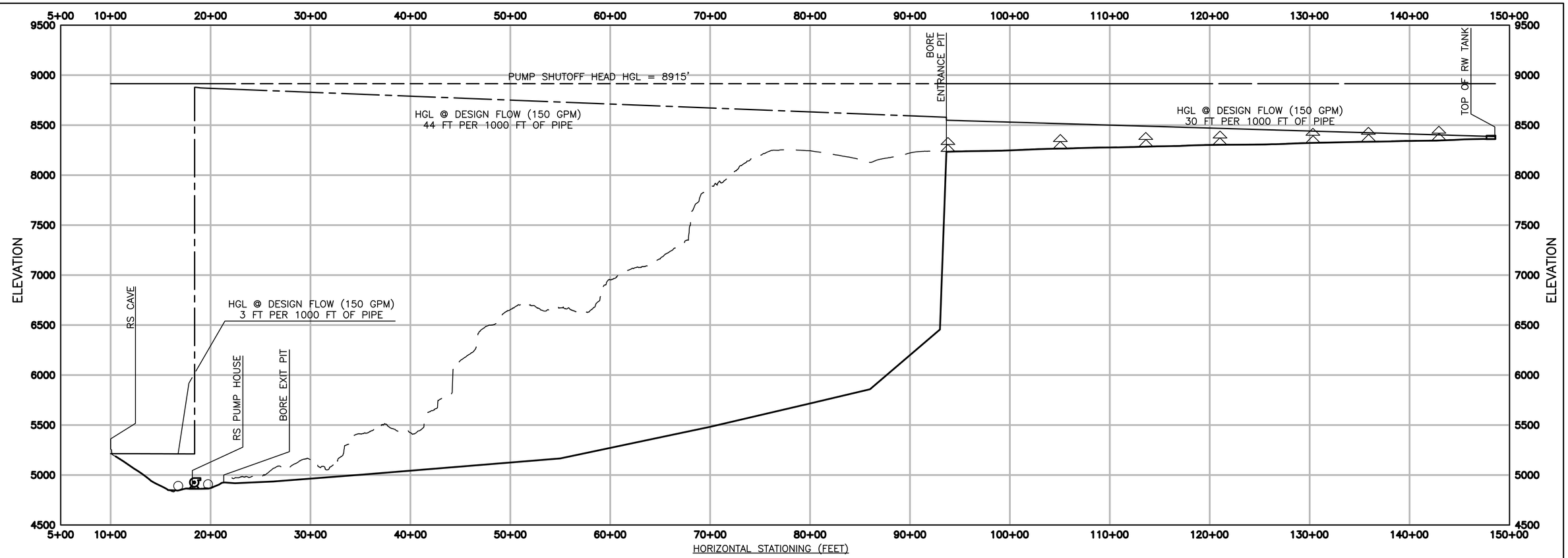

ROARING SPRINGS CAVE TO NORTH RIM WATER TREATMENT PLANT HYDRAULIC PROFILE

STATION	VOLUME STATION	FUNCTION	VALVE TYPE	VALVE SIZE	CONNECTION TYPE	ASME PRESSURE CLASS	OPERATOR TYPE	INSTALLATION DETAILS/NOTES
16+76	509+76	ISOLATION	GATE VALVE	8"	FLANGED	900	HAND WHEEL	PIPELINE BURIED
19+75	512+73	ISOLATION	BALL VALVE	4"	FLANGED	900	HAND WHEEL	PIPELINE IN VALVE VAULT
93+80	102+31	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
105+05	113+56	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
113+62	122+13	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
121+02	129+53	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
130+34	138+85	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
135+90	144+41	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT
142+95	151+46	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	PIPELINE IN VALVE VAULT

ROARING SPRINGS CAVE TO NORTH RIM WATER TREATMENT PLANT PIPELINE VALVE SCHEDULE

STATION	VOLUME STATION	LOCATION
10+00	401+00	RS CAVE
18+18	511+13	RS PUMP HOUSE
18+40	511+35	PUMP
21+32	72+45/501+00	BORE EXIT PIT
94+65	0+00/102+16	BORE ENTRANCE PIT
148+59	156+81	TOP OF RW TANK

ROARING SPRINGS CAVE TO NORTH RIM WATER TREATMENT PLANT PIPELINE LOCATIONS

LEGEND:

- PUMP
- COMBINATION AIR RELEASE AND VACUUM VALVE
- ISOLATION VALVE
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- PUMP SHUTOFF HEAD

500 0 500 1000HORIZ.
500 0 500 1000VERT.
SCALE OF FEET
H: 1"=500' V: 1"=500'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	005 D101	GENERAL RW HYDRAULIC PROFILE AND VALVE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 58 OF 99

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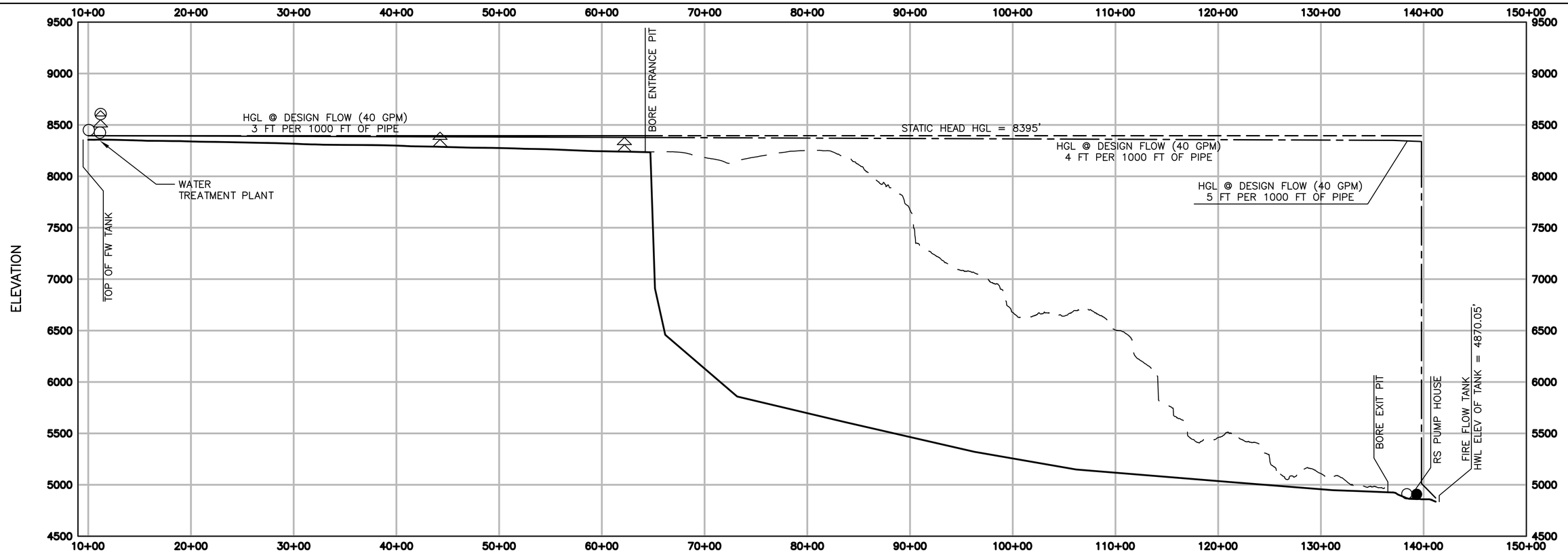

NORTH RIM WATER TREATMENT PLANT TO ROARING SPRINGS PUMPHOUSE HYDRAULIC PROFILE

STATION	VOLUME STATION	FUNCTION	VALVE TYPE	VALVE SIZE	CONNECTION TYPE	ASME PRESSURE CLASS	OPERATOR TYPE	INSTALLATION DETAILS/NOTES
10+15	158+99	ISOLATION	GATE VALVE	12"	FLANGED	150	NUT	WTP WATER VAULT
11+95	157+99	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	125	–	WTP
11+95	157+99	ISOLATION	BUTTERFLY VALVE	4"	FLANGED	150	MANUAL	WTP
11+95	157+99	FLOW CONTROL	BUTTERFLY VALVE	4"	FLANGED	150	ELECTRIC ACTUATOR (FV-68031)	WTP
62+69	103+31	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	125	–	PIPELINE IN VALVE VAULT
44+75	122+13	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	125	–	PIPELINE IN VALVE VAULT
138+89	502+84	ISOLATION	BALL VALVE	4"	FLANGED	900	HAND WHEEL	PIPELINE IN VALVE VAULT
139+79	503+74	PRESSURE REDUCING	AXIAL CONTROL VALVE	2"	FLANGED	900	ELECTRIC ACTUATOR (EDS-53117)	MOUNTED IN RSPH

NORTH RIM WATER TREATMENT PLANT TO RS PUMP HOUSE PIPELINE VALVE SCHEDULE

LEGEND:

- COMBINATION AIR RELEASE AND VACUUM VALVE
- ISOLATION VALVE
- FLOW CONTROL VALVE
- PRESSURE REDUCING VALVE
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- HGL WATERLINE
- STATIC HEAD

	VOLUME STATION	LOCATION
10+00	157+46	TOP OF FW TANK
64+72	0+00/102+16	BORE ENTRANCE PIT
137+05	72+45/501+00	BORE EXIT PIT
139+33	503+29	ROARING SPRINGS PUMP HOUSE
141+17	505+12	TOP OF FIRE FLOW TANK

NORTH RIM WATER TREATMENT PLANT TO RS PUMP HOUSE PIPELINE LOCATIONS

500 0 500 1000HORIZ.
500 0 500 1000VERT.
SCALE OF FEET
H: 1"=500' V: 1"=500'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Re-Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 005 D102	TITLE OF SHEET GENERAL FW HYDRAULIC PROFILE AND VALVE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 59 OF 99

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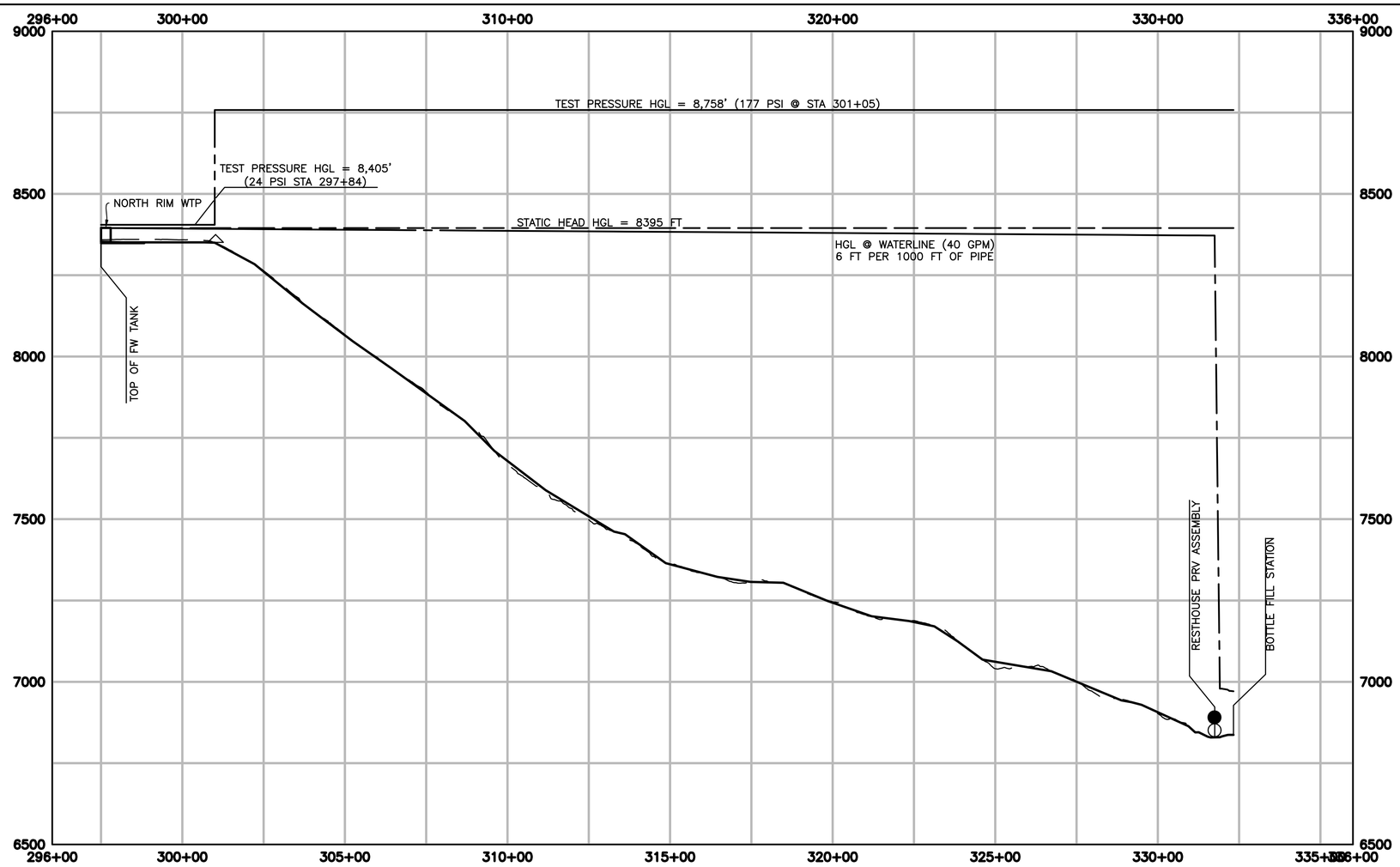


GENERAL NOTE:

A. CONTRACTOR TO SUBMIT TESTING PLAN TO COR PRIOR TO TESTING OF THE STATIC LINE

NORTH RIM WATER TREATMENT PLANT TO SUPAI TUNNEL BOTTLE FILL STATION

STATION	FUNCTION	VALVE TYPE	VALVE SIZE	CONNECTION TYPE	ASME PRESSURE CLASS	OPERATOR TYPE	INSTALLATION DETAILS/NOTES
301+05	COMBINATION AIR RELEASE/VACUUM RELIEF	COMBINATION AIR RELEASE AND VACUUM VALVE	2"	FLANGED	150	–	BURIED IN VALVE VAULT
331+75	PRESSURE REDUCING	PRESSURE REDUCING VALVE	1"	FLANGED	300	–	BURIED IN VALVE VAULT
331+79	ISOLATION	BALL VALVE	1"	FLANGED	150	MANUAL	BURIED IN VALVE VAULT
331+79	ISOLATION	CHECK VALVE	1"	FLANGED	150	–	BURIED IN VALVE VAULT

NORTH RIM WATER TREATMENT PLANT TO SUPAI REST AREA PIPELINE VALVE SCHEDULE

LEGEND:

- △ COMBINATION AIR RELEASE AND VACUUM VALVE
- ○ ISOLATION VALVE
- ● PRESSURE REDUCING VALVE
- —— —— —— EXISTING GROUND AT SURFACE
- ——————— PIPE CENTERLINE
- ——— – ——— WATERLINE HGL
- ——— – – ——— TEST PRESSURE
- – — — — — – STATIC HEAD

STATION	LOCATION
297+84	NORTH RIM WTP
331+75	RESTHOUSE PRV ASSEMBLY
332+33	BOTTLE FILL STATION

NORTH RIM WATER TREATMENT PLANT TO SUPAI TUNNEL PIPELINE LOCATIONS

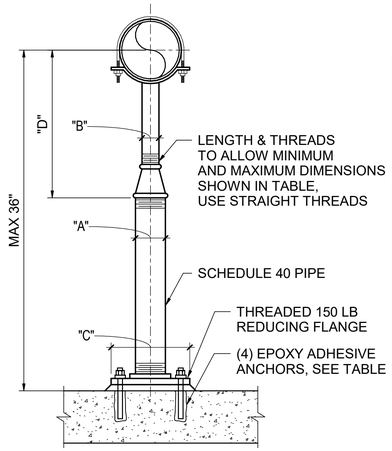
200 0 200 400
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

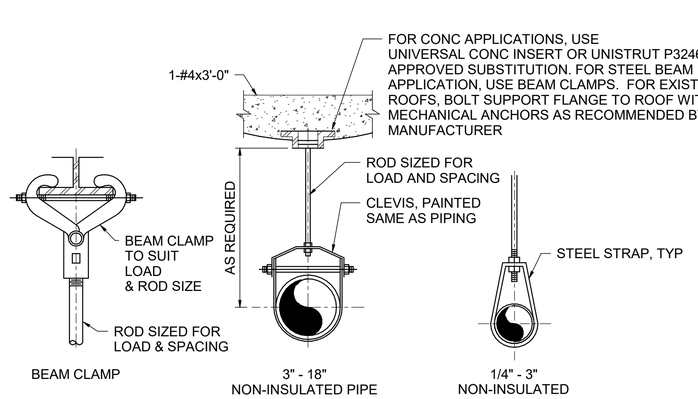
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 005 D103	TITLE OF SHEET GENERAL SUPAI FW HYDRAULIC PROFLE AND VALVE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 60 OF 99

4/21/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\GEN\005-D103.dwg



FLOOR PIPE SUPPORT SCHEDULE DIMENSIONS IN INCHES							
PIPE SIZE	"A"	"B"	"C"	"D"		ANCHORS	
				MINIMUM	MAXIMUM	DIA	EMBED
2	2 1/2	-	9	-	-	5/8	5
2 1/2	2 1/2	-	9	-	-	5/8	5
3	2 1/2	-	9	-	-	5/8	5
3 1/2	2 1/2	-	9	10 1/4	10 1/4	5/8	5
4	4	3	9	9 1/2	14	5/8	5
5	4	3	9	10 1/8	14 5/8	5/8	5
6	4	3	9	10 3/4	15 1/4	5/8	5
8	4	3	9	12 1/4	16 3/4	5/8	5
10	4	3	9	13 3/4	18 1/4	5/8	5
12	4	3	9	14 5/8	19 1/8	5/8	5
14	4	3	11	15 13/16	20 5/16	3/4	6 5/8
16	4	3	11	16 13/16	21 5/16	3/4	6 5/8
18	6	4	13 1/2	19 1/8	23 5/8	3/4	6 5/8
20	6	4	13 1/2	20 1/8	24 5/8	3/4	6 5/8
22	6	4	13 1/2	21 5/16	25 13/16	3/4	6 5/8
24	6	4	13 1/2	23 5/16	27 13/16	3/4	6 5/8
26	6	4	13 1/2	24 5/16	28 13/16	3/4	6 5/8
30	6	4	13 1/2	26 7/16	30 15/16	3/4	6 5/8
32	6	4	13 1/2	27 7/16	31 15/16	3/4	6 5/8
36	6	4	13 1/2	29 7/16	33 15/16	3/4	6 5/8

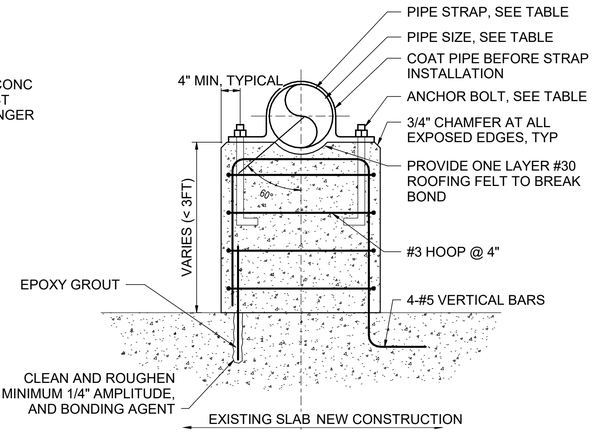
1 005D501 FLOOR PIPE SUPPORT
NOT TO SCALE



NOTES:

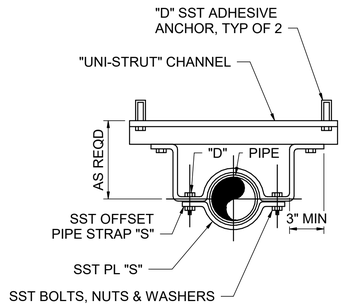
1. FOR INSULATED PIPES, USE GRINNELL FIG 167 OR ELCEN FIG 219 INSULATION PROTECTION SHIELD.
2. TOTAL LOADING ON EACH CONC. INSERT OR OTHER TYPE HANGER ROD ANCHOR SHALL NOT EXCEED MFR'S RECOMMENDED LOADINGS.
3. TOTAL LOADING OF EACH BEAM CLAMP SHALL NOT EXCEED MFR RECOMMENDED LOADINGS.

2 005D501 PIPE HANGER DETAIL
NOT TO SCALE



PIPE SIZE, DIA	PIPE STRAP	ANCHOR BOLTS, DIA	PEDESTAL THICKNESS
6"-12"	1/4"x2"	1/2"	12"
14"-24"	3/8"x2 1/2"	5/8"	16"
30"-48"	1/2"x3"	3/4"	24"

3 005D501 CONCRETE PEDESTAL SUPPORT
NOT TO SCALE

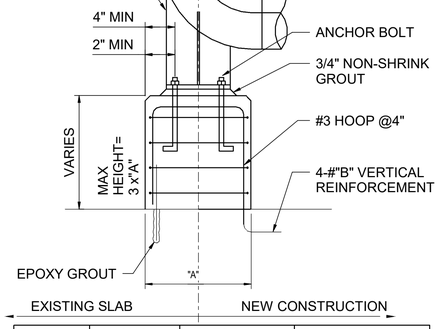


NOTE:

1. SPACING PER SPECIFICATION.
2. THIS DETAIL CAN ALSO BE USED TO SUPPORT PIPE OFF FLOOR.

PIPE SIZE	"D"	"S"
< 2-1/2"	3/8"	1/4"x1"
3"-4"	5/8"	1/4"x1-1/2"
6"-8"	3/4"	3/8"x2"
10"-14"	7/8"	5/8"x2"

4 005D501 OFFSET PIPE SUPPORT
NOT TO SCALE

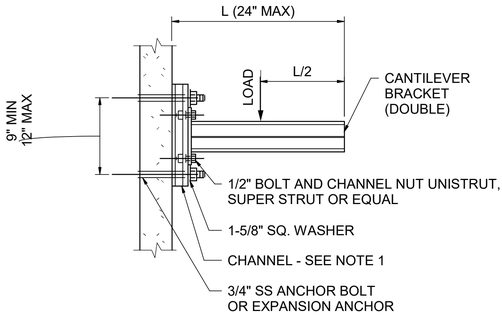


PIPE SIZE, DIA	ANCHOR BOLTS, DIA	PEDESTAL DIMENSION "A"	"B" VERTICAL REINFORCEMENT
TO 6"	1/2"	12"	#5
TO 16"	3/4"	16"	#7
TO 30"	3/4"	20"	#8
36"	7/8"	36"	6-#8

NOTE:

1. USE FOR DUCTILE IRON OR CAST IRON PIPE ONLY.

5 005D501 BASE ELBOW AND TEE PEDESTAL
NOT TO SCALE



NOTES:

1. FOR EXISTING CONCRETE, NEW PRECAST CONCRETE, OR NEW CONCRETE MASONRY UNITS, USE SURFACE MOUNT CHANNEL, UNISTRUT P-1000, SUPER STRUT A-1200, ELCEN 600 OR EQUAL.
2. FOR ALL POURED-IN-PLACE CONCRETE PLACED UNDER THIS CONTRACT, USE EMBEDDED CONCRETE INSERT CHANNEL, UNISTRUT P3200 SERIES WITH END CAPS, OR EQUAL, AND DELETE THE SURFACE MOUNT ANCHOR BOLTS.
3. BRACKET MAY BE USED WITH VARIOUS HANGER, ROLLER, GUIDE, AND CLAMP ASSEMBLIES.

6 005D501 SMALL PIPE (3 INCH) INSTRUMENT CANTILEVER MOUNTING
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	M. HOWARD	005 D501	GENERAL PROCESS DETAILS - PIPE SUPPORTS
	D. DEATERLA		
	TECH. REVIEW: J. GLOVER		
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	61 OF 99

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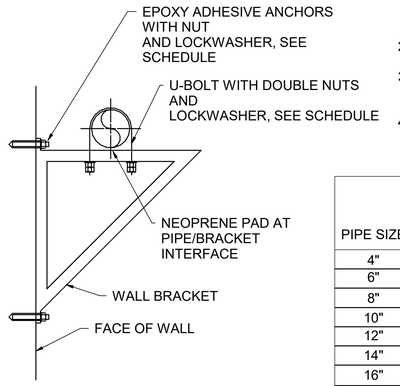
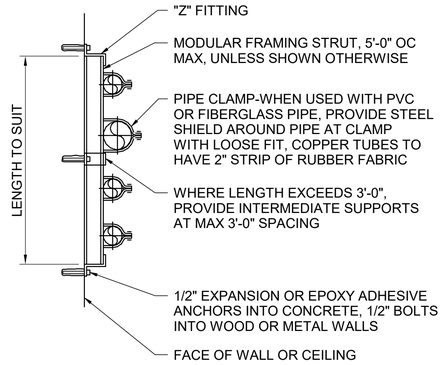


NOTE:

1. WHEN USED WITH PVC OR FIBERGLASS PIPE, PROVIDE STEEL SHIELD AROUND PIPE AT CLAMP WITH LOOSE FIT; COPPER TUBES TO HAVE STRIP OF RUBBER FABRIC.
2. FABRICATE FROM STL, 3/8" MIN THICKNESS.
3. HOT DIP GALVANIZE AFTER FABRICATION: U-BOLT AND NUTS, PIPE SADDLE ASSEMBLY & WALL BRACKET.
4. HOT DIP GALVANIZE AFTER FABRICATION OR USE PAINT SYSTEM FOR SUBMERGED APPLICATIONS PER SPECIFICATIONS.

PIPE SIZE	SPACING	ANCHOR BOLT SIZE FOR STANDARD BRACKETS	U-BOLT DIA
4"	10'	3/4"	1/2"
6"	15'	1"	5/8"
8"	15'	1"	5/8"
10"	20'	1"	3/4"
12"	20'	1"	7/8"
14"	20'	1"	7/8"
16"	20'	1"	7/8"
18"	16'	1"	1"
20"	13'	1"	1"
24"	10'	1"	1"

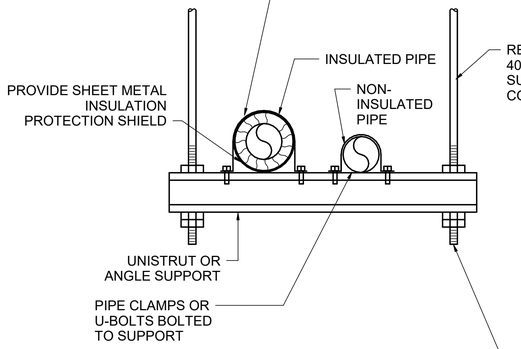
1 005D502 WALL BRACKET PIPE SUPPORT
NOT TO SCALE



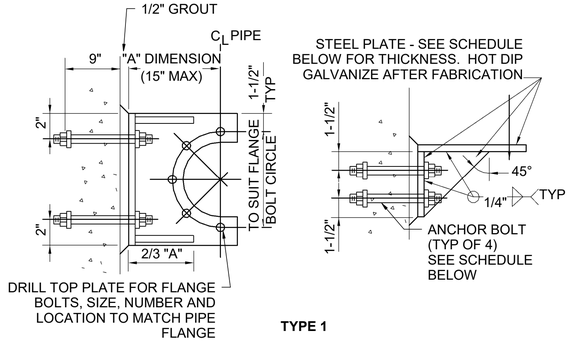
NOTES:

1. UP TO 8" DIAMETER PIPES ONLY.
2. MAXIMUM OF 3-6" PIPES.

2 005D502 SURFACE MOUNT MODULAR STRUT
NOT TO SCALE



3 005D502 TRAPEZE PIPE SUPPORT
NOT TO SCALE

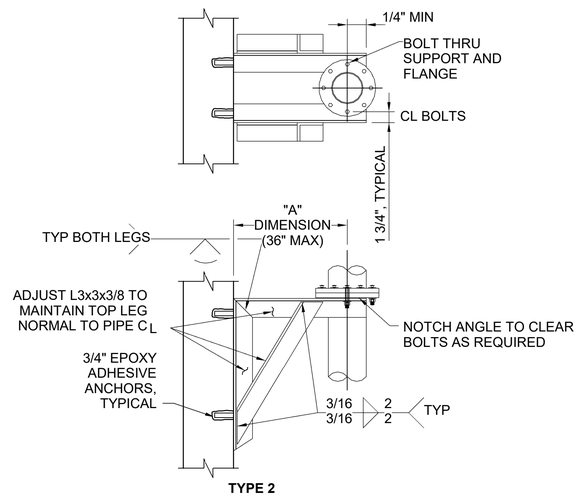


TYPE 1

COMPONENT SCHEDULE		
PIPE SIZE (INCHES)	BOLT SIZE (INCHES)	PLATE THICKNESS (INCHES)
2 & 2-1/2	3/8	1/4
3 TO 6	1/2	3/8
8 TO 12	5/8	3/8
14 TO 16	3/4	1/2
18 TO 20	7/8	5/8
24 TO 30	1	3/4

NOTES:

1. USE TYPE ONLY WHERE SPECIFIED ON THE DRAWINGS.
2. FOR STRUCTURAL ATTACHMENT, OMIT GROUT AND USE BOLTS, NUTS AND LOCK WASHERS IN BOLT SIZE SPECIFIED.
3. USE TYPE 1 FOR DIMENSION "A" <15". USE TYPE 2 FOR DIMENSION "A" >15".



TYPE 2

4 RISER PIPE SUPPORTS
NO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	M. HOWARD	005 D502	GENERAL PROCESS DETAILS - PIPE SUPPORTS CONTINUED	113 182752
	D. DEATERLA			PMIS/PKG NO. 318719
	TECH. REVIEW: J. GLOVER			SHEET 62 OF 99
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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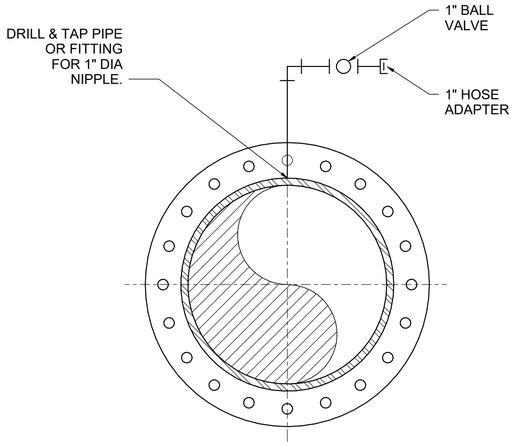
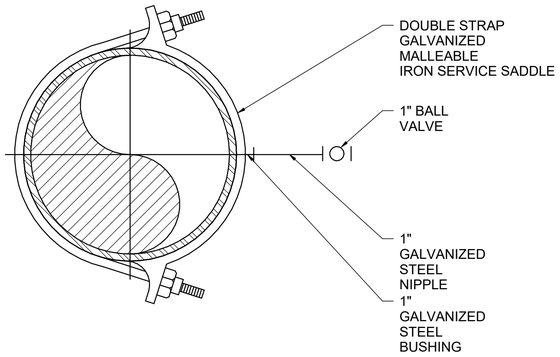
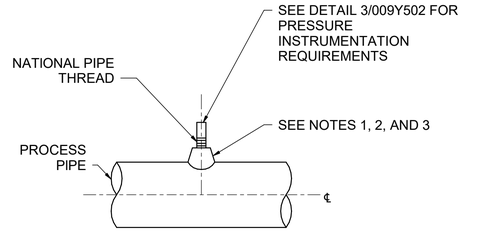


1 005D503 FLUSHING CONNECTION
NOT TO SCALE

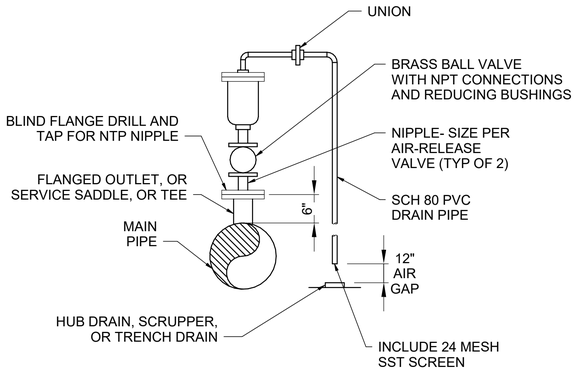


2 005D503 SADDLE TAP
NOT TO SCALE



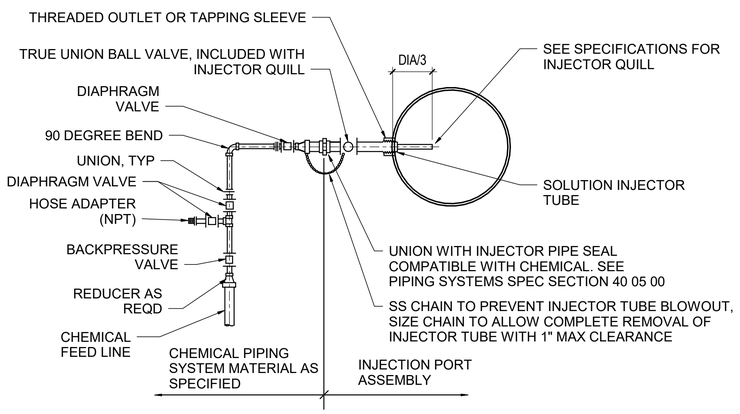
NOTES:
1. FOR STEEL, GALVANIZED STEEL AND PVC 2 1/2" AND SMALLER USE A BUSHING IN A TEE.
2. FOR DUCTILE IRON AND PVC LARGER THAN 2 1/2" ALL SIZES, USE PIPE SADDLE WITH A BUSHING. FOR TAPS SHOWN ON FITTINGS USE TAPPED BOSS.
3. FOR STEEL AND STAINLESS STEEL PIPES 3" AND LARGER, AND PRESSURE VESSELS, USE THREAD-O-LET AS SHOWN.

3 005D503 PIPE TAP
NOT TO SCALE



NOTE:
1. COMPONENTS TO MATCH OR EXCEED PRESSURE RATING OF MAIN PIPE.

4 005D503 AIR RELEASE AND COMBINATION VALVES
NOT TO SCALE



5 005D503 CHEMICAL INJECTOR QUILL
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	M. HOWARD	005 D503	GENERAL PROCESS DETAILS - PIPES AND APPURTENANCES
	D. DEATERLA		
	TECH. REVIEW: J. GLOVER		NORTH RIM - GRAND CANYON NATIONAL PARK
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	63 OF 99

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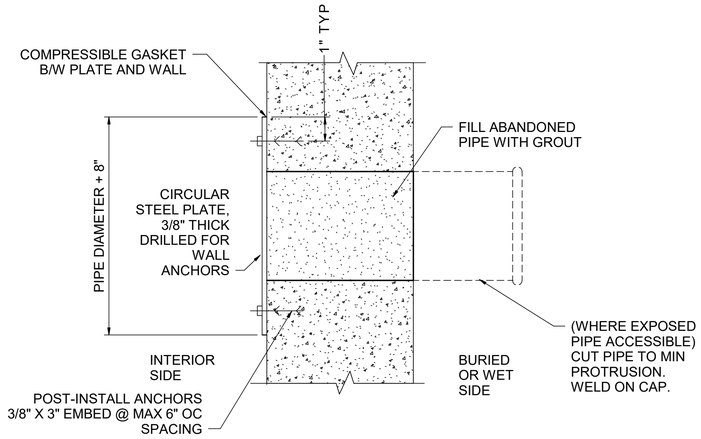
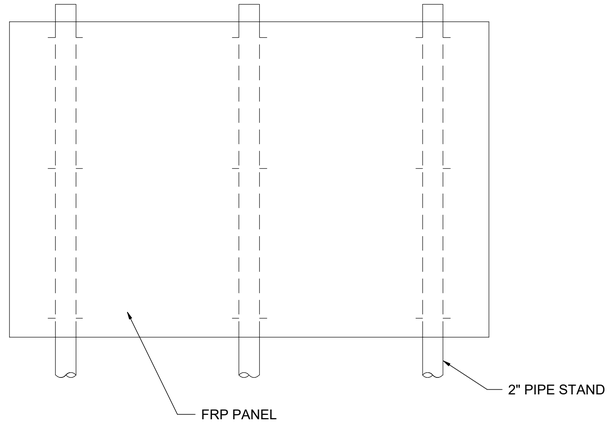
SEE DETAIL 7/003S507

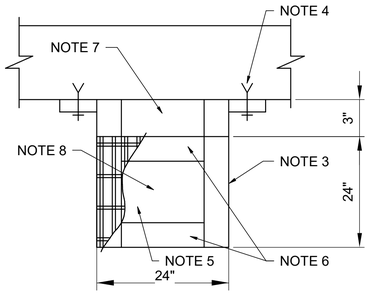


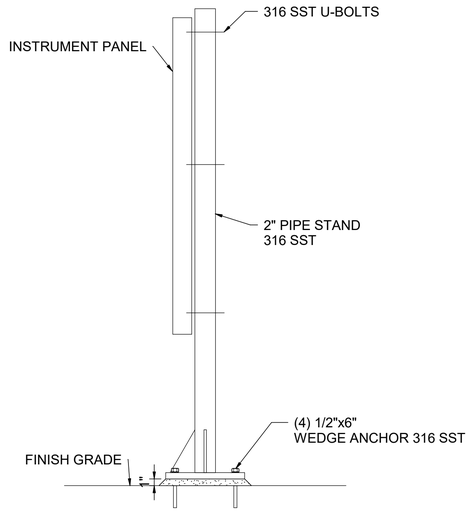
MATERIALS	
WASTEWATER SERVICE	ALL OTHER SERVICE
METALS AND HARDWARE - 304/316 SS ELASTOMERS - EPDM	METALS AND HARDWARE - GALVANIZED OR CARBON STEEL (COATED PER SECTION 09 96 00) ELASTOMERS - BUNA-N

1 005D504 **UNDERSLAB PIPE ENCASEMENT**
NOT TO SCALE



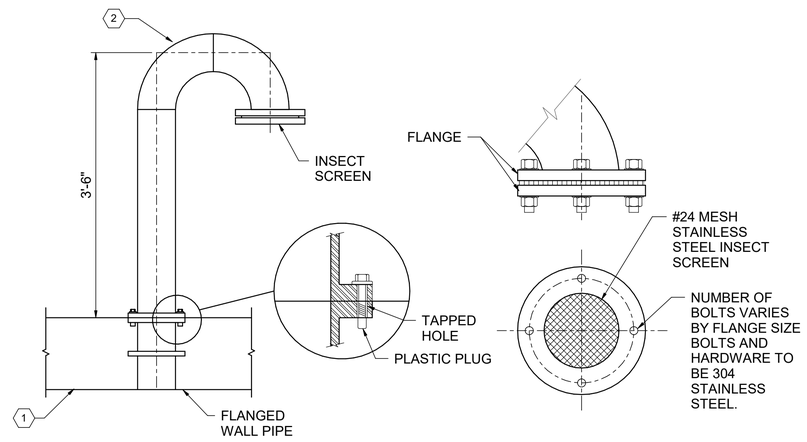
2 005D504 **EXISTING WALL PENETRATION PLUGGING DETAIL**
NOT TO SCALE





5 005D504 **INSTRUMENT PANEL MOUNTING STAND**
NOT TO SCALE

NOTES:
1. MOUNT SHELF MAXIMUM 3 FT ABOVE FINISHED FLOOR
2. PUMPS NOT REQUIRED SHELF SHALL BE MOUNTED ON EQUIPMENT PAD.
3. FIBERGLASS SUPPORT BRACKET (MINIMUM OF 2). SIZE REQUIRING A HANDLE TOTAL WEIGHT OF 75 LBS.
4. 3/4" DIA HOLE FOR 5/8" DIA SST ANCHOR. (TYP 2 EA BRACKET).
5. 1 1/2" FIBERGLASS GRATING
6. L2" x 2" x1/4" FIBERGLASS (TYP).
7. GAP TO ALLOW PIPING PASS THROUGH.
8. ASSEBLE BY FRP LAY-UP OR BOLT TOGETHER WITH FIBERGLASS HARDWARE.



GENERAL NOTES:

A. PRESSURE TESTING OF VENTS NOT REQUIRED.

KEY NOTES:

1. CONCRETE SHOWN FOR PURPOSE OF THIS DETAIL. CONNECT VENT TO OTHER TANK MATERIALS AS NEEDED.
2. VENT MATERIAL PER PRIMARY DESIGNATED PIPE MATERIAL.

3 005D504 **GOOSE NECK VENT DETAIL**
NOT TO SCALE

4 005D504 **PUMP SHELF**
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/26 ARIZONA U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	M. HOWARD
	D. DEATERLA
TECH. REVIEW:	J. GLOVER
DATE:	MAY 2026
SUB SHEET NO.	005 D504
TITLE OF SHEET	GENERAL PROCESS DETAILS - MISCELLANEOUS
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	64 OF 99

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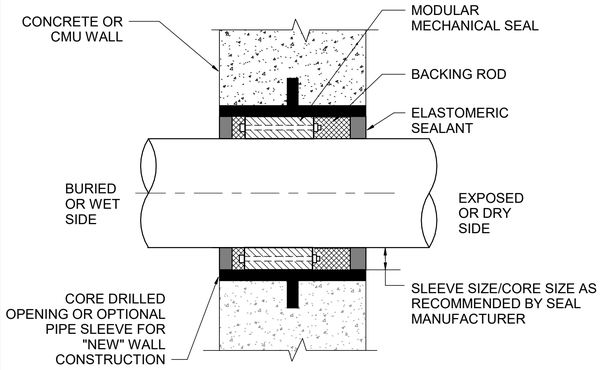
GENERAL NOTES

A. SEE SPEC SECTION 01 73 20 FOR ADDITIONAL REQUIREMENTS.

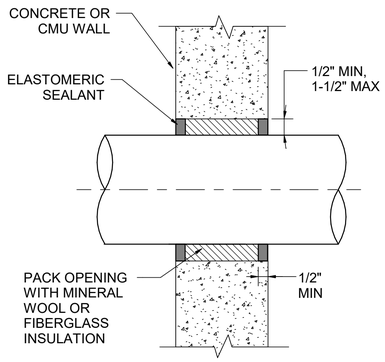
B. SEE DETAILS 6 AND 7 ON 003S507 FOR PIPES THROUGH FOUNDATION STEM WALLS AND FOOTINGS.



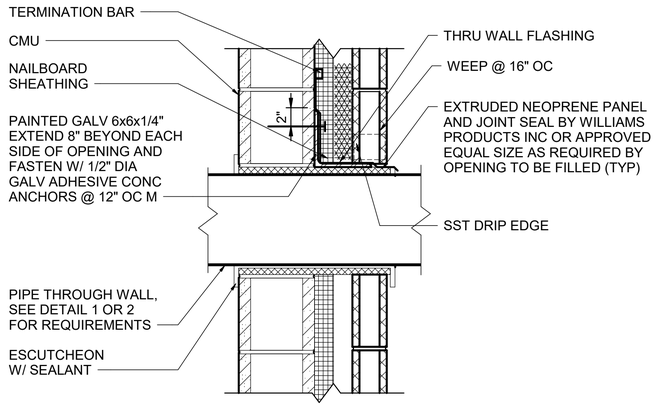
NOTES:

1. PROVIDE DOUBLE ROW OF MODULAR MECHANICAL SEALS FOR PIPES 8" DIAMETER AND LARGER IF CONCRETE THICKNESS PERMITS.
2. BOLT HEADS ON ACCESSIBLE SIDE OF PENETRATION.
3. PROVIDE MECHANICAL SEALS WITH TYPE 316 STAINLESS STEEL BOLTS AND HARDWARE AND EPDM SEALING ELEMENTS.

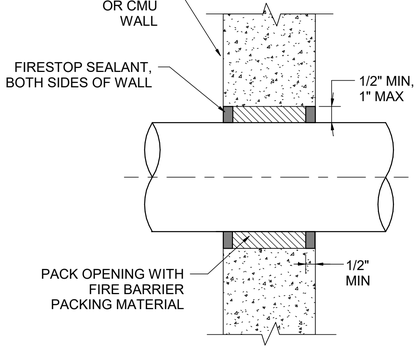
1 / 005D505 PIPE 4" AND GREATER THROUGH NON-RATED WALL
NOT TO SCALE



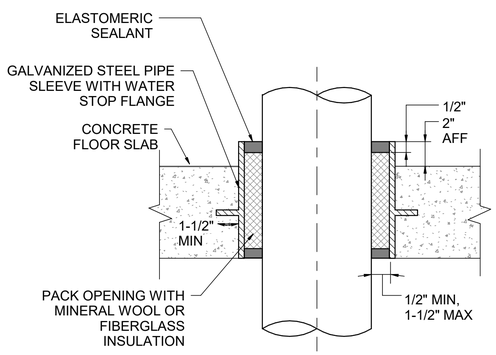
2 / 005D505 PIPE LESS THAN 4" THROUGH NON-RATED WALL
NOT TO SCALE



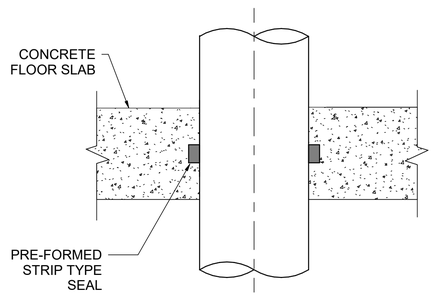
3 / 005D505 PIPE THROUGH EXTERIOR WALL WITH VENEER
NOT TO SCALE



4 / 005D505 PIPE THROUGH FIRE-RATED WALL
NOT TO SCALE



5 / 005D505 PIPE LESS THAN 4" THROUGH FLOOR SLAB
NOT TO SCALE



6 / 005D505 PIPE 4" AND GREATER THROUGH FLOOR SLAB
NOT TO SCALE

SCALE A 12 6 0 12 1-1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/27/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 005 D505	TITLE OF SHEET GENERAL PROCESS DETAILS - PIPE PENETRATION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 65 OF 99

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PIPING SYSTEMS

NG	NATURAL GAS
LPG	LIQUEFIED PROPANE
HWS	HEATING HOT WATER SUPPLY
HWR	HEATING HOT WATER RETURN
GHWS	GLYCOL HEATING HOT WATER SUPPLY
GHWR	GLYCOL HEATING HOT WATER RETURN
CHWS	CHILLED WATER SUPPLY
CHWR	CHILLED WATER RETURN
GCWS	GLYCOL CHILLED WATER SUPPLY
GCWR	GLYCOL CHILLED WATER RETURN
CWS	CONDENSER WATER SUPPLY
CWR	CONDENSER WATER RETURN
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
CD	CONDENSATE DRAIN
CPD	CONDENSATE PUMP DISCHARGE
STM -15	STEAM SUPPLY - PSI
BBD	BOILER BLOW DOWN
BF	BOILER FEED
SV	STEAM VENT
MU	MAKE-UP WATER

PIPING SYMBOLOGY

- PIPE ANCHOR
- PIPE GUIDE
- EXPANSION JOINT
- PRESSURE/ TEMPERATURE PORT (PT)
- THERMOMETER
- THERMOWELL
- PRESSURE GAUGE
- TEMPERATURE GAUGE (T)
- FLEXIBLE PIPING CONNECTION
- WYE STRAINER
- MANUAL AIR VENT
- AUTOMATIC AIR VENT (A)
- METER (WATER, GAS, OTHER) (M)
- FLOOR CLEANOUT (FCO)
- CLEANOUT (CO)
- WALL CLEANOUT (WCO)
- DOUBLE GRADE CLEANOUT (GCO)
- WATER HAMMER ARRESTOR
- EARTHQUAKE VALVE (E)
- CONCENTRIC REDUCER
- ECCENTRIC REDUCER, FLAT ON BOTTOM
- ECCENTRIC REDUCER, FLAT ON TOP
- ELBOW, 90° TURN DOWN
- ELBOW, 90° TURN UP
- TEE, OUTLET UP
- TEE, OUTLET DOWN
- TEE, OUTLET UP W/ 90° TURN
- TEE, OUTLET DOWN W/ 90° TURN
- PIPE BREAK
- PIPE CAP
- BLIND FLANGE
- UNION
- FLOW ARROW
- SHUTOFF VALVE (NORMALLY OPEN)
- SHUTOFF VALVE (NORMALLY CLOSED)
- DRAIN VALVE
- CHECK VALVE
- VACUUM BREAKER (VB)
- AUTOMATIC FLOW CONTROL VALVE
- CALIBRATED MANUAL BALANCING VALVE
- PRESSURE-RELIEF VALVE (OR)
- PRESSURE-REDUCING VALVE (PRV)
- AUTOMATIC CONTROL VALVE, 2-WAY
- AUTOMATIC CONTROL VALVE, 3-WAY
- BFP — BACKFLOW PREVENTER
- XX-1 — PLUMBING FIXTURE

HVAC SYMBOLOGY

- SUPPLY AIR OR OUTSIDE AIR DUCT UP (SECTION CUT, FIRST DIMENSION DUCT WIDTH) (24x18)
- SUPPLY AIR OR OUTSIDE AIR DUCT DOWN (NO SECTION CUT)
- RETURN AIR DUCT UP (SECTION CUT)
- RETURN AIR DUCT DOWN (NO SECTION CUT)
- EXHAUST AIR DUCT UP (NO SECTION CUT)
- EXHAUST AIR DUCT DOWN (NO SECTION CUT)
- ROUND ELBOW UP
- ROUND ELBOW DOWN
- TRANSITION - RECTANGULAR TO ROUND DUCT
- STANDARD BRANCH
- ELBOW - W/TURNING VANE (RECTANGULAR)
- ELBOW - (RECTANGULAR), SMOOTH RADIUS
- RECTANGULAR DUCT OR OPENING SIZE FIRST NUMBER INDICATES SIZE OF SIDE SHOWN (18x24)
- ROUND DUCT SIZE (18ø)
- RECTANGULAR DUCT INCLINE - RISE OR DROP IN RESPECT TO THE AIR FLOW (R OR D)
- ROUND DUCT INCLINE - RISE OR DROP IN RESPECT TO THE AIR FLOW (R OR D)
- HIDDEN DUCT (18x24)
- DUCT/PIPE ELEVATION TAG ABOVE FINISH FLOOR (T 9'-10" B 9'-0", 18x10)
- VOLUME DAMPER
- MOTOR OPERATED DAMPER (M)
- FIRE DAMPER
- SMOKE DAMPER (S)
- SMOKE AND FIRE DAMPER (S)
- FLEXIBLE CONNECTION
- FLEXIBLE DUCT - TWO LINE
- FLEXIBLE DUCT - ONE LINE
- ACOUSTICAL LINING - DUCT DIMENSIONS FOR NET FREE AREA
- UNDERCUT DOOR (UC)
- NEW TO EXISTING CONNECTION
- REMOVE EXISTING UP TO THIS POINT
- HVAC EMERGENCY SHUTDOWN SWITCH

- SUPPLY AIR REGISTER (TAG-SIZE CFM)
- EXHAUST AIR OR RETURN AIR GRILLE (TAG-SIZE CFM)
- EXHAUST AIR OR RETURN AIR GRILLE (TAG CFM)
- SUPPLY AIR ASSEMBLY SQUARE DIFFUSER (TAG CFM)
- SUPPLY AIR ASSEMBLY ROUND DIFFUSER (TAG CFM)

TEMPERATURE CONTROL DIAGRAM SYMBOLOGY

MISCELLANEOUS DEVICE IDENTIFIER (X)

CO	SPACE CARBON MONOXIDE SENSOR
CO2	SPACE CARBON DIOXIDE SENSOR
H	SPACE HUMIDITY SENSOR
S	SPACE TEMPERATURE SENSOR
T	THERMOSTAT OR TEMPERATURE SENSOR
TIT	TEMPERATURE TRANSMITTER
TS	TEMPERATURE SWITCH

CONTROL DEVICE IDENTIFIER (X)

AFS	AIRFLOW MEASURING STATION
AM	AIRFLOW MEASURING SENSOR
CO	CARBON MONOXIDE SENSOR
CO2	CARBON DIOXIDE SENSOR
CS	CURRENT SWITCH
F	FLOW SWITCH
FRZ	FREEZE STAT
H	HUMIDITY SENSOR
HS	HIGH STATIC SWITCH
LS	LOW STATIC SWITCH
NO2	NITROGEN DIOXIDE SENSOR
P	PRESSURE SENSOR
SD	SMOKE DETECTOR
ΔP	DIFFERENTIAL PRESSURE SENSOR

CONTROL INPUT/OUTPUT IDENTIFIER (XX)

AI	ANALOG INPUT
AO	ANALOG OUTPUT
DI	DIGITAL INPUT
DO	DIGITAL OUTPUT

COIL IDENTIFIERS (XX/)

CC	CHILLED WATER COOLING COIL
DG	DIRECT FIRED GAS BURNER
DX	DIRECT EXPANSION COOLING COIL
EH	ELECTRIC HEATING COIL
ER	DIRECT EVAPORATIVE COIL
FC	FACE AND BYPASS COOLING COIL
FH	FACE AND BYPASS HEATING COIL
HC	HEATING HOT WATER HEATING COIL
IG	INDIRECT FIRED GAS BURNER
ST	STEAM HEATING COIL

CONTROL ABBREVIATIONS

BTU	BTU METER
COMM	COMMUNICATION
CS	CURRENT SWITCH
EC	ELECTRONICALLY COMMUTATED
FM	FLOW METER
MD	MOTOR DAMPER
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
TCC	TEMPERATURE CONTROL CONTRACTOR
VFC	VARIABLE FREQUENCY CONTROLLER

- INDICATING LIGHT, PANEL MOUNTED
- HAND SWITCH, PANEL MOUNTED, 2 OR 3 POSITION (HOA OR ON/OFF)
- DUCT MOUNTED SMOKE DETECTOR (S)
- TEMPERATURE AVERAGING SENSOR (T)
- MOTOR OPERATED DAMPER (M)
- MOTOR OPERATED SINGLE BLADE DAMPER (M)
- BACKDRAFT DAMPER (BDD)
- AIRFLOW MEASURING STATION (AFS)
- INTAKE/ EXHAUST LOUVER
- FILTER
- CONTROL WIRING
- POWER WIRING (120VAC OR HIGHER)
- AIR SAMPLING LINE
- CENTRIFUGAL OR INLINE FAN (M)
- PUMP

ABBREVIATIONS

AAV	AUTOMATIC AIR RELEASE VENT
AC	AIR CONDITIONING UNIT
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AHU	AIR HANDLING UNIT
AI	ANALOG INPUT
AO	ANALOG OUTPUT
APD	AIR PRESSURE DROP
BAS	BUILDING AUTOMATION SYSTEM
BDD	BACK DRAFT DAMPER
BHP	BRAKE HORSE POWER
BOD	BOTTOM OF DUCT
BOE	BOTTOM OF EQUIPMENT
BRD	BAROMETRIC RELIEF DAMPER
BTUH	BRITISH THERMAL UNITS PER HOUR
CAV	CONSTANT AIR VOLUME
CFH	CUBIC FEET PER HOUR
CFM	CUBIC FEET PER MINUTE
CO	CLEAN OUT
COMM	COMMUNICATIONS
COP	COEFFICIENT OF PERFORMANCE
CRAC	COMPUTER ROOM AIR CONDITIONER
CU	CONDENSING UNIT
DB	DRY BULB
DDC	DIRECT DIGITAL CONTROL
DI	DIGITAL INPUT
DMP	DAMPER
DO	DIGITAL OUTPUT
DP	DIFFERENTIAL PRESSURE
DX	DIRECT EXPANSION
EA	EXHAUST AIR
EC	ELECTRICALLY COMMUTATED
ECU	EVAPORATIVE COOLING UNIT
ECM	ELECTRONIC COMMUTATED MOTOR
EDH	ELECTRIC DUCT HEATER
EER	ENERGY EFFICIENCY RATIO
EG	EXHAUST GRILLE
ESO	EMERGENCY SHUT OFF
ESP	EXTERNAL STATIC PRESSURE
ESS	EMERGENCY SHUTOFF SWITCH
EUH	ELECTRIC UNIT HEATER
EF	EXHAUST FAN
EXH	EXHAUST
FA	FOUL AIR
FACP	FIRE ALARM CONTROL PANEL
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FDC	FLEXIBLE DUCT CONNECTOR
FLA	FULL LOAD AMPS
FPM	FEET PER MINUTE
FRP	FIBERGLASS REINFORCED PLASTIC
FS	FLOW SWITCH
FSD	FIRE SMOKE DAMPER
GE	GRAVITY EXHAUST
GI	GRAVITY INTAKE
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GV	GRAVITY VENTILATOR
HEX	HAZARDOUS EXHAUST
HP	HEAT PUMP
I/O	INPUT/OUTPUT
IAQ	INDOOR AIR QUALITY
IPLV	INTEGRATED PART LOAD VALUE
LAT	LEAVING AIR TEMPERATURE
LPG	LIQUEFIED PETROLEUM GAS
LWT	LEAVING WATER TEMPERATURE
MAU	MAKE-UP AIR UNIT
MBH	THOUSAND BTUH
MCC	MOTOR CONTROL CENTER
MFR	MANUFACTURER
MOCP	MOTOR OVER CURRENT PROTECTION
NC	NOISE CRITERIA
NO	NUMBER
OA	OUTSIDE AIR
OAT	OUTSIDE AIR TEMPERATURE
OCF	ODOR CONTROL FAN
OS&Y	OUTSIDE SCREW AND YOKE
PD	PRESSURE DROP
POC	POINT OF CONNECTION TO EXISTING
POD	POINT OF DISCONNECT FROM EXISTING
P&T	PRESSURE AND TEMPERATURE
RA	RETURN AIR
RAT	RETURN AIR TEMPERATURE
RD	RETURN DIFFUSER
RG	RETURN GRILLE
RH	RELATIVE HUMIDITY
SA	SUPPLY AIR
SAT	SUPPLY AIR TEMPERATURE
SCFM	STANDARD CUBIC FEET PER MINUTE
SD	SUPPLY DIFFUSER
SEER	SEASONAL ENERGY EFFICIENCY RATIO
SF	SUPPLY FAN
SG	SUPPLY GRILLE
SP	STATIC PRESSURE
TCC	TEMPERATURE CONTROL CONTRACTOR
TCP	TEMPERATURE CONTROL PANEL
TD	TEMPERATURE DIFFERENTIAL
TG	TRANSFER GRILLE
TP	TRAP PRIMER
TSP	TOTAL STATIC PRESSURE
UH	UNIT HEATER
V&C	VALVE AND CAP
VAV	VARIABLE AIR VOLUME
VD	VOLUME DAMPER OR MANUAL ISOLATION DAMPER
VRF	VARIABLE REFRIGERANT FLOW
VTR	VENT THROUGH ROOF
WB	WET BULB
WC	WATER COLUMN
WPD	WATER PRESSURE DROP

GENERAL NOTES

GENERAL MECHANICAL NOTES

1. THESE NOTES ARE NOT ALL INCLUSIVE. REFER TO DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.
2. THIS IS A STANDARD MECHANICAL SYMBOLOGY AND ABBREVIATIONS SHEET. LISTING OF SYMBOLS AND ABBREVIATIONS DOES NOT IMPLY ALL SYMBOLS AND ABBREVIATIONS HAVE BEEN USED ON THIS PROJECT.
3. VALVE SYMBOLS SHOWN HERE ARE APPLICABLE ONLY TO MECHANICAL SHEETS. PROVIDE ALL MATERIALS, LABOR, AND EQUIPMENT FOR COMPLETE AND OPERABLE SYSTEMS AS INDICATED ON THE DRAWINGS AS SPECIFIED, OR AS REQUIRED BY CODE.
4. MECHANICAL INSTALLATION SHALL COMPLY WITH THE ADA/ABA ACCESSIBILITY GUIDELINES.
5. DETAILS APPLY TO THE ENTIRE PROJECT AND ARE ONLY REFERENCED TO PROVIDE CLARITY IF THERE ARE MULTIPLE DETAILS THAT COULD APPLY TO A PARTICULAR PROJECT CONDITION.
6. COORDINATE LOCATION OF ALL MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING WITH OTHER TRADES BEFORE PROCEEDING WITH WORK. DO NOT INSTALL MECHANICAL EQUIPMENT, DUCTWORK, OR PIPING ABOVE ELECTRICAL EQUIPMENT WHERE PROHIBITED BY ELECTRICAL CODES (SWITCHBOARDS, PANELS, ETC.).
7. LIGHT LINE WEIGHT INDICATES EXISTING PIPING, DUCTWORK, AND/OR EQUIPMENT TO REMAIN. BOLD LINE WEIGHT INDICATES NEW WORK TO BE INSTALLED AS WORK OF THIS CONTRACT.
8. COORDINATE INSTALLATION OF OUTSIDE AIR INTAKE WITH INSTALLATION OF PLUMBING VENTS, FLUES AND EXHAUST/RELIEF OUTLETS TO MAINTAIN 10' SEPARATION.
9. ALL WORK IN FINISHED SPACES SHALL BE LOCATED ABOVE CEILINGS, IN CHASES OR OTHER CONCEALED ACCESSIBLE LOCATIONS UNLESS NOTED OTHERWISE. LOCATE AND ARRANGE VALVES, DRAIN FITTINGS, ETC. TO BE ACCESSIBLE THROUGH LAY-IN CEILINGS, ACCESS PANELS OR ACCESS DOORS. PROVIDE AN ACCESS PANEL OR DOOR FOR ALL NON-ACCESSIBLE INSTALLATIONS. COORDINATE LOCATION OF ACCESS PANELS OR DOORS WITH THE ARCHITECT/ENGINEER AND OTHER TRADES.
10. ALL MATERIALS LOCATED IN PLENUM SHALL BE RATED FOR PLENUM INSTALLATION.
11. ALL DUCTWORK, PIPING, AND EQUIPMENT SUPPORTED FROM STRUCTURAL STEEL SHALL BE COORDINATED WITH ALL TRADES. ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS OR AS SHOWN ON THE MECHANICAL OR STRUCTURAL DRAWINGS. WELDING TO STRUCTURAL MEMBERS SHALL NOT BE PERMITTED.
12. ALL MISCELLANEOUS METALS AND MATERIALS REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN IN DETAILS FOR PIPING, DUCTWORK, AND EQUIPMENT (UNLESS OTHERWISE NOTED) SHALL BE PROVIDED BY THE INSTALLING CONTRACTOR.
13. PROVIDE DIELECTRIC UNIONS AT ALL CONNECTIONS OF DISSIMILAR METALS (SUCH AS COPPER TO GALVANIZED STEEL).
14. PROVIDE ISOLATION VALVES AT EACH PIECE OF EQUIPMENT. ALSO PROVIDE ISOLATION VALVES ON EACH BRANCH AND/OR RISER SERVING MULTIPLE PIECES OF EQUIPMENT OR FIXTURES AND ELSEWHERE AS INDICATED. INSTALL VALVES AS CLOSE TO MAIN AS POSSIBLE.

GENERAL HVAC NOTES

1. DUCTWORK DIMENSIONS: FIRST NUMBER INDICATES SIDE OF DUCTWORK SHOWN. ALL DIMENSIONS ARE IN INCHES AND ARE INSIDE CLEAR DIMENSIONS.
2. VOLUME DAMPERS ABOVE PLASTER OR GYPBOARD CEILINGS SHALL HAVE EXTENSION RODS AND CHROME-PLATED ESCUTCHEON PLATES.
3. COORDINATE ALL GRILLE, REGISTER AND DIFFUSER LOCATIONS WITH REFLECTED CEILING PLAN, LIGHTING, AND ALL OTHER CEILING MOUNTED DEVICES.
4. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK AND PIPING SYSTEMS CONNECTED TO FANS, PUMPS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION.
5. PROVIDE ACCESSIBLE VOLUME DAMPERS OR OTHER MEANS OF AIRFLOW ADJUSTMENT AT ALL DUCT RUN-OUTS TO DIFFUSERS AND GRILLES.
6. PROVIDE DUCT ACCESS DOORS AT OUTSIDE AIR INTAKE PLENUMS.
7. ALL DUCT RUN-OUTS TO DIFFUSERS AND GRILLES SHALL BE THE SAME AS THE DIFFUSER OR GRILLE NECK SIZE UNLESS NOTED OTHERWISE.
8. WALL-MOUNTED THERMOSTATS, TEMPERATURE SENSORS AND SWITCHES SHALL BE MOUNTED AT 4'-0" ABOVE FINISHED FLOOR.
9. REFER TO PROCESS DETAILS IN SERIES 005, AND SPECIFICATION SECTION 01 73 20 FOR PIPE PENETRATION DETAILS THROUGH WALLS AND FLOOR SLABS.

GENERAL HVAC CONTROL SYSTEM NOTES:

1. IN GENERAL, HVAC SYSTEMS SHALL BE PROVIDED WITH MANUFACTURER'S OR THIRD PARTY MICROPROCESSOR CONTROLLERS TO PROVIDE ALL CONTROL AND MONITORING CAPABILITY AS INDICATED. WHERE NECESSARY, PROVIDE ADDITIONAL CONTROLLERS OR CONTROL COMPONENTS AS REQUIRED TO ACCOMPLISH THE SEQUENCES OF OPERATION. COORDINATE INDICATED SCADA CONNECTIONS WITH THE FACILITY CONTROL SYSTEM PROVIDER.
2. ALL ELECTRICAL AND CONTROL COMPONENTS SHALL BE RATED FOR THE ELECTRICAL CLASSIFICATION THEY ARE LOCATED IN.
3. ALL WIRING SHALL BE RUN IN A NEAT AND WORKMANLIKE MANNER, PARALLEL AND PERPENDICULAR TO BUILDING LINES. WIRING SHALL BE RUN ON J-HOOKS IN ACCESSIBLE LOCATIONS, IN CONDUIT IN EXPOSED LOCATIONS, ABOVE GYP CEILINGS, AND BELOW 8 FOOT (ALL WIRING WITHIN WALLS TO BE RUN IN CONDUIT).
4. WIRING AND CONDUIT SHALL BE PROVIDED AND INSTALLED IN ACCORDANCE WITH NEC AND DIVISION 26 SPECIFICATIONS.
5. CONTROL PANELS SHALL BE UL LISTED AND FABRICATED AND INSTALLED IN ACCORDANCE WITH NEC AND PROJECT SPECIFICATIONS.
6. ALL WIRING, CONTROL COMPONENTS, CONTROL PANELS, DEVICES AND PROGRAMMING SHOWN ON THE HVAC CONTROL DRAWINGS SHALL BE PROVIDED BY THE TEMPERATURE CONTROLS CONTRACTOR (TCC), UNLESS SPECIFICALLY NOTED OTHERWISE.
7. TCC SHALL PROVIDE ALL CONTROL COMPONENTS AND ACCESSORIES AS REQUIRED FOR EQUIPMENT TO BE CONTROLLED IN ACCORDANCE WITH THE SEQUENCE OF OPERATION REGARDLESS OF WHETHER ALL CONTROL COMPONENTS OR POINTS ARE SHOWN IN THE ASSOCIATED CONTROL DIAGRAM.
8. DUCT SMOKE DETECTORS SHALL BE PROVIDED UNDER DIVISION 28 SPECIFICATIONS. PROVIDE ALL WIRING BETWEEN THE FIRE ALARM CONTROL PANEL (IF APPLICABLE) AND THE EQUIPMENT RELAYS OR VFDS AS REQUIRED FOR EQUIPMENT SHUTDOWN.
9. EACH CONTROL PANEL SHALL HAVE A LAMINATED COPY OF THE APPLICABLE SEQUENCE OF OPERATION AND CONTROL DIAGRAM INDICATING THE POINTS, COMPONENTS AND OPERATION OF EQUIPMENT ASSOCIATED WITH EACH PANEL.
10. CONTROL DIAGRAMS ARE SCHEMATIC IN NATURE AND MAY NOT SHOW ALL REQUIRED CONTROL DEVICES AND COMPONENTS. REFER TO FLOOR PLANS, FLOW DIAGRAMS AND DETAILS FOR ADDITIONAL CONTROL DEVICES, COMPONENTS AND REQUIREMENTS NOT SHOWN ON THE CONTROL DRAWINGS.

Professional Engineer Seal — CERTIFICATE 66199 — REGISTERED — 5-31-26 — ARIZONA U.S.A.

A/E FIRM	
PRIME:	HDR ENGINEERING, INC. DENVER, CO 80202
DESIGNED:	A. PUTMAN
DRAWN:	B. BAXTER
TECH. REVIEW:	H. PEDRAZA
DATE:	MAY 2026
SUB SHEET NO.	006 M001
TITLE OF SHEET	GENERAL HVAC LEGENDS, NOTES, AND ABBREVIATIONS
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	66 OF 99

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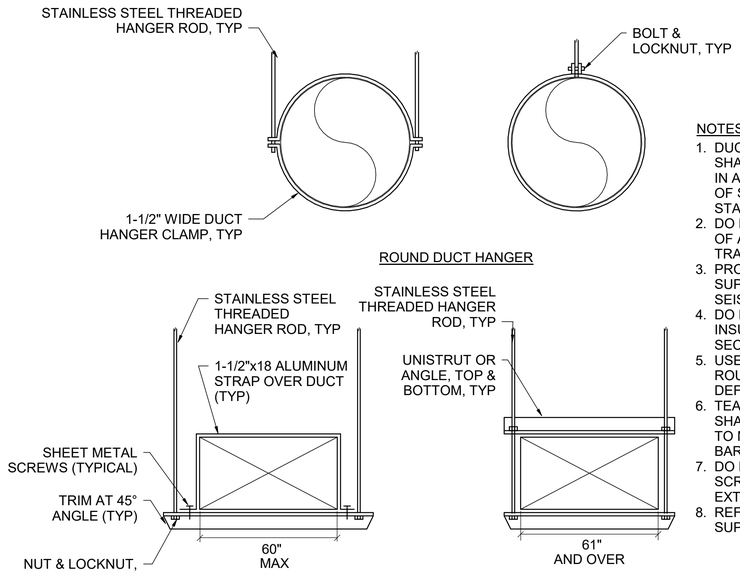
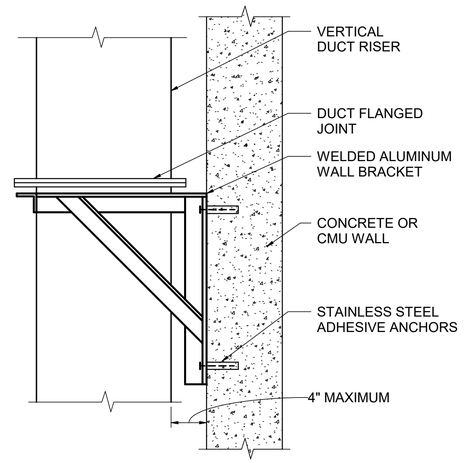


NOTES:

1. DUCT SUPPORTS AND SEISMIC BRACING SHALL BE DESIGNED BY THE CONTRACTOR IN ACCORDANCE WITH THE LATEST EDITION OF SMACNA DUCT CONSTRUCTION STANDARDS.
2. DO NOT EXCEED ALLOWABLE LOAD LIMITS OF ANY OR ALL OF THE INDIVIDUAL TRAPEZE PARTS.
3. PROVIDE SEISMIC BRACING FOR DUCT SUPPORTS IN ACCORDANCE WITH SMACNA SEISMIC RESTRAINT MANUAL.
4. DO NOT PENETRATE DUCT OR DUCT INSULATION VAPOR BARRIER WHEN SECURING HANGERS TO DUCTWORK.
5. USE HANGER CLAMPS OR BANDS FOR ROUND DUCT AS ROUND DUCT TENDS TO DEFORM ON TRAPEZE HANGERS.
6. TEARS OR PUNCTURES IN INSULATION SHALL BE REPAIRED WITH TAPE OR MASTIC TO MAINTAIN THE INTEGRITY OF THE VAPOR BARRIER.
7. DO NOT USE ANY STRAP HANGERS WITH SCREWS ON ANY DUCTWORK THAT IS EXTERNALLY WRAPPED.
8. REFER TO SPECIFICATIONS FOR DUCT SUPPORT MATERIALS OF CONSTRUCTION.

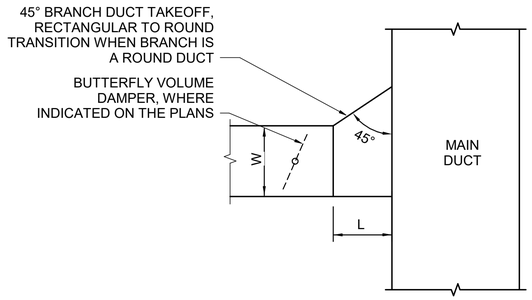
1 006M501 DETAIL - DUCT HANGERS
NOT TO SCALE



NOTES:

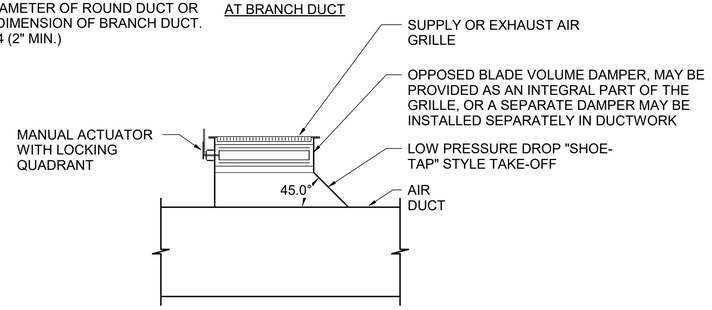
1. WALL BRACKET DESIGN AND SUPPORT SPACING SHALL BE BASED ON SMACNA HVAC DUCT CONSTRUCTION STANDARDS.
2. KEEP VERTICAL DUCTS AS TIGHT AGAINST THE WALL AS POSSIBLE.
3. REFER TO SPECIFICATIONS FOR DUCT SUPPORT MATERIALS OF CONSTRUCTION.

2 006M501 DETAIL - DUCT RISER SUPPORT
NOT TO SCALE



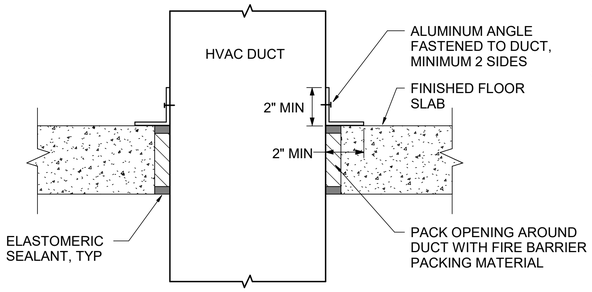
W = DIAMETER OF ROUND DUCT OR PLAN DIMENSION OF BRANCH DUCT.
L = W/4 (2" MIN.)

AT BRANCH DUCT

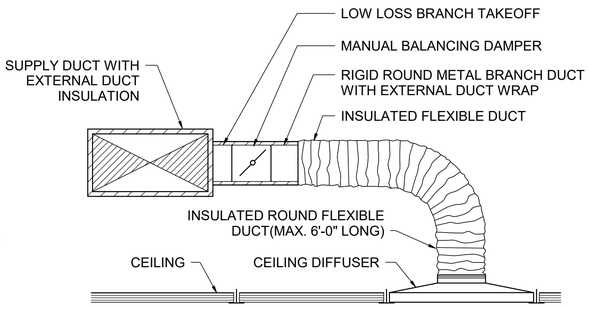


AT SUPPLY OR EXHAUST GRILLE

3 006M501 DETAIL - BRANCH DUCT TAKEOFF FITTING
NOT TO SCALE



4 006M501 DETAIL - FLOOR SLAB DUCT PENETRATION
NOT TO SCALE



5 006M501 DETAIL - CEILING DIFFUSER
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DENVER, CO 80202

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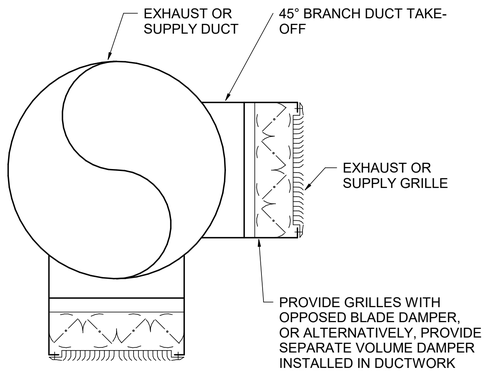
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 006 M501

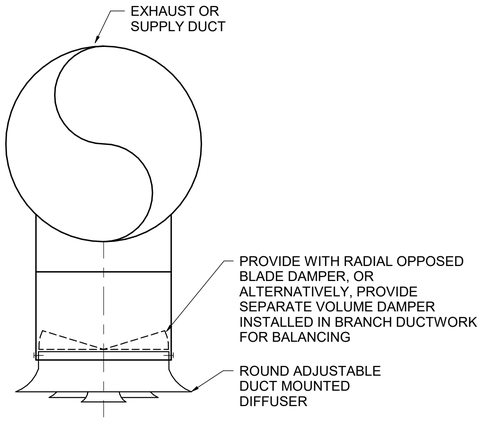
TITLE OF SHEET
GENERAL
HVAC DETAILS - DUCTS

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 67 OF 99



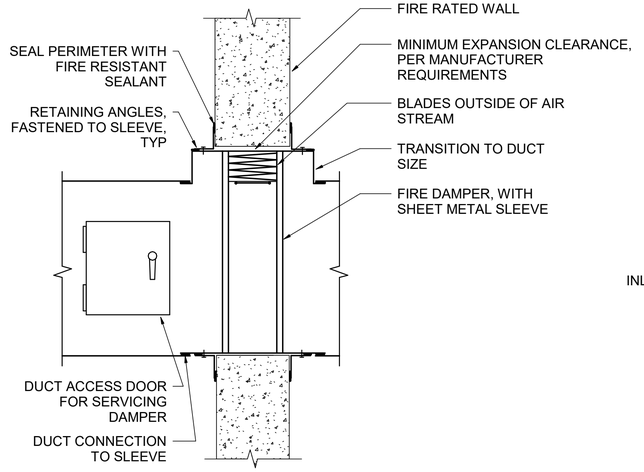
RECTRANGULAR GRILLES



ROUND DIFFUSER

1 006M502 DETAIL - DUCT MOUNTED EXHAUST GRILLE OR SUPPLY DIFFUSER

NOT TO SCALE

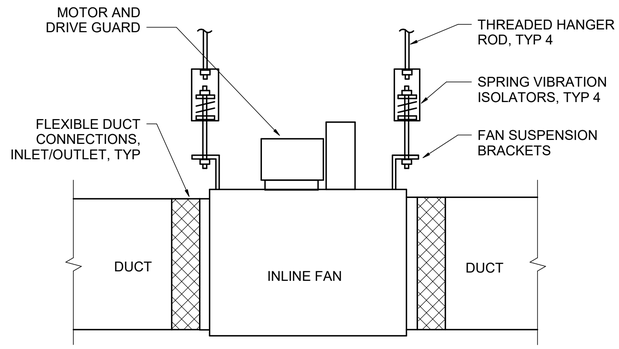


NOTES:

1. INSTALL FIRE DAMPERS IN ACCORDANCE WITH MANUFACTURER REQUIREMENTS AND NFPA 90A.

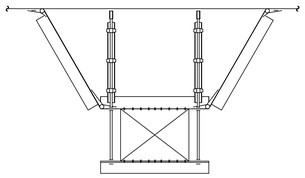
2 006M502 DETAIL - FIRE DAMPER

NOT TO SCALE

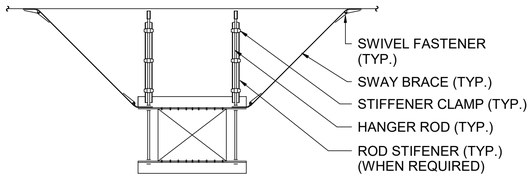


3 006M502 DETAIL - SUSPENDED INLINE FAN

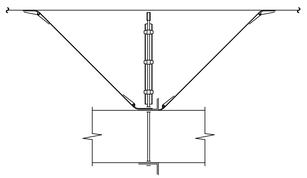
NOT TO SCALE



TRANSVERSE SOLID BRACE



TRANSVERSE CABLE BRACE



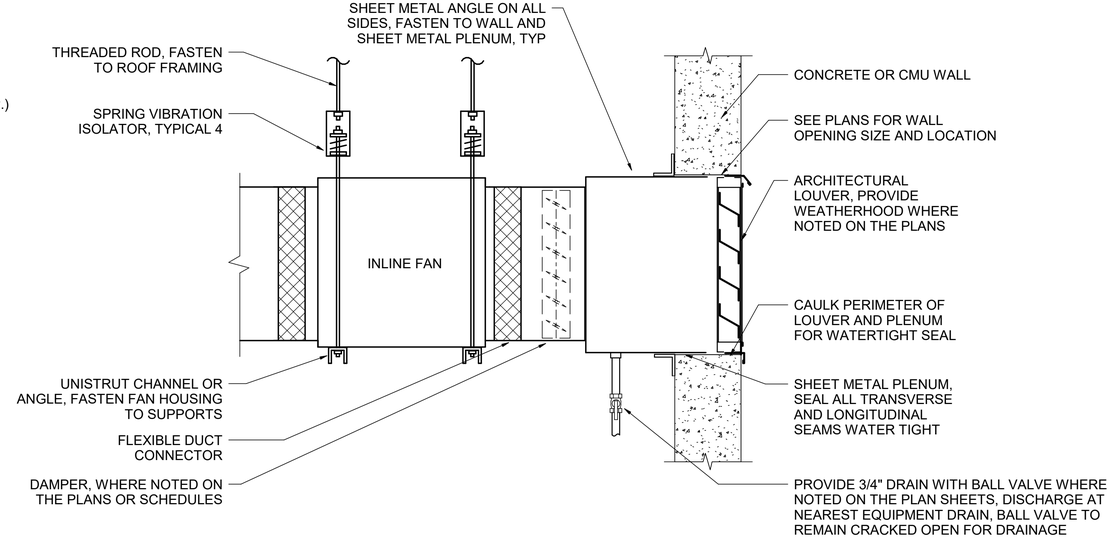
LONGITUDINAL BRACE

NOTES:

1. GENERAL REQUIREMENTS FOR SEISMIC RESTRAINT OF DUCTWORK ARE SHOWN. SPECIFIC REQUIREMENTS, INCLUDING ATTACHMENT TO BUILDING STRUCTURE, SHALL BE DETERMINED BY THE SEISMIC RESTRAINT DESIGNER.
2. ALL HANGERS SHALL BE ATTACHED TO STRUCTURAL STEEL OR CONCRETE FLOOR/ROOF.

4 006M502 DETAIL - SEISMIC BRACING OF SUSPENDED DUCT WORK

NOT TO SCALE



5 006M502 DETAIL - WALL-MOUNTED INLINE FAN

NOT TO SCALE

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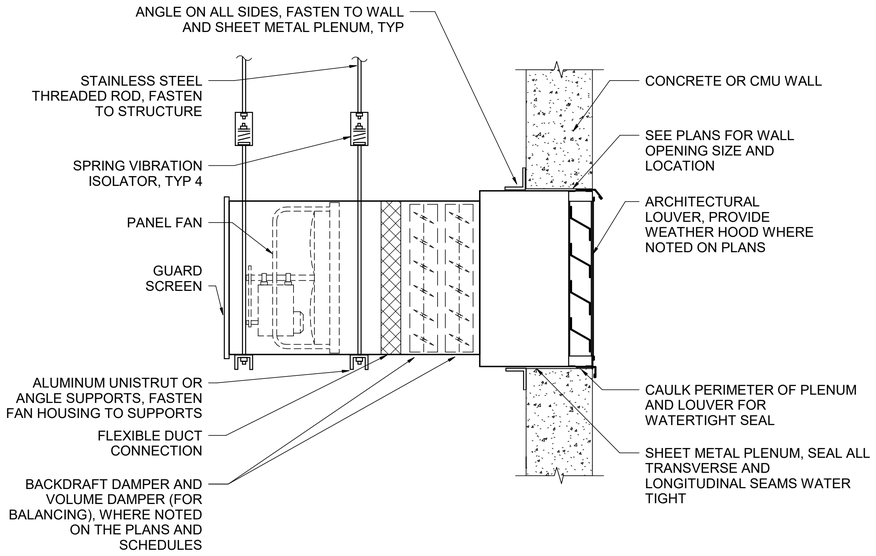
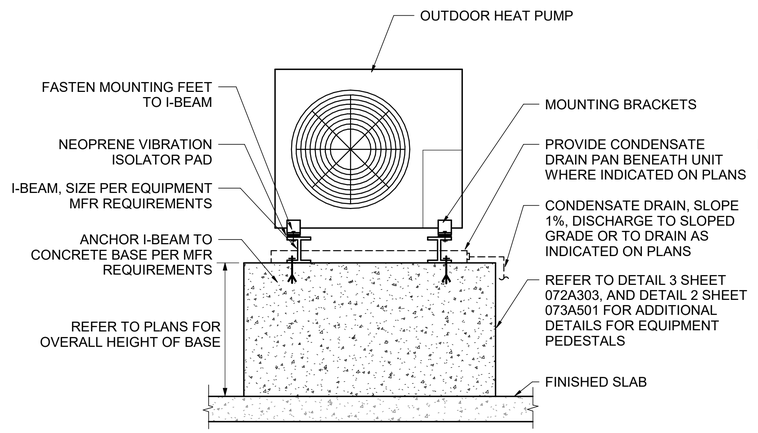
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 006 M502	TITLE OF SHEET GENERAL HVAC DETAILS - DUCTS CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 68 OF 99

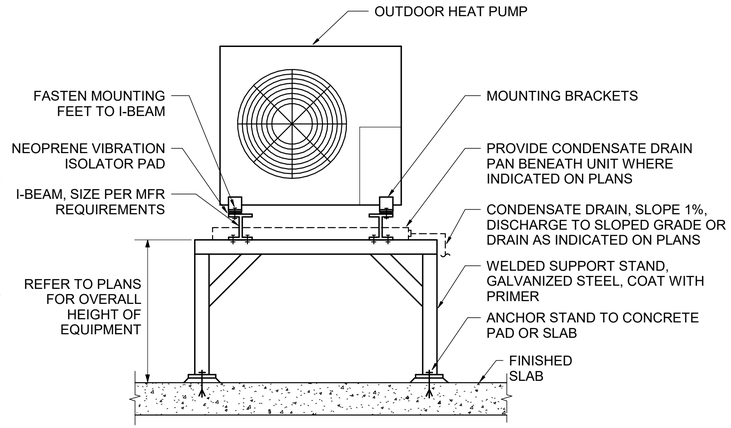


1 006M503 DETAIL - WALL-MOUNTED PROP FAN

NOT TO SCALE



EQUIPMENT PEDESTAL

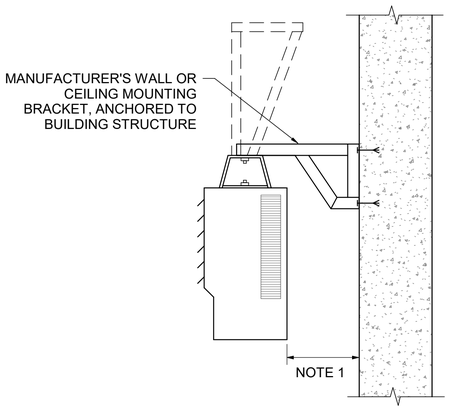


SUPPORT STAND

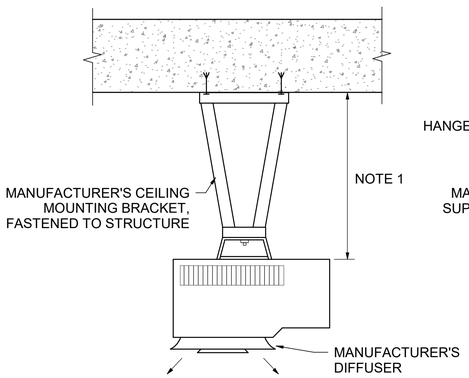
NOTES:
1. INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER REQUIRMENTS.
2. FASTENERS, ANCHORS, BOLTS AND HARDWARE SHALL BE TYPE 316 STAINLESS STEEL.
3. COAT SUPPORT STANDS AND I-BEAMS WITH CORROSION RESISTANT PRIMER PER SPEC 09 96 00.

2 006M503 DETAIL - HEAT PUMP SUPPORTS

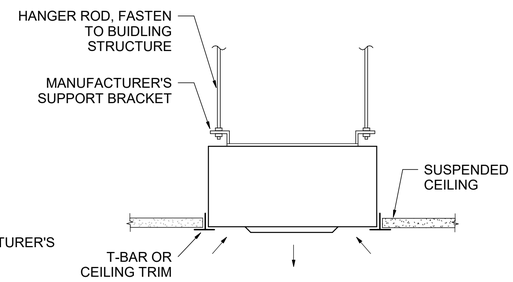
NOT TO SCALE



HORIZONTAL DISCHARGE



VERTICAL DISCHARGE



VERTICAL DISCHARGE (CEILING RECESSED)

NOTES:
1. MAINTAIN MANUFACTURERS REQUIRED CLEARANCES TO SURFACES AND TO COMBUSTIBLE MATERIALS.
2. PROVIDE MISCELLANEOUS METAL, ANGLE, UNISTRUT, AND HARDWARE AS REQUIRED TO SPAN BETWEEN STRUCTRAL ELEMENTS FOR SUPPORT.
3. MOUNT HEATERS AT ELEVATIONS INDICATED ON THE PLAN SHEETS.

3 006M503 DETAIL - ELECTRIC UNIT HEATERS

NOT TO SCALE

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4/27/2026 11:53:36 AM

Professional Engineer Illinois CERTIFICATE 62-61999 Arda Jastman 5-31-26 REGISTERED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 006 M503	TITLE OF SHEET GENERAL HVAC DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 69 OF 99

PLUMBING LEGENDS, NOTES & ABBREVIATIONS

GENERAL

SYMBOL	DESCRIPTION
	EXISTING TO REMAIN
	DEMOLITION
	NEW
	90 ELBOW
	UNION
	FLANGE
	CAP OR PLUG
X	PIPE ANCHOR
	PIPE DROP
	PIPE RISE
	TEE FITTING - SIDE BRANCH CONNECTION
	CONNECTION OUT OF BOTTOM
	TEE FITTING - TOP BRANCH CONNECTION
	CONNECTION OF TOP
	CONCENTRIC INCREASER
	ECCENTRIC INCREASER
	NEW CONNECTION TO EXISTING
S-1	PLUMBING FIXTURE & DESIGNATION SEE FIXTURE CONNECTION SCHEDULE
LA	LAB COMPRESSED AIR
LV	LAB VACUUM

DOMESTIC WATER SUPPLY

CW	POTABLE COLD WATER
HW	POTABLE HOT WATER SUPPLY
HWC	POTABLE HOT WATER RETURN
NPW	NON POTABLE WATER
	EXPANSION JOINT
	WYE STRAINER
	WALL HYDRANT OR HOSE BIBB
	SHOWER HEAD (RISER - PLAN)
	THERMOMETER
TW	TEPID WATER
WHA	WATER HAMMER ARRESTORS
RW	RECYCLED WATER

VALVES

SYMBOL	DESCRIPTION
	OUTSIDE SCREW AND YOKE GATE VALVE
	GATE VALVE
	CHECK VALVE
	PRESSURE REGULATING VALVE
	BALANCING VALVE
	BALL VALVE
	GLOBE VALVE
	ANGLE VALVE
R	RELIEF VALVE
S	SOLENOID VALVE
	PRESSURE & TEMPERATURE RELIEF VALVE
	VALVE IN RISER
P	PRESSURE GAUGE
P	PRESSURE GAUGE W/ COCK
	AUTOMATIC AIR VENT
	AQUASTAT

RISER DIAGRAMS

SYMBOL	DESCRIPTION
	WALL MOUNTED URINAL OR WATER CLOSET
	LAVATORY OR SINK, P-TRAP
	WATER HAMMER ARRESTOR
	FLOOR DRAIN
	WALL CLEANOUT
	SHUTOFF VALVE
	CHECK VALVE
	BALANCE VALVE
	NEW CONNECTION TO EXISTING
	CLEANOUT IN FLOOR

SANITARY SEWER/WASTE VENT (PLAN VIEW)

SYMBOL	DESCRIPTION
(DIAGRAM) (PLAN)	2-WAY CLEANOUT. MOUNT IN CONCRETE PAD FLUSH W/ FINISH GRADE
CO	CLEANOUT EXPOSED
YCO	YARD CLEANOUT
FCO	CLEANOUT IN FLOOR
V	SANITARY VENT
SAN	SANITARY WASTE
D	DRAIN
SPD	SUMP PUMP DISCHARGE LINE
FD	FLOOR DRAIN

ABBREV.

ABBREV.	DESCRIPTION
ABBREV.	ABBREVIATION
ABV	ABOVE
AC	ABOVE CEILING
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
&	AND
AP	ACCESS PANEL
APPROX	APPROXIMATE
ARCH	ARCHITECTURAL
ASA	AMERICAN STANDARDS ASSOCIATION
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS
ASSY	ASSEMBLY
ASTM	AMERICAN SOCIETY FOR TESTING MATERIALS
AUX	AUXILIARY
AV	ACID VENT
AVTR	ACID VENT THRU ROOF
AWWA	AMERICAN WATER WORKS ASSOCIATION
BA	BREATHING AIR
BFF	BELOW FINISHED FLOOR
BFP	BACK FLOW PREVENTER
BLDG	BUILDING
BTU	BRITISH THERMAL UNIT
CAP	CAPACITY
CD	CONDENSATE DRAIN
CI	CAST IRON
CL	CENTER LINE
CLG	CEILING
CLO	CLOSET
CO	CLEANOUT
CONN	CONNECTION
CONT	CONTINUATION
CTG	CLEANOUT TO GRADE (EXTERIOR CO)
CU	COPPER
CW	COLD WATER
D	DRAIN
DN	DOWN
DIA	DIAMETER
DWBP	DOMESTIC WATER BOOSTER PUMP SKID
DWG	DRAWING
EC	ELECTRICAL CONTRACTOR
ED	EQUIPMENT DRAIN
ELEC	ELECTRICAL
ELEV	ELEVATION (S)
ENGR	ENGINEERING
EQ	EQUAL
EQUIP	EQUIPMENT
ERW	NON POTABLE WATER
ESEW	EMERGENCY SHOWER EYEWASH
EX	EXISTING
FAB	FABRICATE
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
FDTN	FOUNDATION

ABBREV. DESCRIPTION

FF	FINISH FLOOR
FLG	FLANGE
FLS	FLOOR SINK
FL. MTD.	FLOOR MTD.
GC	GENERAL CONTRACTOR
GAL	GALLON(S)
GALV	GALVANIZED
GCO	GRADE CLEAN OUT
GPH	GALLONS PER HOUR
HB	HOSE BIB
HD	HUB DRAIN
HP	HORSEPOWER
HR	HOSE RACK
HW	HOT WATER
HWC	HOT WATER RE-CIRCULATION
HWP	HOT WATER RE-CIRCULATING PUMP
IE	INVERT ELEVATION
LA	LAB COMPRESSED AIR
LAV	LAVATORY
LBS	POUNDS
LS	LAB SINK
LV	LAB VACUUM
MAX	MAXIMUM
MECH	MECHANICAL
MGZA	MEDICAL GAS ZONE ALARM
MGZV	MEDICAL GAS ZONE VALVE
MIN	MINIMUM
MS	MOP SINK
MV	METHANE VENT
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NTS	NOT TO SCALE
NO	NORMALLY OPEN
NPW	NON-POTABLE WATER
OC	ON CENTER
OH	OVERHEAD
PCT	PERCENT (%)
PD	PUMP DISCHARGE
PLBG	PLUMBING
POC	POINT OF CONNECTION
PRESS	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSIG	POUNDS PER SQUARE INCH, GAUGE
PWC	POTABLE COLD WATER
PWH	POTABLE HOT WATER
PWHR	POTABLE HOT WATER RETURN
R	RELOCATE
RD	ROOF DRAIN
REQ	REQUIRED
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE BACKFLOW PREVENTER
RW	RECYCLED WATER
SK	SINK
SAN	SANITARY SEWER
SHW	SHOWER
SMP	SUMP PUMP
SO	SOIL AND/OR WASTE PIPE
SP	SOIL PIPE
SPEC	SPECIFICATION
SSTL	STAINLESS STEEL
ST	STORM PIPE
STL	STEEL
TC	TECHNOLOGY CONTRACTOR
TCC	TEMPERATURE CONTROL CONTRACTOR
TK	TANK
TMV	TEPID MIXING VALVE
TP	TRAP PRIMER
TW	TEPID WATER
TYP	TYPICAL
UF	UNDER FLOOR
UG	UNDERGROUND
UL	UNDERWRITERS LABORATORIES
UR	URINAL
UTIL	UTILITY
V	VENT PIPE
VTR	VENT THRU ROOF
W/	WITH
WB	WALL BOX
WC	WATER CLOSET
WCO	WALL CLEAN OUT
WH	WATER HEATER
WHA	WATER HAMMER ARRESTOR
WST	WASTE PIPE
YCO	YARD CLEAN OUT

GENERAL NOTES:

A. THIS IS A STANDARD PROCESS, PLUMBING SYMBOLOGY AND ABBREVIATIONS SHEET. ALL SYMBOLS ARE NOT NECESSARILY USED ON THIS PROJECT.

B. BASIS OF DESIGN PRODUCT BRANDS ARE FOR INFORMATIONAL PURPOSES ONLY AND SHALL NOT BE CONSTRUED AS THE ONLY PRODUCT AVAILABLE.

C. PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS TO SERVICE CONTROL DEVICES, VALVES, AND OTHER CONCEALED PLUMBING EQUIPMENT. ACCESS PANEL INSTALLATION SHALL BE COORDINATED WITH ARCHITECTURAL WORK TO MATCH MATERIALS AND FINISHES. REFER TO SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

D. MAINTAIN THE CLEARANCE SPACES AROUND EQUIPMENT AND SYSTEM COMPONENTS FOR OBSERVATION, OPERATION, AND MAINTENANCE.

E. INSTALL PLUMBING VENTS NO CLOSER THAN 10 FEET FROM ANY OUTSIDE AIR INTAKE.

F. PROVIDE DIELECTRIC UNIONS AT ALL CONNECTIONS OF DISSIMILAR METALS (SUCH AS COPPER TO GALVANIZED STEEL).

G. PLUMBING INSTALLATION SHALL COMPLY WITH APPLICABLE PLUMBING CODE AND THE ADA GUIDELINES FOR ACCESSIBILITY TO PLACES OF PUBLIC ACCOMMODATION AND COMMERCIAL FACILITIES BY INDIVIDUALS WITH DISABILITIES.

H. CLEANOUTS, VALVES, GAUGES, ETC. SHALL BE LOCATED TO MAINTAIN ACCESSIBILITY. WALL CLEANOUTS SHALL BE 18" A.F.F. AND FLOOR CLEANOUTS NO LESS THAN 12" FROM WALL, WHERE POSSIBLE.

I. PLANS ARE DIAGRAMMATIC. CONTRACTOR SHALL FURNISH AND INSTALL ALL EQUIPMENT, MATERIALS, AND LABOR AS SHOWN AND AS NECESSARY FOR A COMPLETE WORKABLE SYSTEM.

J. PRIOR TO CUTTING WALL OPENINGS, CONTRACTOR SHALL COORDINATE WORK WITH ALL DISCIPLINES BEFORE START OF WORK.

K. CONTRACTOR SHALL MAKE PROVISIONS TO ROUTE DUCTWORK AND PIPING TO AVOID CONFLICTS WITH OTHER PIPING, DUCTWORK, EQUIPMENT, CONDUITS, STRUCTURE, ETC.

L. CONTRACTOR IS RESPONSIBLE FOR PROVIDING AND INSTALLING ALL NECESSARY SUPPORTS AND SEISMIC BRACING FOR THE EQUIPMENT, DUCTWORK, PIPING, ETC. AS REQUIRED BY THE SPECIFICATIONS, THE MANUFACTURER, AND ALL LOCAL CODES.

M. WHERE PIPING CROSSES SEISMIC EXPANSION JOINT, CONTRACTOR TO PROVIDE FLEXIBLE EXPANSION JOINTS IN COMPLIANCE WITH ALL LOCAL CODES.

N. CONTRACTOR SHALL ROUTE CONDENSATE DRAIN PIPING WITH 1% SLOPE TOWARD THE POINT OF DISPOSAL, AND SHALL PROVIDE A PIPE CLEANOUT AT EVERY CHANGE IN DIRECTION GREATER THAN 90 DEGREES.

O. CONTRACTOR TO PROVIDE CLEANOUTS ON PLUMBING DRAINAGE PIPING AS REQUIRED BY THE PLUMBING CODE REGARDLESS OF WHETHER ALL CLEANOUTS ARE SHOWN ON PLANS.

P. PROVIDE AIR RELEASE VALVES AT ALL HIGH POINT ELEVATIONS IN PLUMBING FORCE-MAINS, TO RELEASE AIR POCKETS / VENT TRAPPED AIR.

Q. UNLESS NOTED OTHERWISE ON THE PLANS, GRAVITY DRAINAGE AND SEWER PIPING SMALLER THAN 3" SHALL BE SLOPED AT 1/4" PER FOOT, PIPES 3" AND LARGER MAY BE SLOPED AT 1/8" PER FOOT.

R. PROVIDE BACKFLOW PREVENTERS IN ACCORDANCE WITH THE LOCAL CODES. PROVIDE AIR GAP FITTINGS FOR ALL RPZ BACKFLOW PREVENTERS AND ROUTE DRAIN PIPING TO NEAREST FLOOR DRAIN OR FLOOR SINK OR AS SHOWN ON DRAWINGS.

S. REFER TO PROCESS DETAILS IN SERIES 005, AND SPECIFICATION SECTION 01 73 20 FOR PIPE PENETRATION DETAILS THROUGH WALLS AND FLOOR SLABS.

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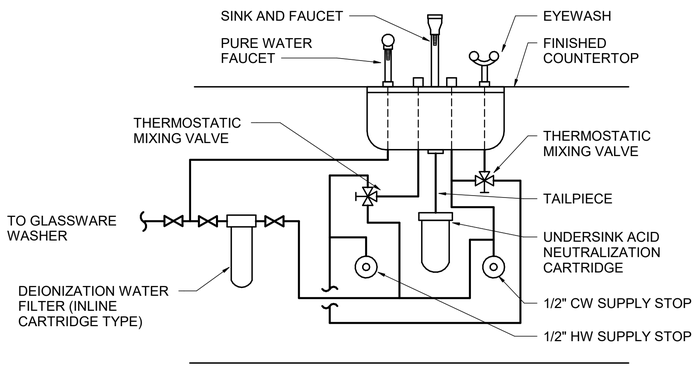
A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: A. PUTMAN
DRAWN: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 007 P001

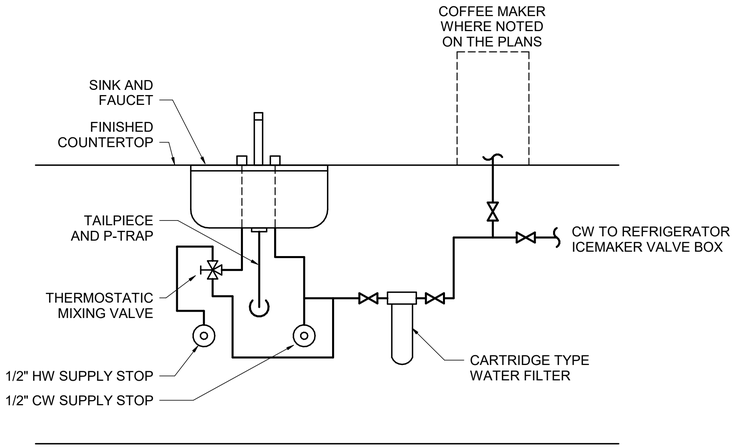
TITLE OF SHEET
GENERAL
PLUMBING LEGENDS, NOTES, AND ABBREVIATIONS
NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 70 OF 99



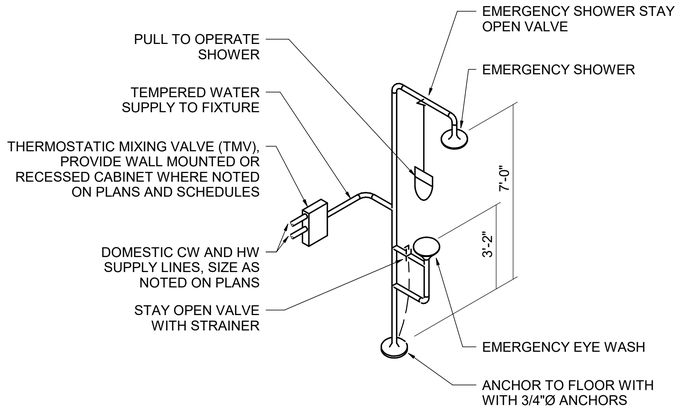
1 007P501 DETAIL - LAB SINK

NOT TO SCALE



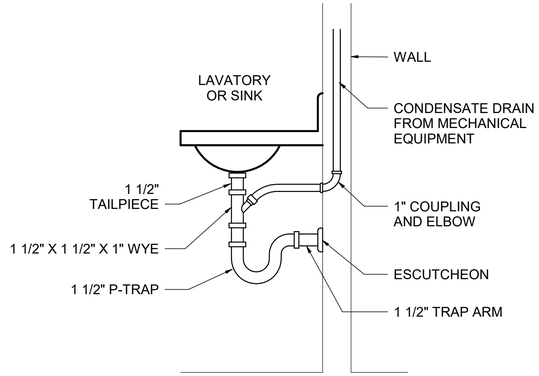
2 007P501 DETAIL - BREAK ROOM SINK

NOT TO SCALE



3 007P501 DETAIL - EMERGENCY SHOWER EYE WASH

NOT TO SCALE



NOTES:

1. REFER TO PLAN SHEETS FOR LINE SIZES

4 007P501 DETAIL - LAVATORY TAILPIECE

NOT TO SCALE

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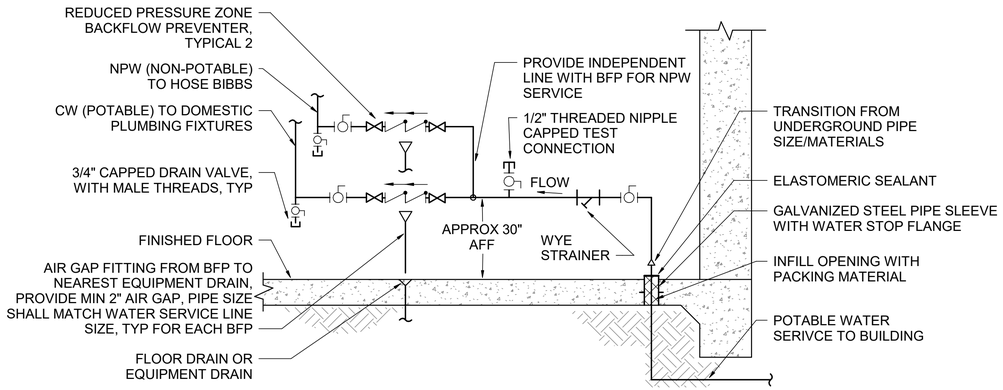
PLUMBING INTERNATIONAL Engineering, Inc. CERTIFICATE 86199 SIGNED 4-27-26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 007 P501	TITLE OF SHEET GENERAL PLUMBING DETAILS - FACILITIES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 71 OF 99

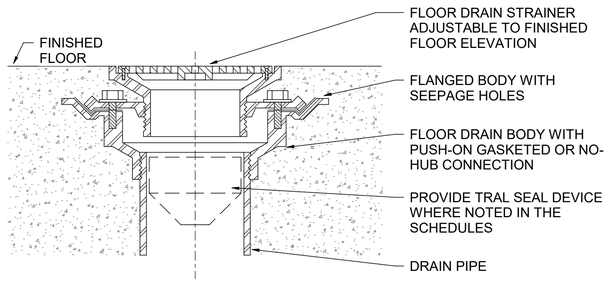


NOTES:

1. REFER TO PLAN SHEETS FOR LINE SIZES.

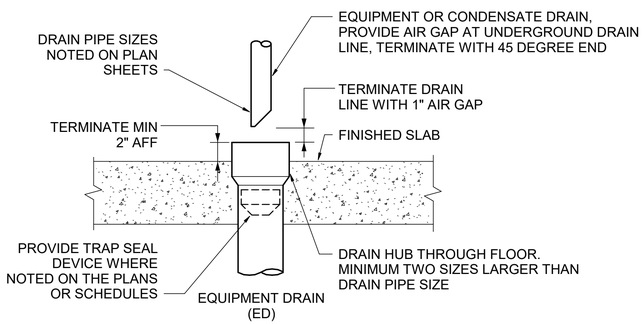
1 / 007P502 DETAIL - WATER SERVICE ENTRANCE

NOT TO SCALE



2 / 007P502 DETAIL - FLOOR DRAIN

NOT TO SCALE



3 / 007P502 DETAIL - EQUIPMENT DRAIN

NOT TO SCALE

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_000_GENERAL.rvt
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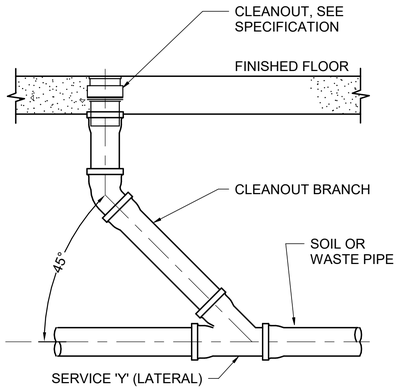
National Institute of Engineering, Inc. CERTIFICATE 86199 REGISTERED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 007 P502	TITLE OF SHEET GENERAL PLUMBING DETAILS - FACILITIES CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 72 OF 99

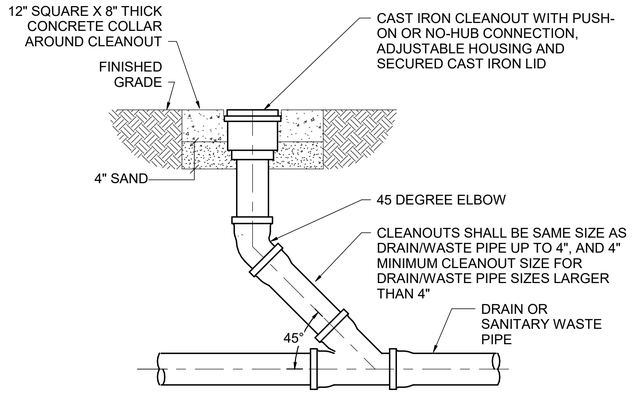


NOTES:

1. CLEANOUTS SHALL BE RECESSED 1/2" BELOW FINISHED SURFACE. COVER SHALL BE ACCESSIBLE FOR GENERAL MAINTENANCE.

1 007P503 DETAIL - CLEANOUT

NOT TO SCALE

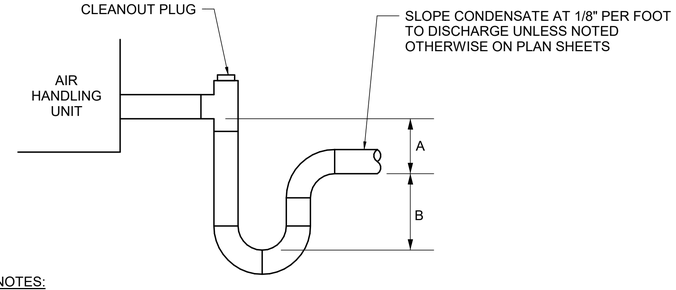


NOTES:

1. CLEANOUTS SHALL BE RECESSED OR FLUSH WITH FINISHED SURFACE. COVER SHALL BE ACCESSIBLE FOR GENERAL MAINTENANCE.

2 007P503 DETAIL - CLEANOUT TO GRADE

NOT TO SCALE

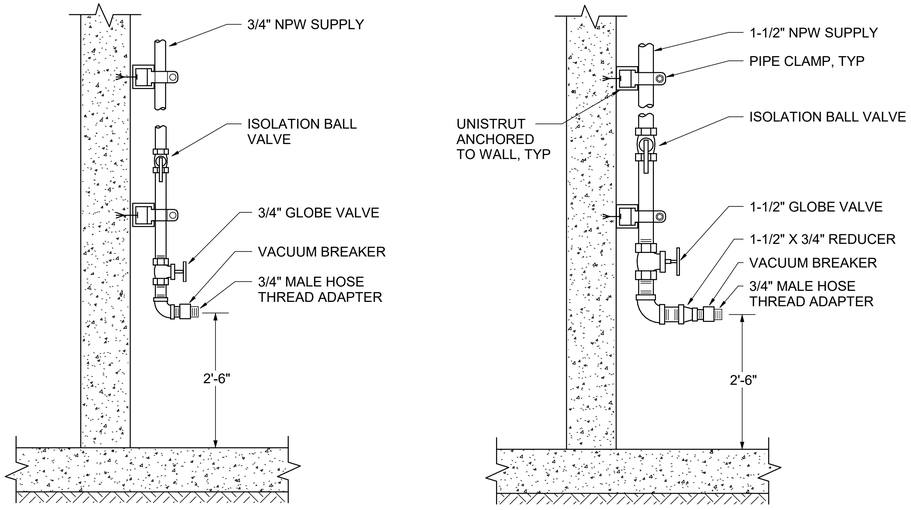


NOTES:

1. ON DRAW-THRU UNITS, THE (A) DIMENSION (IN INCHES) SHALL BE 1.5 TIMES THE AIR HANDLING UNITS SUCTION PRESSURE AT OPERATING CONDITIONS, BUT IN NO CASE LESS THAN 2-INCHES.
2. ON BLOW-THRU UNITS, THE (A) DIMENSION (IN INCHES) CAN BE AS SMALL AS 2 INCHES, BUT THE (B) DIMENSION (IN INCHES) SHALL BE 1.5 TIMES THE AIR HANDLING UNITS TOTAL STATIC PRESSURE BUT IN NO CASE LESS THAN 2 INCHES.
   - A = DISTANCE BETWEEN THE DRAIN PAN OUTLET AND THE TRAP
   - B = TRAP DEPTH

3 007P503 DETAIL - CONDENSATE P-TRAP

NOT TO SCALE

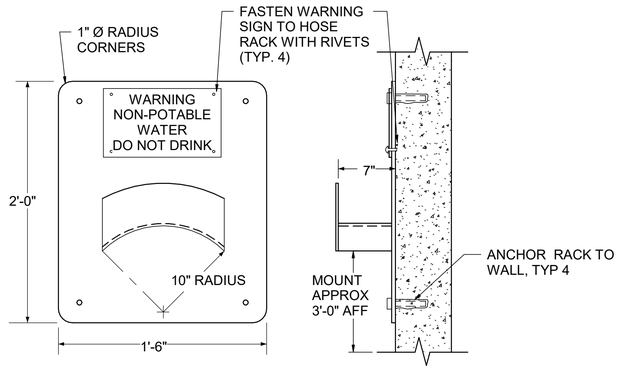


NOTES:

1. PROVIDE HOSE RACK AT HOSE BIBBS WHERE INDICATED ON THE PLANS.
2. INSULATE PIPING PER SPECIFICATIONS.

4 007P503 DETAIL - HOSE BIBBS

NOT TO SCALE



NOTES:

1. ALL HOSE RACKS AND MOUNTINGS SHALL BE WELDED CONSTRUCTION. MATERIAL SHALL BE 3/16" THICK TYPE 304 STAINLESS STEEL PLATE. MOUNTING HARDWARE AND ANCHORS SHALL BE TYPE 316 STAINLESS STEEL.

5 007P503 DETAIL - HOSE RACK (HR-1)

NOT TO SCALE

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 007 P503	TITLE OF SHEET GENERAL PLUMBING DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 73 OF 99

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ONE-LINE, POWER, AND LIGHTING SYMBOLOGY

X 20A 3P OR X 100AF 80AT 3P — LOW VOLTAGE CIRCUIT BREAKER (CB). RATING AND NO. OF POLES AS SHOWN. WHEN SPECIFIC TYPE, OTHER THAN MCCB, IS REQUIRED, X INDICATES TYPE.

TYPES:
- MCCB - MOLDED CASE
- ICCB - INSULATED CASE
- LVP - LOW VOLTAGE POWER
- MCP - MOTOR CIRCUIT PROTECTOR (RATING PER CONNECTED LOAD)

TRIP UNIT:
- L - LONG TIME PICKUP
- S - SHORT TIME PICKUP
- I - INSTANTANEOUS PICKUP
- G - GROUND FAULT PICKUP
- A - ARC ENERGY REDUCTION MODE

X — INTERLOCK: X - INDICATES TYPE

TYPES:
- E - ELECTRICAL
- M - MECHANICAL
- K - KEY

GFP — GROUND FAULT PROTECTION

52 — MEDIUM VOLTAGE CIRCUIT BREAKER

FUSE, RATING, AND NUMBER OF FUSES AS NOTED

FUSED CUTOUT, CURRENT RATING, FUSE RATING, AND QUANTITY AS NOTED

FUSIBLE SWITCH, CURRENT RATING, FUSE RATING, AND QUANTITY AS NOTED (3 POLE UON)

NON-FUSED SWITCH, CURRENT RATING, AND NUMBER OF POLES AS NOTED (3 POLE UON)

NON-FUSED SWITCH

INDICATES PART OR ALL OF A CIRCUIT RUNS IN UNDERGROUND CONDUIT

INDICATES PART OR ALL OF A CIRCUIT RUNS IN CABLE TRAY

DISCONNECT OR DRAWOUT CONNECTION

A B & A B — MAGNETIC MOTOR STARTER AND SEPARATELY MOUNTED COMBINATION MAGNETIC MOTOR STARTER

D C & D C — MOTOR/LOAD CONTROLLER AND SEPARATELY MOUNTED MOTOR/LOAD CONTROLLER WITH SHORT CIRCUIT PROTECTION AND DISCONNECT

MOTOR STARTER AND CONTROLLER SUBSCRIPTS:

A - MAGNETIC STARTER NEMA SIZE

B - STARTER TYPE
- NONE - FULL VOLTAGE NON-REVERSING (FVNR)
- FVR - FULL VOLTAGE REVERSING
- 2S - TWO SPEED
- RVAT - REDUCED VOLTAGE AUTO TRANSFORMER

C - CONTROL DIAGRAM OR CONTROLS SCHEDULE NUMBER (IF REQUIRED)

D - CONTROLLER TYPE
- VFD - VARIABLE FREQUENCY DRIVE
- SS - SOLID STATE
- CONT - CONTACTOR

SEPARATELY MOUNTED COMBINATION MOTOR STARTER OR CONTROLLER; SEE ELECTRICAL ONE - LINE DIAGRAM OR SCHEDULE FOR DESCRIPTION

SEPARATELY MOUNTED MOTOR STARTER OR CONTROLLER; SEE ELECTRICAL ONE-LINE DIAGRAM OR SCHEDULE FOR DESCRIPTION.

X — NON-FUSED SAFETY SWITCH, 30A, 3P, X INDICATES AMP RATING GREATER THAN 30A

X/Y — FUSED SAFETY SWITCH, 3P, X INDICATES AMP RATING GREATER THAN 30A, Y INDICATES FUSE SIZE

CB — SEPARATELY MOUNTED CIRCUIT BREAKER; SEE ELECTRICAL ONE - LINE DIAGRAM OR SCHEDULE FOR DESCRIPTION

7 1/2 OR HP — MOTOR WITH DESIGN HORSEPOWER (WHEN INDICATED)

G — GENERATOR

ATS — TRANSFER SWITCH, CURRENT RATING, AND NUMBER OF POLES AS NOTED

ATS - AUTOMATIC
MTS - MANUAL

TRANSFORMER
- 3-PHASE, 3-WIRE DELTA CONNECTION
- 3-PHASE, 4-WIRE GROUNDED WYE CONNECTION

INTERNAL MEDIUM VOLTAGE TRANSFORMER FUSE

LP100 208/120V 3Ø 4W — SWITCHBOARD OR PANELBOARD; NAME, VOLTAGE, PHASE, NUMBER OF WIRES WHEN INDICATED

100 KVA — NON-MOTOR LOAD WITH DESIGN KVA, KW, OR AMP

MEDIUM VOLTAGE SHUNT REACTOR

VOLTAGE TRANSFORMER (VT, PT, OR CPT)

CURRENT TRANSFORMER (CT)

M WH — UTILITY WATT-HOUR METER PER UTILITY REQUIREMENTS

DMP — DIGITAL METERING PACKAGE

GROUND

LA — LIGHTNING ARRESTER

G	GROUND	SB	SHORTING BLOCK
IC	INSTRUMENTATION / CONTROL DEVICE	SPD	LOW VOLTAGE SURGE PROTECTIVE DEVICE
N	NEUTRAL	SS	SELECTOR SWITCH
PB	PUSHBUTTON	SV	SOLENOID VALVE
R	RELAY	TB	TERMINAL BLOCK

CONTROL PANEL INTEGRAL OR PROVIDED WITH ASSOCIATED EQUIPMENT

CONTROL PANEL WITH DISCONNECT SWITCH INTEGRAL OR PROVIDED WITH ASSOCIATED EQUIPMENT

J OR J — JUNCTION OR PULL BOX

PANELBOARD (250V TO 600V)

PANELBOARD (LESS THAN 250V)

X — ELECTRICAL EQUIPMENT ENCLOSURE: SWITCHBOARD, MOTOR CONTROL CENTER, CONTROL PANEL, TRANSFORMER OR OTHER EQUIPMENT AS INDICATED. ESTIMATED SIZE AS INDICATED. WHEN USED X INDICATES EQUIPMENT TYPE.

EQUIPMENT TYPES:
- ATS - AUTOMATIC TRANSFER SWITCH
- CP - CONTROL PANEL
- MTS - MANUAL TRANSFER SWITCH
- MCC - MOTOR CONTROL CENTER
- UPS - UNINTERRUPTIBLE POWER SUPPLY
- VFD - VARIABLE FREQUENCY DRIVE
- SB - SWITCHBOARD
- SG - SWITCHGEAR
- T - TRANSFORMER

Z X/Y — CEILING/PENDANT/BOLLARD MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED

CEILING/PENDANT/BOLLARD MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED, EMERGENCY (INTERNAL OR EXTERNAL POWER SOURCE AS INDICATED)

WALL MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED

WALL MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED, EMERGENCY (INTERNAL OR EXTERNAL POWER SOURCE AS INDICATED)

WALL MOUNTED FLOOD LUMINAIRE, LAMP TYPE AS SPECIFIED

POLE/STANCHION MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED

POLE/STANCHION MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED, EMERGENCY (INTERNAL OR EXTERNAL POWER SOURCE AS INDICATED

POLE/STANCHION MOUNTED FLOOR LUMINAIRE, LAMP TYPE AS SPECIFIED

CEILING/PENDANT MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED

WALL MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED

CEILING/PENDANT MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED, ALL OR PARTIAL EMERGENCY (INTERNAL OR EXTERNAL POWER SOURCE AS INDICATED

WALL MOUNTED LUMINAIRE, LAMP TYPE AS SPECIFIED, ALL OR PARTIAL EMERGENCY (INTERNAL OR EXTERNAL POWER SOURCE AS INDICATED

EMERGENCY LIGHT, NUMBER OF ATTACHED HEADS AS SHOWN

EMERGENCY LIGHT, REMOTE MOUNTED HEAD

DOUBLE-FACED CEILING OR WALL MOUNTED EXIT LIGHT; DIRECTIONAL ARROWS (IF REQUIRED) AS INDICATED ON PLANS

SINGLE-FACED CEILING OR WALL MOUNTED EXIT LIGHT; DIRECTIONAL ARROWS (IF REQUIRED) AS INDICATED ON PLANS

LIGHTING FIXTURE SUBSCRIPTS:
- X - INDICATES LUMINAIRE TYPE PER LUMINAIRE SCHEDULE
- Y - INDICATES CIRCUIT NUMBER FROM PANELBOARD
- Z - INDICATES CONTROLLING SWITCH (IF REQUIRED)
- NL - NIGHT LIGHT UNSWITCHED

Y/X — WALL SWITCH

SUBSCRIPTS:
X - INDICATES TYPE
- NONE - SINGLE POLE
- 2 - DOUBLE POLE
- 3 - THREE-WAY
- 4 - FOUR-WAY
- K - KEY SWITCH
- P - PILOT LIGHT
- L - LIGHTED HANDLE
- DM - DIMMING
- MC - MOMENTARY CONTACT
- T - TIMER

Y - INDICATES CONTROLLING SWITCH (IF REQUIRED)

Y/X — MANUAL MOTOR STARTER

SUBSCRIPTS:
X - INDICATES TYPE
- HP - HORSEPOWER RATED
- TE - HORSEPOWER RATED WITH THERMAL ELEMENT
- FT - HORSEPOWER RATED WITH FUSETRON FUSE

Y - INDICATES SWITCH TYPE
- NONE - TOGGLE SWITCH TYPE
- R - ROTARY SWITCH TYPE

PC — PHOTOCELL

TC — TIME CLOCK

OSX OR OSX — LIGHTING CONTROL OCCUPANCY SENSOR, WALL MOUNTED, X INDICATES SPECIFIC TYPE AS SPECIFIED

OSX — LIGHTING CONTROL OCCUPANCY SENSOR, CEILING MOUNTED, X INDICATES SPECIFIC TYPE AS SPECIFIED

X — ROOM/AREA LIGHTING CONTROL TYPE, SEE LIGHTING CONTROL SCHEDULE FOR REQUIREMENTS

2[a] — LOW VOLTAGE DIGITAL WALL SWITCH, NUMBER INDICATES QUANTITY OF PUSH BUTTONS PER SINGLE GANG PLATE, LETTER INDICATES CONTROL ZONE WHEN SHOWN

PLUG-IN RECEPTACLE STRIP, QUANTITY AND SPACING OF RECEPTACLES AS NOTED OR SPECIFIED

X/Y — SPECIAL-PURPOSE RECEPTACLE AS DEFINED ON PLANS

X/Y OR X/Y — TWO RECEPTACLES IN 2-GANG BOX UNDER COMMON COVER PLATE

X/Y — DUPLEX RECEPTACLE

X/Y — SIMPLEX RECEPTACLE

X/Y — RECESSED FLOOR MOUNTED BOX, QUANTITY AND TYPE OF RECEPTACLES AS INDICATED

SUBSCRIPTS:

X - INDICATES TYPE
- GFCI - GROUND FAULT CIRCUIT INTERRUPTER
- IG - ISOLATED GROUND
- TR - TAMPER RESISTANT
- PLH - PLUG LOAD HALF CONTROLLED (*)
- PLD - PLUG LOAD DUAL CONTROLLED (*)
- PLF - PLUG LOAD FULL CONTROLLED (*)
- USB - USB CHARGING STATION
- SPD - SURGE PROTECTIVE DEVICE
- AC - ABOVE COUNTER
- WP - WEATHERPROOF
- WPI - WEATHERPROOF IN-USE

Y - INDICATES CIRCUIT NUMBER FROM PANELBOARD
(*) - REFER TO PLUG LOAD CONTROL DETAIL #3 ON SHEET 008-E504.

CONDUIT TURNING UP

CONDUIT TURNING DOWN

(#X) — HOMERUN TO SOURCE (E.G. PANELBOARD, MCC)
NUMBER IN PARENTHESES REPRESENTS CONDUCTOR SIZE OTHER THAN #12
SINGLE PHASE: 2#12, 1#12G IN 3/4"C
THREE PHASE: 3#12, 1#12G IN 3/4"C
UNLESS OTHERWISE NOTED, CONDUCTOR SIZE IS FOR ENTIRE CIRCUIT, SOURCE TO LAST DEVICE. ALSO, SEE ONE LINE DIAGRAM FOR CIRCUIT REQUIREMENTS

CONDUIT CONNECTION TO EQUIPMENT

CIRCUIT RUN BETWEEN DEVICES EXPOSED IN NON-ARCHITECTURALLY FINISHED AREAS; CONCEALED IN ARCHITECTURALLY FINISHED AREAS. CONDUIT AND CONDUCTOR SIZES SHALL BE THE SAME AS THE HOMERUN FOR THE CIRCUIT.

CONDUIT RUN BETWEEN DEVICES CONCEALED IN NON-ARCHITECTURALLY FINISHED AREAS OR UNDER FLOOR SLAB. CONDUIT AND CONDUCTOR SIZES SHALL BE THE SAME AS THE HOMERUN FOR THE CIRCUIT.

(#X) — CIRCUIT HASH MARKS (WHEN INDICATED); LONG, SHORT, SINGLE DOT, AND DOUBLE DOT REPRESENT PHASE, NEUTRAL, EQUIPMENT GROUND, AND ISOLATED EQUIPMENT GROUND, RESPECTIVELY. X REPRESENTS CONDUCTOR SIZE OTHER THAN #12 IN 3/4" CONDUIT.

CIRCUIT CONTINUATION

CONDUIT STUBBED OUT AND CAPPED

CORD AND PLUG CONNECTION

XXX OR XXX — CONDUIT TAG OR CIRCUIT NUMBER - WIRE AND CONDUIT SIZE AS SPECIFIED IN CIRCUIT SCHEDULE ON THE SHEETS

GROUND CABLE

GROUND ROD

GENERAL NOTES:

1. THIS IS A STANDARD ELECTRICAL SYMBOLOGY SHEET. NOT ALL SYMBOLS MAY BE USED ON THIS PROJECT.
2. SCREENING OR SHADING OF WORK IS USED TO INDICATE EXISTING COMPONENTS OR TO DE-EMPHASIZE PROPOSED IMPROVEMENTS TO HIGHLIGHT SELECTED TRADE WORK. REFER TO CONTEXT OF EACH SHEET FOR USAGE.
3. SEE P&ID LEGEND SHEET FOR PROJECT SPECIFIC EQUIPMENT SYMBOLS, EQUIPMENT ABBREVIATIONS, AND PIPING SYSTEM ABBREVIATIONS.

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 1/14/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	T. MOORE	008 E001	GENERAL ELECTRICAL LEGEND - ONE-LINE, POWER, AND LIGHTING SYMBOLOGY
	DRAWN: N. LYNCH		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: P. DAVILA		
	DATE: AUG 2026		

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 74 OF 99

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COMMUNICATION SYMBOLOGY

- WALL MOUNTED TELEPHONE OUTLET
- WALL MOUNTED DATA OUTLET
- WALL MOUNTED COMBINATION TELEPHONE AND DATA OUTLET
- RECESSED FLOOR MOUNTED TELEPHONE OUTLET
- RECESSED FLOOR MOUNTED DATA OUTLET
- RECESSED FLOOR MOUNTED COMBINATION TELEPHONE AND DATA OUTLET
- WIRELESS ACCESS POINT

AUDIO/VISUAL SYMBOLOGY

- TV — TELEVISION OUTLET
- S — CEILING MOUNT SPEAKER
- S X — WALL MOUNT SPEAKER
  SPEAKER SUBSCRIPTS:
  X - INDICATES HEIGHT
- H — HORN TYPE TRANSDUCER
- VC — VOLUME CONTROL
- PAHE — HEAD END EQUIPMENT
- M — FLOOR MOUNTED MICROPHONE JACK
- M — WALL MOUNTED MICROPHONE JACK

SECURITY SYMBOLOGY

- DOOR POSITION SWITCH
- COMBINATION ELECTRIC DOOR STRIKE AND POSITION SWITCH
- R P — PROXIMITY CARD READER
- R K — PROXIMITY CARD READER WITH KEYPAD
- M — DUAL TECHNOLOGY MOTION DETECTOR
- REQUEST TO EXIT MOTION DETECTOR
- REQUEST TO EXIT PUSH BUTTON
- G — GLASS BREAK DETECTOR
- PTZ — CCTV CAMERA
  PAN/TILT/ZOOM WHEN INDICATED
- SEC — SECURITY EQUIPMENT CABINET
- R — REMOTE KEYPAD/CONTROL STATION

EMERGENCY ALARM SYMBOLOGY

- E — ALARM BELL
- E — ALARM HORN
- E — ALARM FLASHING LIGHT
- E — ALARM BELL AND FLASHING LIGHT COMBINATION UNIT
- E — ALARM HORN AND FLASHING LIGHT COMBINATION UNIT
- E — PUSHBUTTON OR PULLSTATION

SITE SYMBOLOGY

- EXTERIOR PAD MOUNTED TRANSFORMER
- POLE - MOUNTED TRANSFORMER
- X / Y — ELECTRICAL HANDHOLE OR MANHOLE
  X - INDICATES SEQUENCE NUMBER
  Y- MHX OR HHX
- X / Y — POLE/STANCHION MOUNTED FLOOD LUMINAIRE, LAMP TYPE AS SPECIFIED
- X / Y — POLE MOUNTED AREA OR ROADWAY LUMINAIRE, LAMP TYPE AS SPECIFIED
- X / Y — HIGH MAST LIGHTING, NUMBER OF LUMINAIRES AS SPECIFIED
  LIGHTING FIXTURE SUBSCRIPTS:
  X - INDICATES LUMINAIRE TYPE PER LUMINAIRE SCHEDULE
  Y - INDICATES CIRCUIT NUMBER FROM PANELBOARD
- POWER POLE
- DOWNGUY
- UE — UNDERGROUND (UNO) ELECTRICAL AND COMMUNICATION SYSTEMS PATHWAY
- OHE — OVERHEAD ELECTRICAL AND COMMUNICATION SYSTEMS PATHWAY
- FO — FIBER OPTIC PATHWAY
- UE — EXISTING UNDERGROUND (UNO) ELECTRICAL AND COMMUNICATION SYSTEMS PATHWAY
- OHE — DEMO OVERHEAD ELECTRICAL
- EXISTING POWER POLE
- EXISTING GUY AND ANCHOR

CONTROL SYMBOLOGY

- ELECTRICAL CONNECTION
- OR OR — NO ELECTRICAL CONNECTION
- SV OR — SOLENOID VALVE
- X / Y — CONTROL/RELAY COIL:
  X-INDICATES TYPE
  Y-INDICATES LOOP NUMBER, WHEN USED
  TYPES:
  CR-CONTROL RELAY — TC-TIME CLOCK
  PC-PHOTOCELL — LC-LIGHTING CONTACTOR
  DP-DEFINITE PURPOSE — TR-TIMING RELAY
  M-MOTOR STARTER
- NORMALLY OPEN CONTACT (N.O.)
- NORMALLY CLOSED CONTACT (N.C.)
- NORMALLY OPEN TIME DELAY RELAY CONTACT WITH TIME DELAY ON CLOSING AFTER COIL IS ENERGIZED
- NORMALLY CLOSED TIME DELAY RELAY CONTACT WITH TIME DELAY ON OPENING AFTER COIL IS ENERGIZED
- NORMALLY OPEN TIME DELAY RELAY CONTACT WITH TIME DELAY ON OPENING AFTER COIL IS DE-ENERGIZED
- NORMALLY CLOSED TIME DELAY RELAY CONTACT WITH TIME DELAY ON CLOSING AFTER COIL IS DE-ENERGIZED
- NORMALLY OPEN TEMPERATURE SWITCH; CLOSE ON RISING TEMPERATURE
- NORMALLY CLOSED TEMPERATURE SWITCH; OPEN ON RISING TEMPERATURE
- NORMALLY OPEN FLOW SWITCH; CLOSE ON INCREASING FLOW
- NORMALLY CLOSED FLOW SWITCH; OPEN ON INCREASING FLOW
- NORMALLY OPEN LEVEL SWITCH, CLOSE ON RISING LEVEL
- NORMALLY CLOSED LEVEL SWITCH, OPEN ON RISING LEVEL
- NORMALLY OPEN PRESSURE SWITCH, CLOSE ON INCREASING PRESSURE
- NORMALLY CLOSED PRESSURE SWITCH, OPEN ON INCREASING PRESSURE
- NORMALLY OPEN LIMIT SWITCH, CLOSE ON REACHING LIMIT
- NORMALLY CLOSED LIMIT SWITCH, OPEN ON REACHING LIMIT
- MICROPROCESSOR (PLC, RTU, ETC.) OUTPUT
- MICROPROCESSOR (PLC, RTU, ETC.) INPUT
- FIELD WIRING EXTERNAL TO CONTROL PANEL
- OFF / HAND / AUTO — XOO / OOX — 3 POSITION SELECTOR SWITCH, MAINTAINED CONTACTS; UNLESS OTHERWISE NOTED, 2-POSITION SIMILAR
- NORMALLY OPEN PUSHBUTTON, MOMENTARY CONTACT UNLESS OTHERWISE NOTED
- NORMALLY CLOSED PUSHBUTTON, MOMENTARY CONTACT UNLESS OTHERWISE NOTED

CONTROL SYMBOLOGY

- X — INDICATING LIGHT: X INDICATES LENS COLOR
- X — PUSH TO TEST INDICATING LIGHT: X INDICATES LENS COLOR
  LENS COLORS:
  R - RED — Y - YELLOW
  G - GREEN — W - WHITE
  B - BLUE — A - AMBER
- THERMAL OVERLOAD ELEMENT
- OL / (X) — THERMAL OVERLOAD RELAY CONTACT. WHEN SHOWN X INDICATES QUANTITY.
- CPT — CONTROL POWER TRANSFORMER (CPT)
- RTM — RUN TIME METER

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — ELECTRICAL — ARIZONA, U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	T. MOORE
DRAWN:	N. LYNCH
TECH. REVIEW:	P. DAVILA
DATE:	MAY 2026
SUB SHEET NO.	008 E002
TITLE OF SHEET	GENERAL ELECTRICAL LEGEND - CONTROLS, COMMS, AUDIO, SECURITY AND SITE
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	75 OF 99

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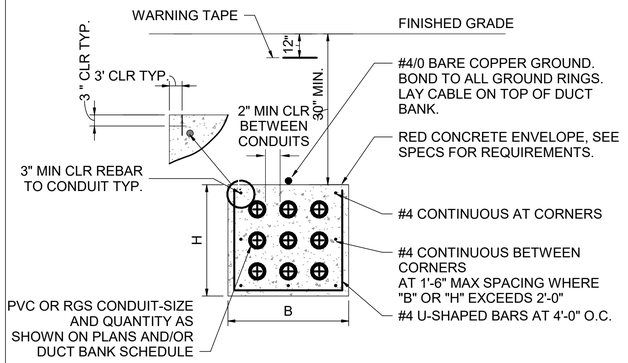
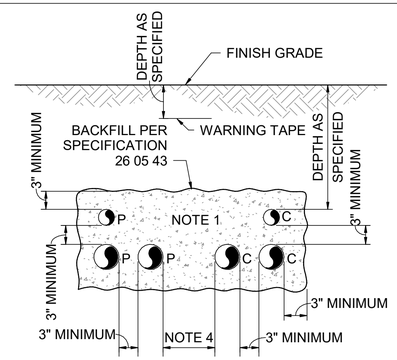


NOTES:

1. REINFORCED CONCRETE:
   A. REFER TO REQUIREMENTS IN DIVISION 3 AND 31 .
   B. ASTM A615, GRADE 60, REINFORCING STEEL.
   C. MINIMUM 4500 PSI CONCRETE.
2. REFER TO DIVISION 31 FOR TRENCHING, BACKFILLING, AND COMPACTING REQUIREMENTS.
3. PROVIDE CARLON PLASTIC DUCT SPACERS AT MAXIMUM INTERVALS OF 10 FEET. DRIVE #3 REINFORCING BAR DOWN THRU THE INTERNAL VERT. CHANNELS AT EACH SIDE OF SPACERS & 12" MIN. INTO TRENCH BOTTOM TO ANCHOR DUCT RUN WHILE CONCRETE IS POURED.
4. ALL DUCT BANKS SHALL BE FORMED TO THE DIMENSIONS INDICATED. DUCT BANKS SHALL NOT BE CONSTRUCTED BY FILLING AN UNFORMED TRENCH WITH CONCRETE.

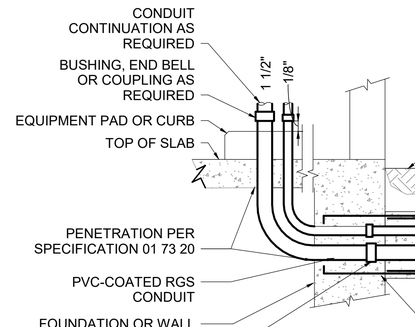
1 TYPICAL LV DUCTBANK SECTION
1 / 008E501
NOT TO SCALE



NOTES:

1. NUMBER OF CONDUITS AS REQUIRED FOR THE APPLICATION.
2. P SUBSCRIPT ELECTRICAL POWER OR CONTROL CONDUIT.
3. C SUBSCRIPT COMMUNICATION (TELEPHONE, DATA, INSTRUMENTATION) CONDUIT.
4. 6" MINIMUM WHEN POWER CONDUIT CONTAINS LESS THAN 1000V. 12" MINIMUM WHEN POWER CONDUIT CONTAINS MORE THAN 1000V.

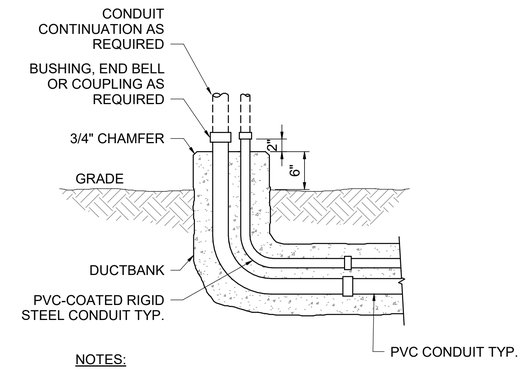
2 DIRECT BURIED CONDUIT(S) SECTION
2 / 008E501
NOT TO SCALE



NOTE:

1. SEE DUCTBANK DETAIL FOR ADDITIONAL REQUIREMENTS.

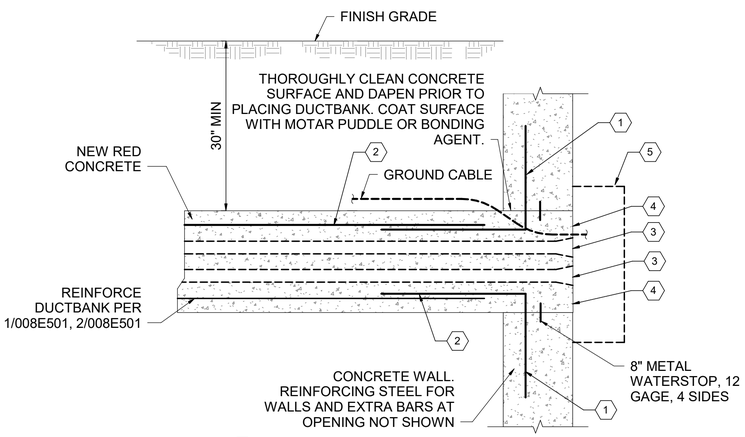
3 CONDUIT TRANSITION TO ABOVE GRADE (INTERIOR)
3 / 008E501
NOT TO SCALE



NOTES:

1. SEE DUCTBANK DETAIL FOR ADDITIONAL REQUIREMENTS.

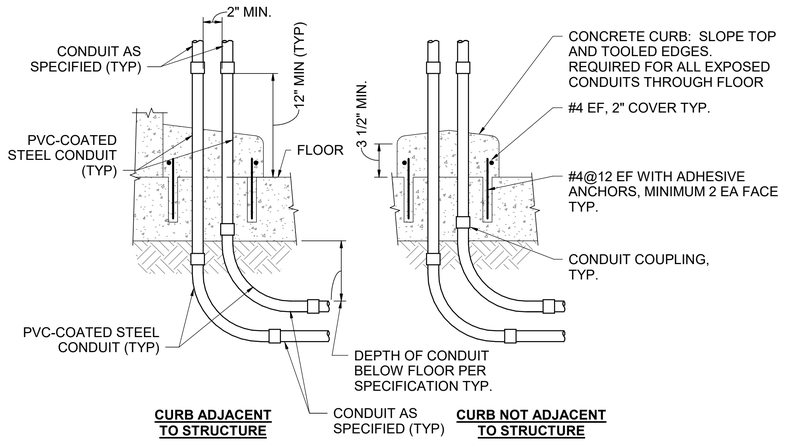
4 CONDUIT TRANSITION TO ABOVE GRADE (EXTERIOR)
4 / 008E501
NOT TO SCALE



KEY NOTES: (X)

1. BLOCK-OUT IN STRUCTURE WALL WITH REINFORCING BARS EXTENDED THROUGH.
2. CUT AND BEND WALL REINFORCING INTO DUCTBANK AND SPLICE TO REINFORCING BARS IN DUCT BANK.
3. END BELLS ON DUCTBANK CONDUITS.
4. DUCTBANK CONCRETE ENVELOPE POURED THROUGH OPENING, FLUSH WITH INSIDE OF WALL.
5. PULLBOX WHERE SHOWN ON DRAWINGS.

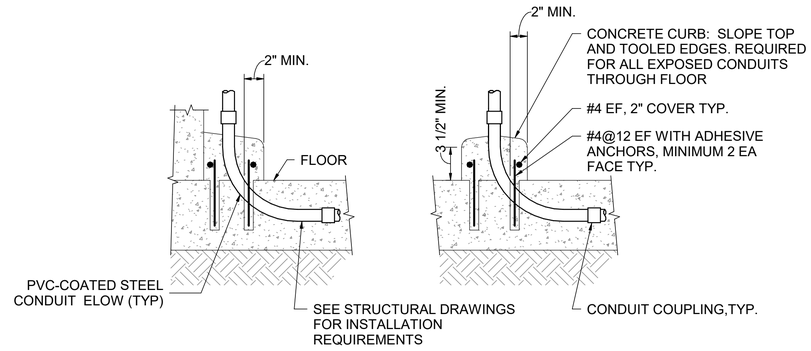
5 UNDERGROUND DUCTBANK FOR BUILDING ENTRY
5 / 008E501
NOT TO SCALE



NOTES

1. CONCRETE: SEE DIVISION 3 SPECIFICATIONS.

6 CONDUIT CURB - BELOW FLOOR CONDUIT
6 / 008E501
NOT TO SCALE



NOTES

1. CONDUIT MATERIAL: SEE SPECIFICATIONS.
2. CONCRETE: SEE DIVISION 3 SPECIFICATIONS.

7 CONDUIT CURB - IN FLOOR CONDUIT
7 / 008E501
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE ARIZONA, U.S.A. Date Signed 2/18/2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE CADD: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E501	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - UNDERGROUND CONDUIT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 76 OF 99

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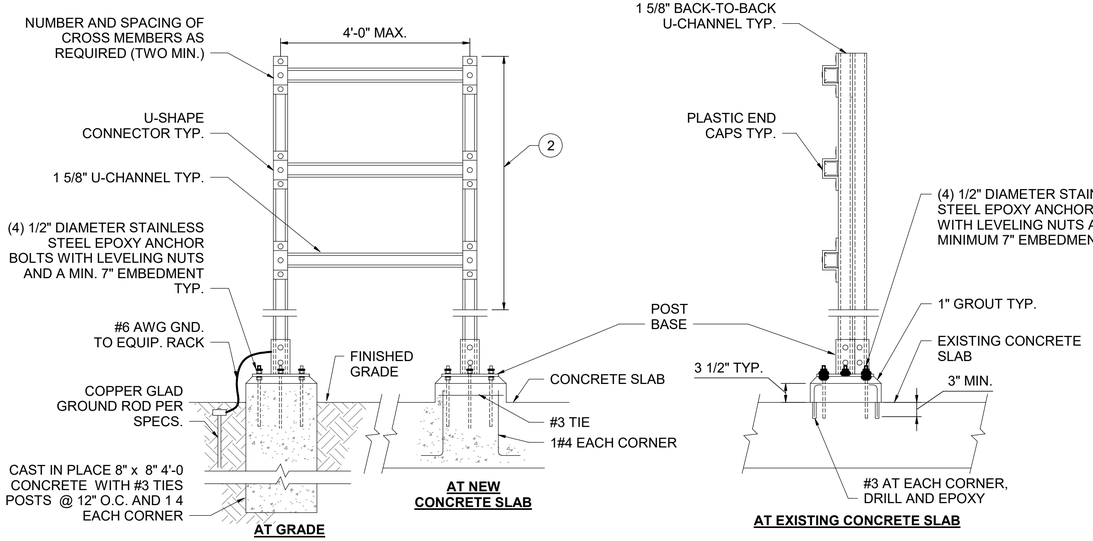
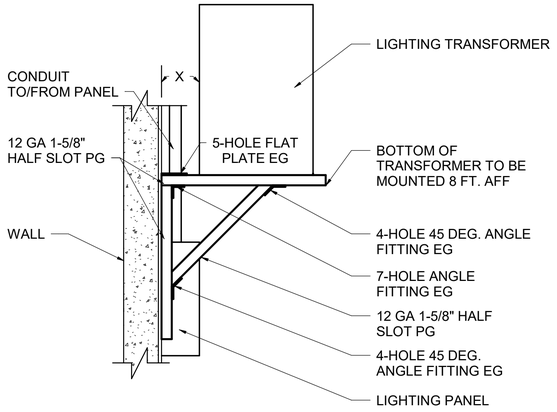


NOTES:

1. EQUIPMENT RACK SHALL BE UTILIZED FOR MOUNTING THE FOLLOWING:
   A. ONE EQUIPMENT ITEM WITH MOUNTING FOOTPRINT GREATER THAN 150 SQ-IN .
   B. TWO EQUIPMENT ITEMS WITH COMBINED MOUNTING FOOTPRINT GREATER THAN 130 SQ-IN .
   C. THREE OR MORE EQUIPMENT ITEMS.
2. WHERE MULTIPLE EQUIPMENT ITEMS WITH SMALL FOOTPRINT AREAS ARE MOUNTED ON THE RACK, A 1/4" PLATE FOR EQUIPMENT MOUNTING CAN BE UTILIZED SPANNING FROM THE TOP TO THE BOTTOM CROSS MEMBER IN LIEU OF MULTIPLE CROSS MEMBERS.
3. EQUIPMENT SHALL BE MOUNTED SO THAT INDICATION, ADJUSTMENTS, OR OPERATING HANDLES ARE FOUR TO FIVE FEET ABOVE FLOOR OR PLATFORM.
4. REFER TO STRUCTURAL DRAWINGS AND SPECIFICATION FOR ANCHORAGE MATERIALS AND METHOD REQUIREMENTS.
5. MATERIALS AND HARDWARE SHALL CONFORM TO THE REQUIREMENTS OF SPECIFICATION 26 05 00.

1 008E502 TYPICAL EQUIPMENT RACK

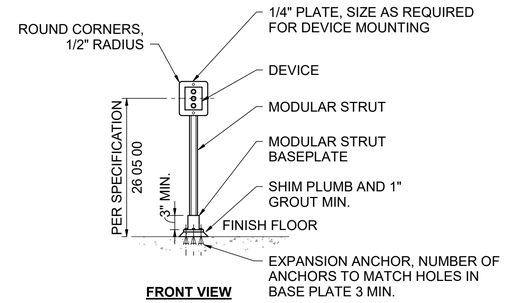
NOT TO SCALE



NOTE:
1. X = DISTANCE SHALL BE AT LEAST THE DEPTH OF THE ASSOCIATED PANEL TO ALLOW FOR UPWARDS CABLE ROUTING FROM THE PANEL.

2 008E502 WALL MOUNTED UNISTRUT

NOT TO SCALE

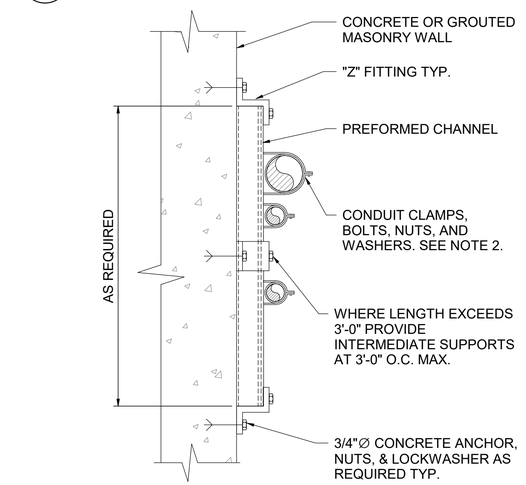


NOTES

1. DEVICE LOAD SHALL NOT EXCEED 25LBS.
2. PEDESTAL ASSEMBLY MATERIAL: GALVANIZED STEEL PER SPECIFICATION 26 05 00.
3. MODULAR STRUT WIDTH: 1 5/8"
4. ANCHORS: STAINLESS STELL, 1/2" DIAMETER, 3 1/2" EMBEDMENT, PER SPECIFICATION 03 15 19.
5. PROTECT SURFACES WITH DISSIMILAR MATERIALS IN ACCORDANCE WITH SPECIFICATION 09 96 00.
6. REPAIR CUT ENDS AND DAMAGED SURFACES IN ACCORDANCE WITH SPECIFICATION 05 50 00.

3 008E502 MODULAR DEVICE PEDESTAL

NOT TO SCALE



NOTES:

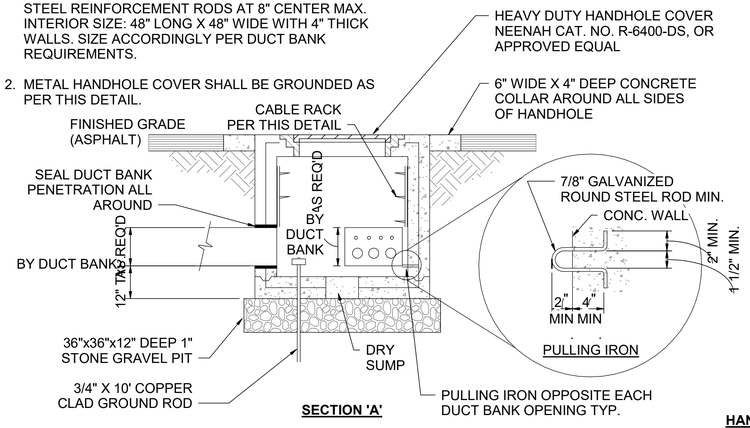
1. THIS DETAIL TYPICAL FOR BOTH VERTICAL AND HORIZONTAL MOUNTING.
2. MATERIAL FOR PREFORMED CHANNEL, FITTINGS, AND CLAMPS SHALL BE AS SPECIFIED IN SECTION 26 05 00.
3. SUPPORTS TO BE SPACED IN ACCORDANCE WITH NEC REQUIREMENTS FOR THE SMALLEST CONDUIT ATTACHED.

4 008E502 CONDUIT WALL SUPPORT

NOT TO SCALE

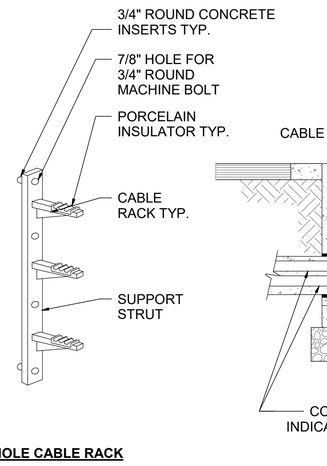
NOTES:

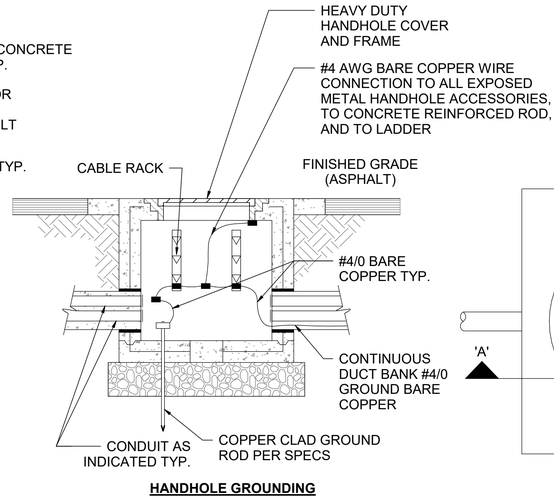
1. MINIMUM REQUIREMENTS: CONCRETE BOX WITH #16 STEEL REINFORCEMENT RODS AT 8" CENTER MAX. INTERIOR SIZE: 48" LONG X 48" WIDE WITH 4" THICK WALLS. SIZE ACCORDINGLY PER DUCT BANK REQUIREMENTS.
2. METAL HANDHOLE COVER SHALL BE GROUNDED AS PER THIS DETAIL.

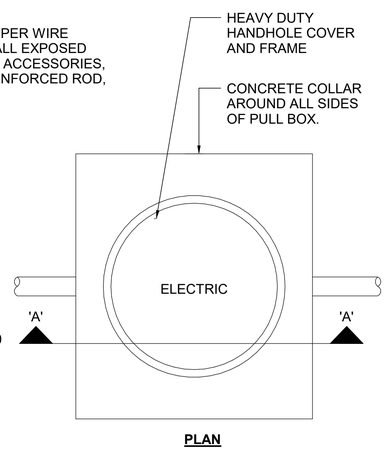


5 008E502 CONCRETE HANDHOLE DETAIL

NOT TO SCALE







REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
58037
TRAVIS T. MOORE
ELECTRICAL
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E502	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - UNDERGROUND AND EQUIPMENT MOUNTING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 77 OF 99

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_000_GENERAL.rvt
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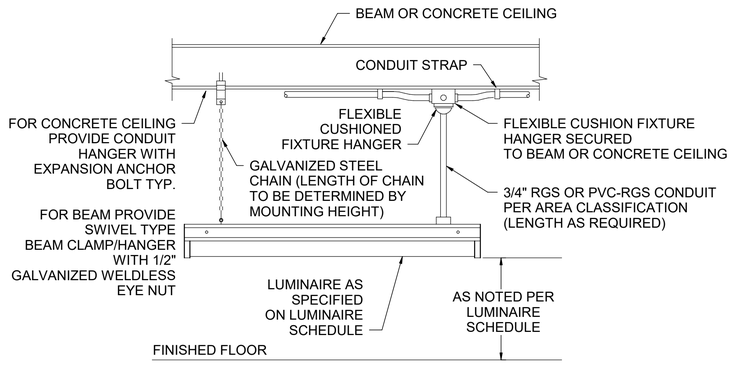
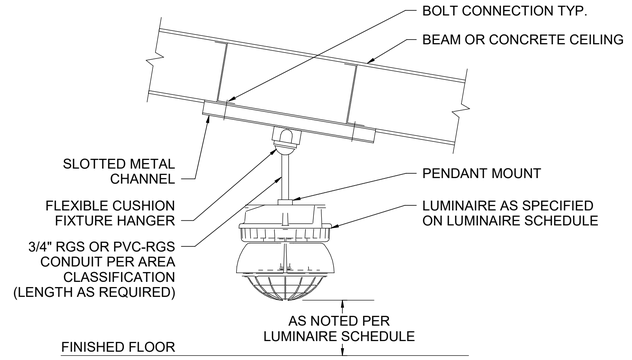


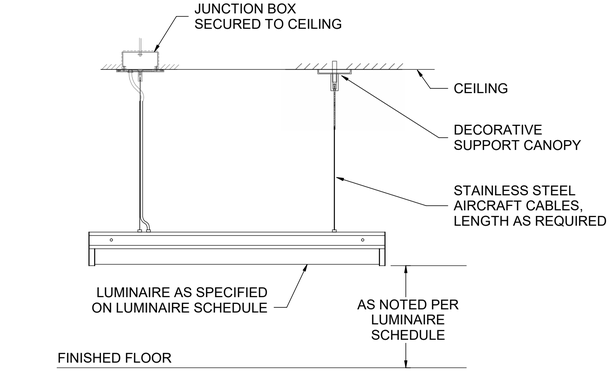
1 008E503 LINEAR LUMINAIRE MOUNTING DETAIL - TYPE 'Ms'

NOT TO SCALE



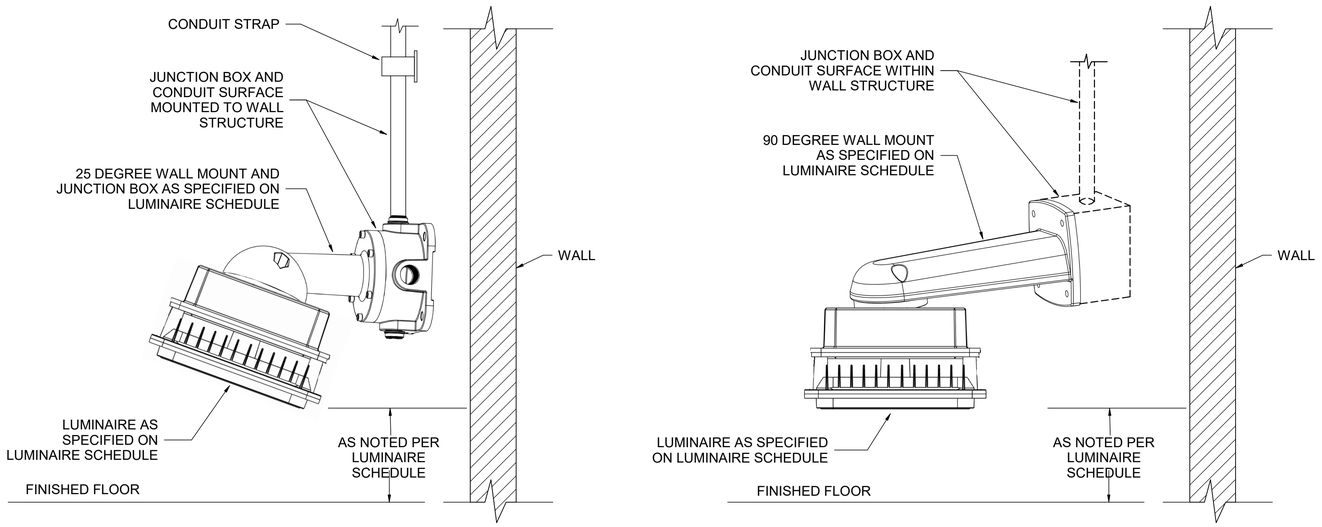
2 008E503 PENDANT LUMINAIRE MOUNTING DETAIL - TYPES 'Bs', 'Cs', & 'Ns'

NOT TO SCALE



3 008E503 DECORATIVE LINEAR LUMINAIRE MOUNTING DETAIL - TYPE 'Ds' & 'Ks'

NOT TO SCALE



4 008E503 WALL LUMINAIRE MOUNTING DETAIL - TYPE N's

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 04/27/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E503	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - LUMINAIRE MOUNTING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 78 OF 99

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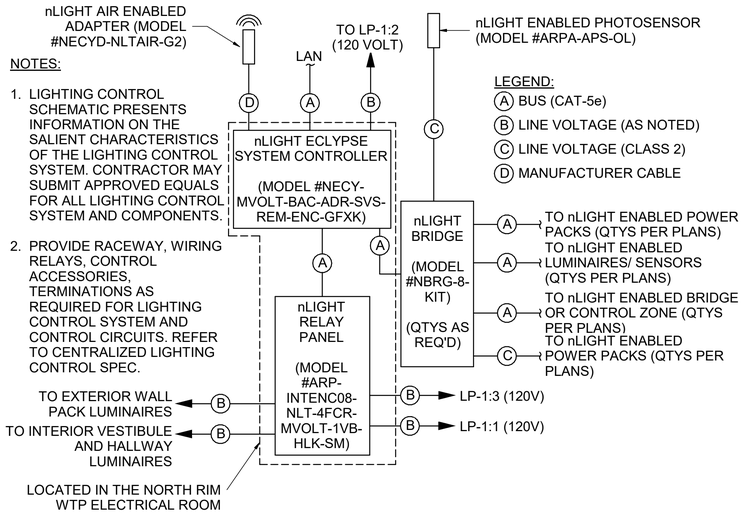
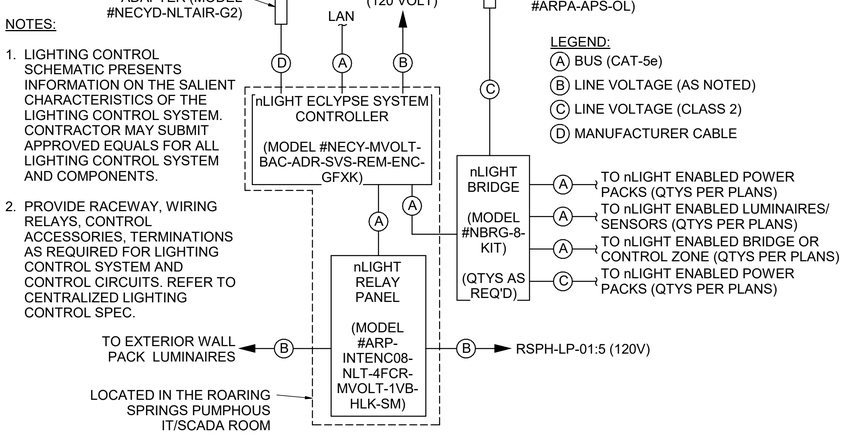


1 008E504 LUMINAIRE CONTROL SCHEMATIC 'LCP-01' (NORTH RIM WTP)
NOT TO SCALE

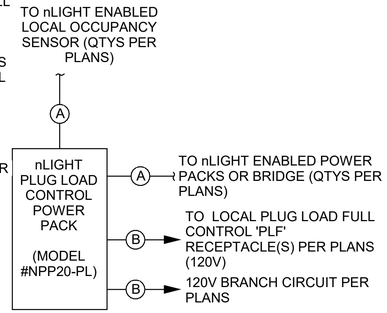


2 008E504 LUMINAIRE CONTROL SCHEMATIC 'RSPH-LCP-01' (ROARING SPRINGS PUMPHOUSE)
NOT TO SCALE

NOTES:

1. PLUG LOAD CONTROL SCHEMATIC PRESENTS INFORMATION ON THE SALIENT CHARACTERISTICS OF THE PLUG LOAD CONTROL POWER PACK. CONTRACTOR MAY SUBMIT APPROVED EQUALS FOR ALL PLUG LOAD CONTROL COMPONENTS.
2. PROVIDE RACEWAY, WIRING RELAYS, CONTROL ACCESSORIES, TERMINATIONS AS REQUIRED FOR PLUG LOAD CONTROL POWER PACK.
3. PLUG LOAD CONTROL POWER PACK TO CONTROL (ON/OFF) LOCAL PLUG LOAD FULL CONTROL 'PLF' RECEPTACLES VIA THE LOCAL OCCUPANCY SENSOR. REFER TO LUMINAIRE CONTROL SEQUENCE SCHEDULE.

LEGEND:
A BUS (CAT-5e)
B LINE VOLTAGE (AS NOTED)



3 008E504 PLUG LOAD CONTROL SCHEMATIC
NOT TO SCALE

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REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 4/24/2026 ARIZONA, U.S.A.

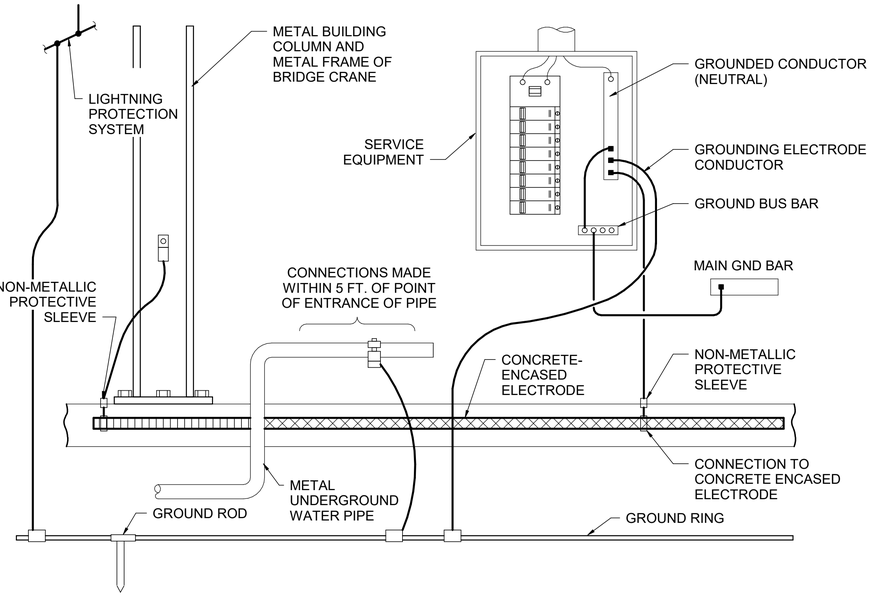
A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: T. MOORE
DRAWN: N. LYNCH
TECH. REVIEW: P. DAVILA
DATE: MAY 2026

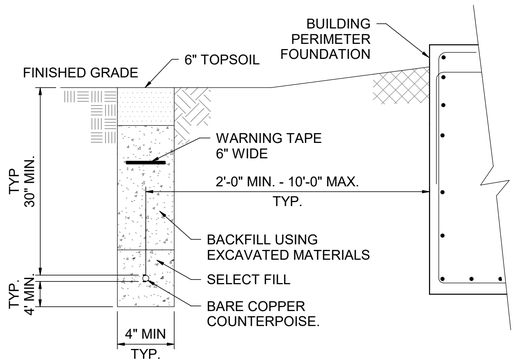
SUB SHEET NO. 008 E504

TITLE OF SHEET
GENERAL
ELECTRICAL DETAILS - LUMINAIRE CONTROL SCHEMATICS
NORTH RIM - GRAND CANYON NATIONAL PARK

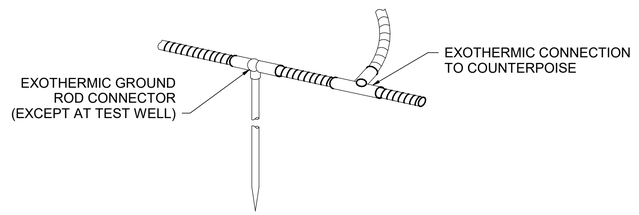
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 79 OF 99



1 008E505 GROUNDING ELECTRODE SYSTEM
NOT TO SCALE



2 008E505 TYPICAL GROUND RING INSTALLATION DETAIL
NOT TO SCALE



3 008E505 GROUNDING CONNECTION DETAIL
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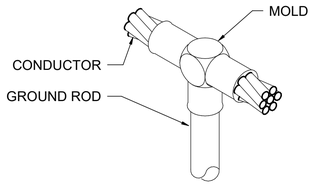
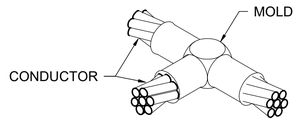
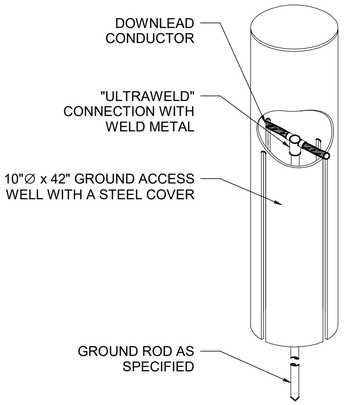


EXOTHERMIC WELD - GROUND ROD/ CONDUCTOR MOLD

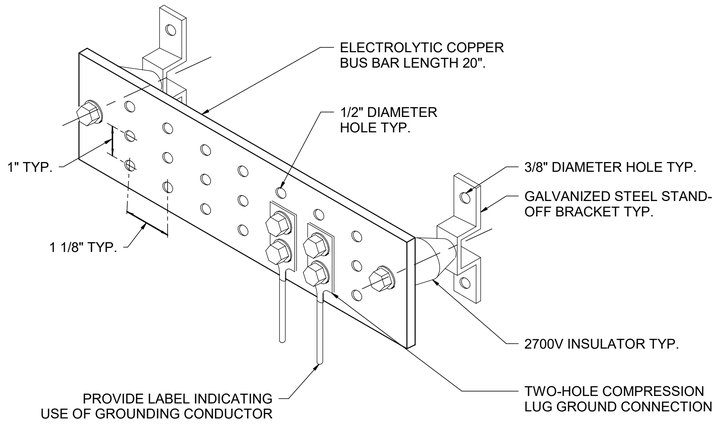


EXOTHERMIC WELD - INTERSECTION MOLD

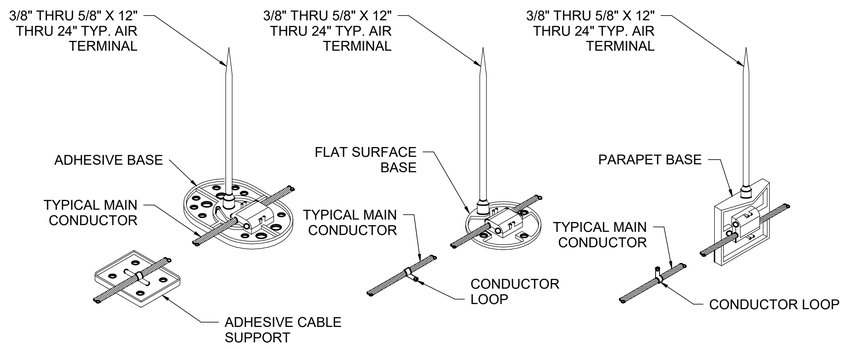
5 008E505 TYPICAL CABLE EXOTHERMIC WELD DETAIL
NOT TO SCALE



6 008E505 GROUND TEST WELL DETAIL
NOT TO SCALE

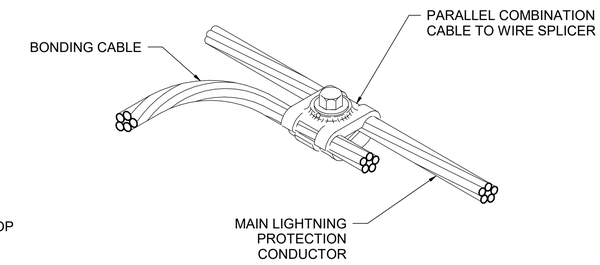


4 008E505 TYPICAL GROUNDING BUS BAR DETAIL
NOT TO SCALE

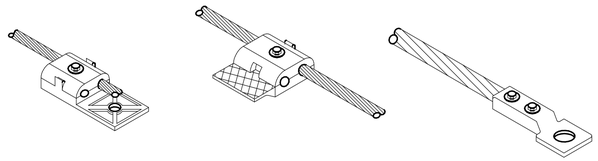


152 - ADHESIVE BASE **FLAT SURFACE BASE** **PARAPET BASE**

8 008E505 TYPICAL ROOF MOUNTED AIR TERMINAL DETAIL
NOT TO SCALE



9 008E505 TYPICAL CABLE SPLICE DETAIL
NOT TO SCALE



BONDING LUG **CABLE TO METAL CONNECTION** **BUS TERMINAL**

7 008E505 TYPICAL CABLE TERMINATIONS/CONNECTIONS DETAIL
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 04/30/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E505	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - GROUNDING AND LIGHTNING PROTECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 80 OF 99

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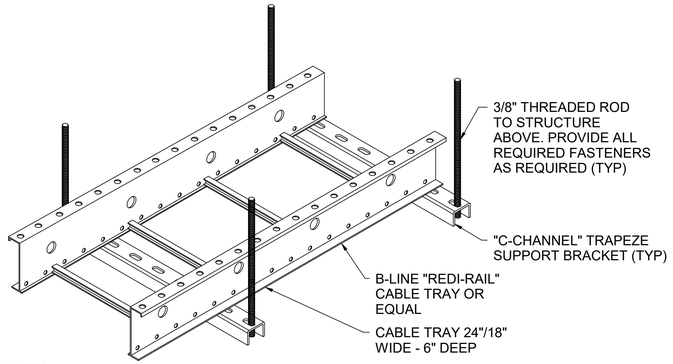
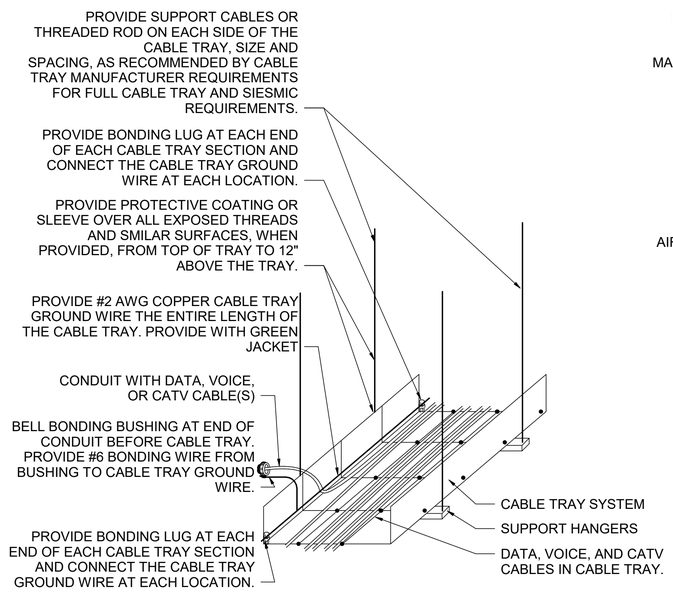


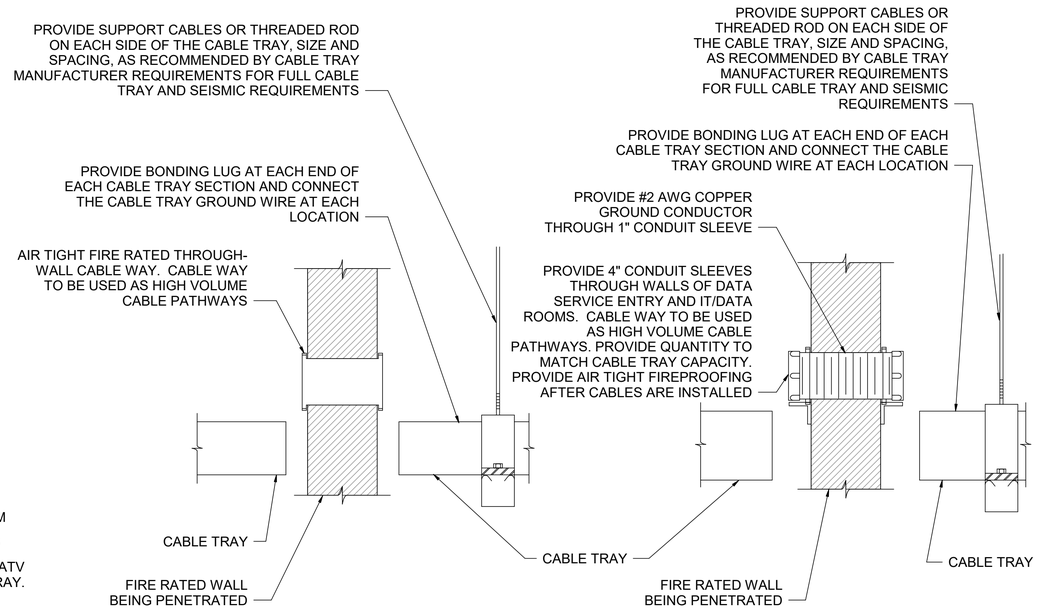
NOTES:

1. PRIMARY PATHWAY CABLE SUPPORTS ARE TO BE USED AS NECESSARY FOR CABLES TO COMPLY WITH NEMA REQUIREMENTS FOR CABLE SUPPORT.
2. CABLE TRAY SHALL BE MOUNTED ABOVE ACOUSTICAL TILE CEILING IN CORRIDORS. CABLE TRAY SHALL BE SUPPORTED WITH UNI-STRUT AND THREADED RODS IN A TRAPEZE CONFIGURATION. THREADED RODS SHALL BE SECURELY FASTENED TO STRUCTURE ABOVE. SUPPORT CABLE TRAY AT INTERVALS AS REQUIRED AND RECOMMENDED BY THE CABLE TRAY MANUFACTURER BASED ON MAXIMUM CAPACITY OF CABLES WITHIN THE CABLE TRAY.
3. REFER TO MANUFACTURER'S RECOMMENDATIONS FOR MAX LOADING, APPLICATION REQUIREMENTS, AND SECURING TO STRUCTURE.
4. ALL CABLE SUPPORTS SHALL BE SPECIFICALLY MANUFACTURED TO SUPPORT HIGH SPEED COMMUNICATIONS CABLES.
5. PROVIDE SOLID BOTTOM TO CABLE TRAY AT ANY POINT CABLE TRAY IS ROUTED ABOVE ANY HARD CEILING AREA.

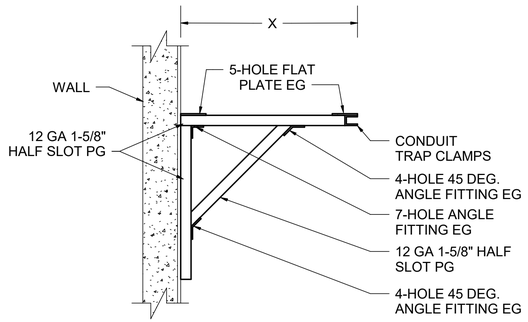
1 / 008E506 CABLE TRAY
NOT TO SCALE



2 / 008E506 TYPICAL CABLE TRAY MOUNTING AND BONDING
NOT TO SCALE

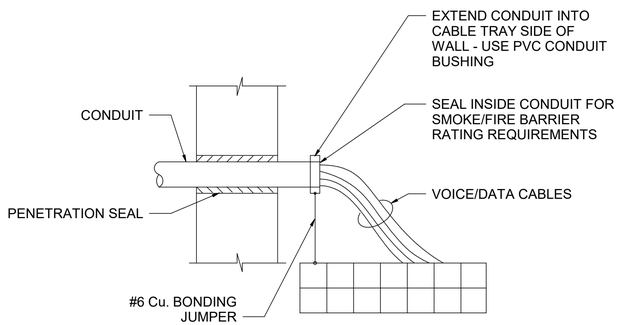


3 / 008E506 TYPICAL CABLE TRAY WALL PENETRATION
NOT TO SCALE

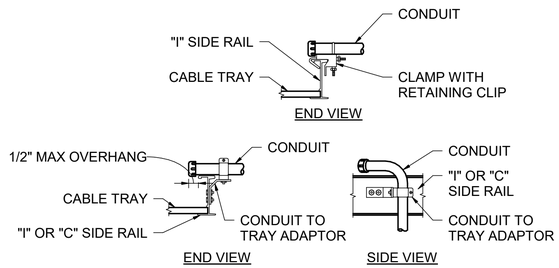


NOTE:
1. X = DISTANCE WILL BE SO THE WALL SIDE OF CONDUIT SHALL CLEAR EXISTING EXPANSION JOINT BY 8".

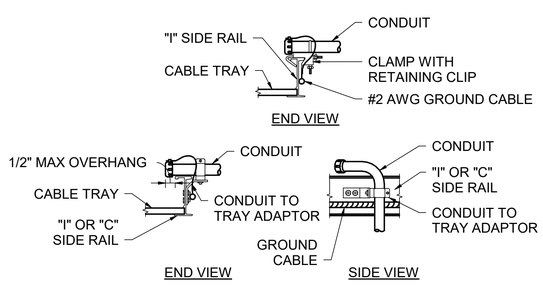
4 / 008E506 WALL MOUNTED UNISTRUT
NOT TO SCALE



5 / 008E506 CONDUIT CABLE TRAY TRANSITION
NOT TO SCALE



6 / 008E506 CONDUIT TO CABLE TRAY
NOT TO SCALE



7 / 008E506 CONDUIT TO CABLE TRAY WITH GROUNDING
NOT TO SCALE

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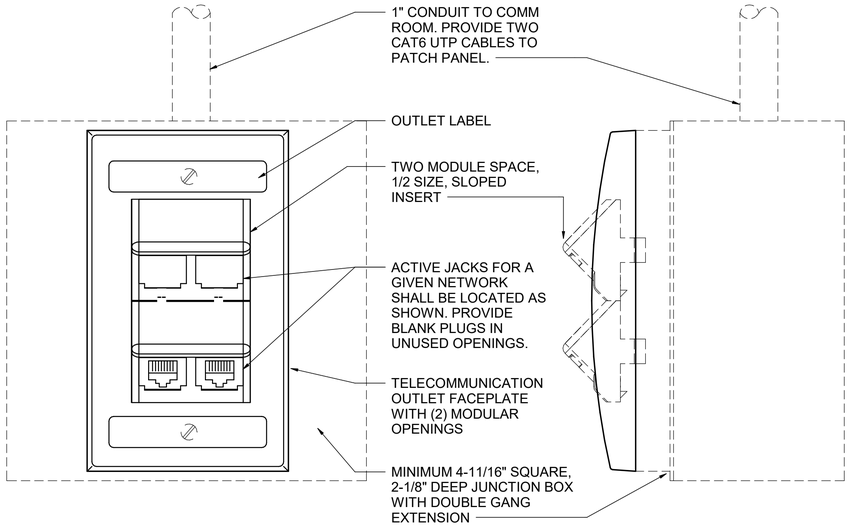
4/27/2026 11:57:35 AM

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 4/27/2026 ARIZONA, U.S.A.

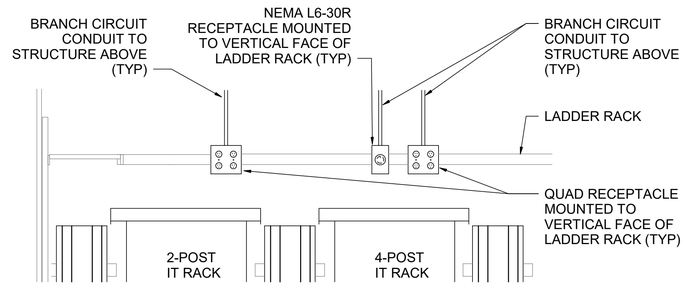
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E506	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - CABLE TRAY NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 81 OF 99



SYMBOL:

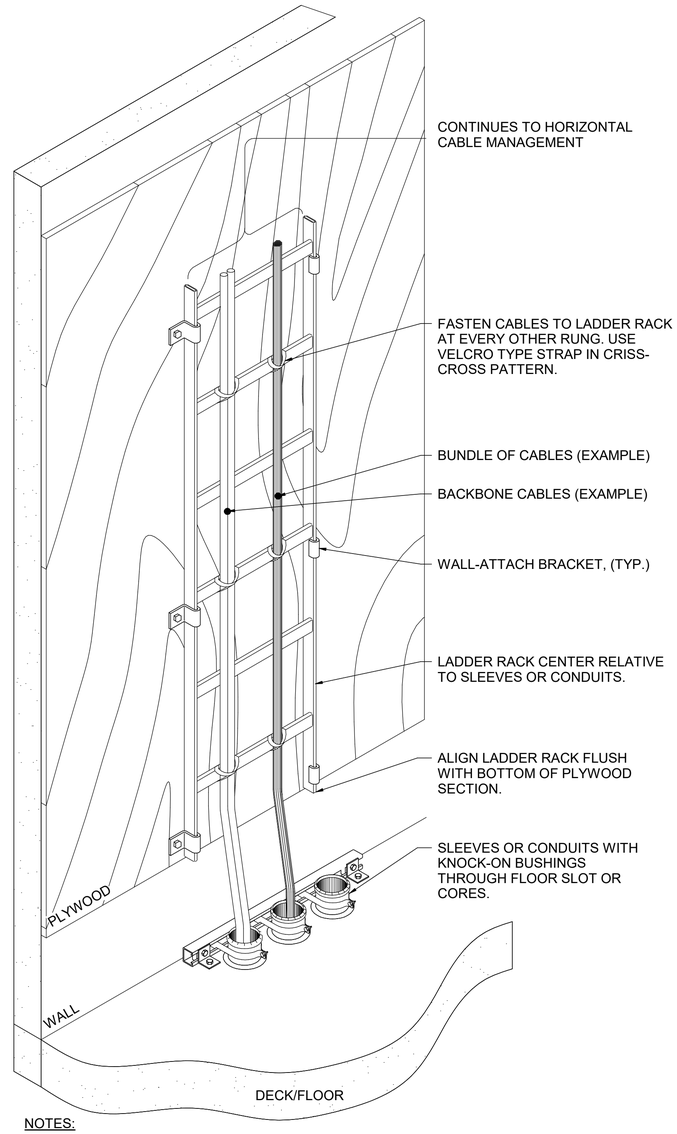
1 008E507 TYPICAL VOICE AND DATA OUTLET

NOT TO SCALE



2 008E507 LADDER RACK RECEPTACLES DETAIL

NOT TO SCALE

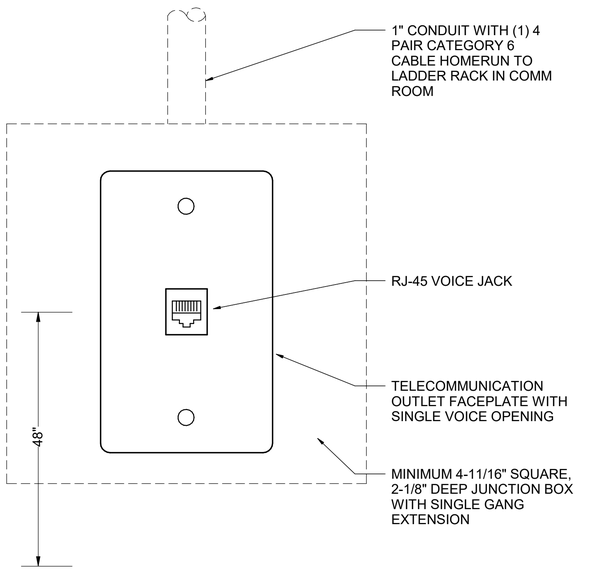


NOTES:

1. REFER TO ROOM PLANS AND DETAILS FOR CABLE RUNWAY SIZE AND LOCATION.
2. REFER TO ROOM PLANS AND DETAILS FOR SLEEVE/CONDUIT SIZES AND QUANTITIES.
3. REFER TO ARCHITECTURAL DRAWINGS FOR WALL AND DECK TYPE.

5 008E507 VERTICAL TRANSITION THROUGH FLOOR

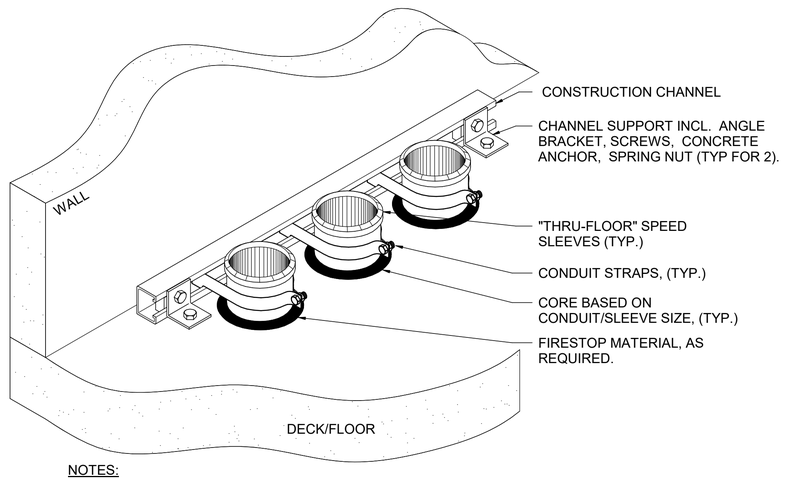
NOT TO SCALE



SYMBOL:

3 008E507 VOICE WALL OUTLET

NOT TO SCALE



NOTES:

1. REFER TO PATHWAYS RISER DIAGRAM SLEEVES/CONDUITS SIZE AND QUANTITY.
2. REFER TO ROOM PLANS FOR RISER SLEEVES/CONDUITS LOCATION.
3. BUSH AND SUPPORT SLEEVES/CONDUITS AT BOTH ENDS.
4. REFER TO ARCHITECTURAL DRAWINGS FOR WALL AND DECK TYPE.

4 008E507 RISER CONDUIT INSTALLATION

NOT TO SCALE

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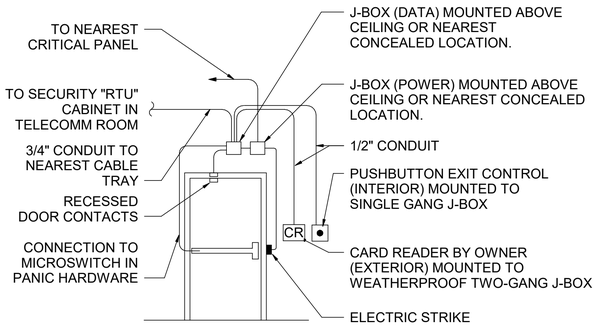
4/27/2026 11:57:35 AM

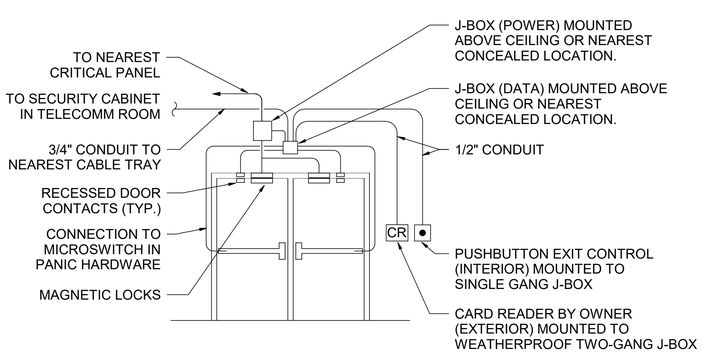
REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO.
58037
TRAVIS T. MOORE
STATE OF ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E507	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - SPECIAL SYSTEMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 82 OF 99



1 008E508 SINGLE DOOR ACCESS CONTROL

NOT TO SCALE



2 008E508 DOOR ACCESS CONTROL DIAGRAM 5

NOT TO SCALE

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_000_GENERAL.rvt
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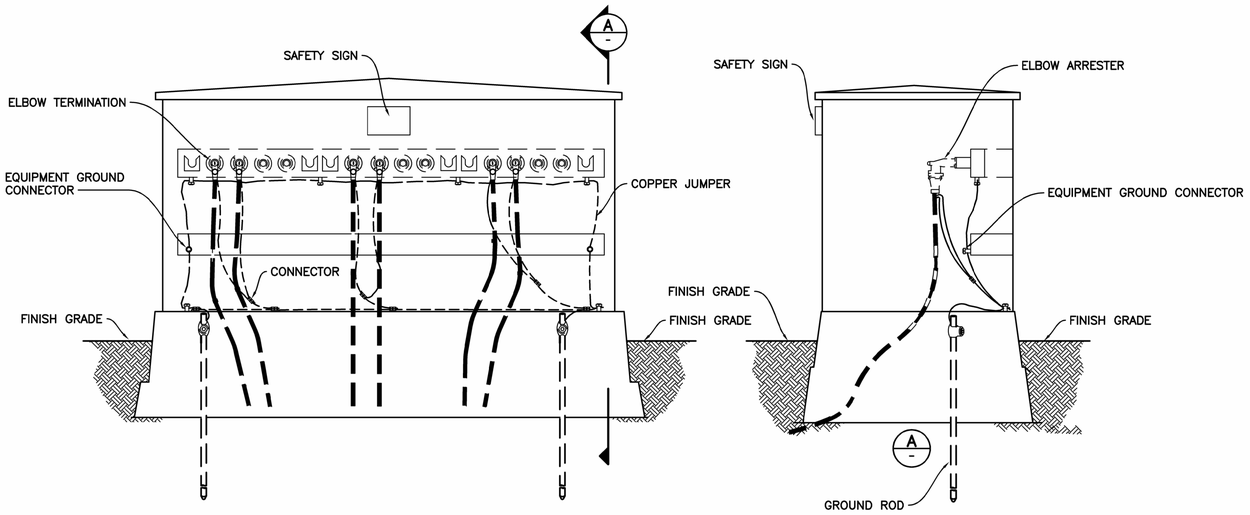
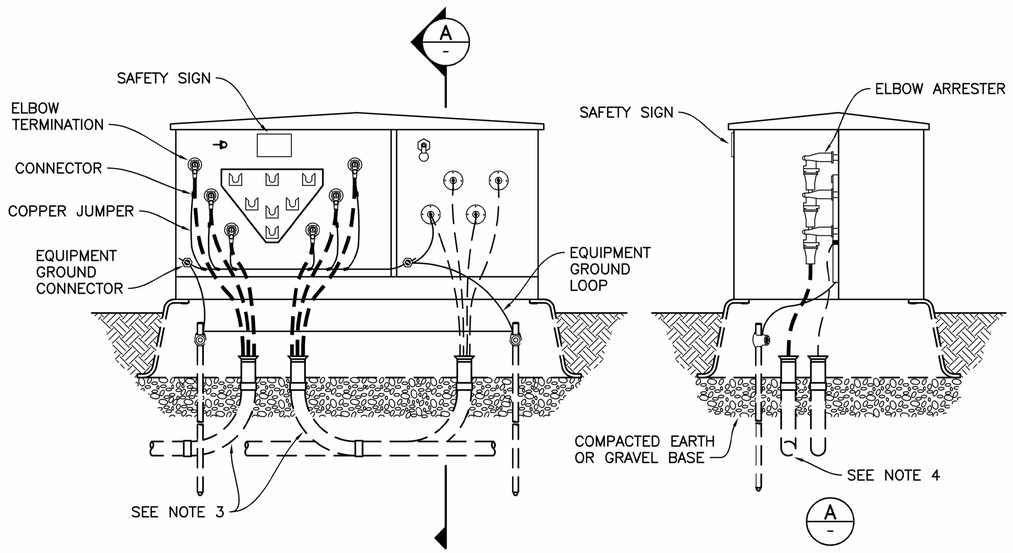
REGISTERED PROFESSIONAL ENGINEER – ELECTRICAL – CERTIFICATE NO. 58037 TRAVIS T. MOORE – STATE OF ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE CADD N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E508	TITLE OF SHEET GENERAL ELECTRICAL DETAILS - SPECIAL SYSTEMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 83 OF 99



1 008E509 TYPICAL THREE–PHASE PADMOUNTED JUNCTION ENCLOSURE

NOT TO SCALE

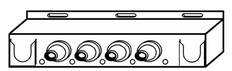


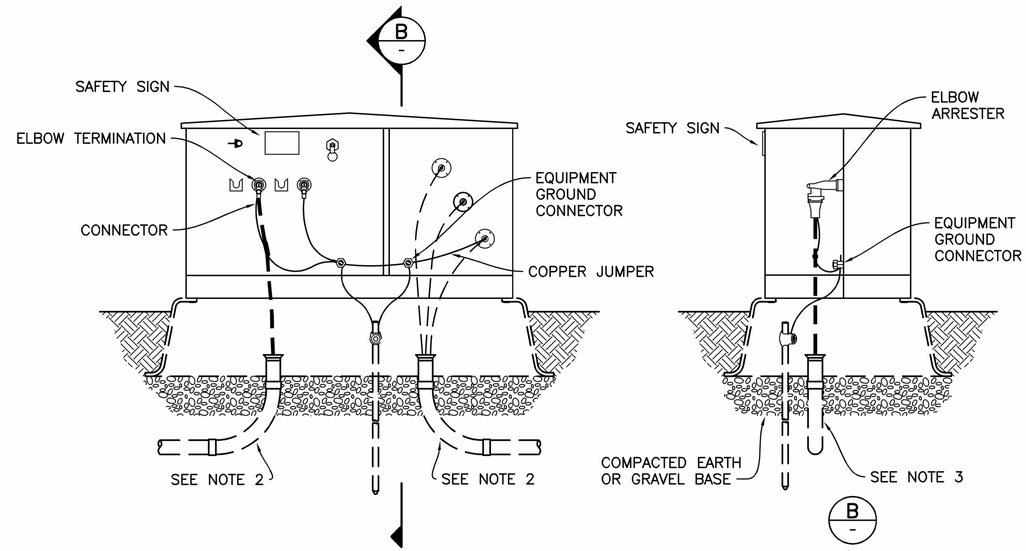
NOTES:

1. PROVIDE SUFFICIENT PRIMARY NEUTRAL PIGTAIL AND CABLE SLACK TO PERMIT READY DISCONNECTION OF ELBOW AND MOUNTING ON PARKING STAND. TRAIN CABLES AS SHOWN.
2. INSTALL SAFETY SIGNS ON TRANSFORMER INSIDE AND OUTSIDE ENCLOSURE ACCORDINGLY.
3. NON–METALLIC SWEEPS SHOULD BE STABILIZED WITH CONCRETE ENCASEMENTS IF PULLING TENSIONS DICTATE.
4. LARGER RADIUS SWEEPS, WHICH MAY BE REQUIRED BY PULLING TENSIONS, MAY INCREASE BURIAL DEPTHS.

2 008E509 THREE–PHASE, LOOP FEED, PADMOUNTED TRANSFORMER

NOT TO SCALE





NOTES:

1. PROVIDE SUFFICIENT PRIMARY NEUTRAL PIGTAIL AND CABLE SLACK TO PERMIT READY DISCONNECTION OF ELBOW AND MOUNTING ON PARKING STAND. TRAIN CABLES AS SHOWN.
2. NON–METALLIC SWEEPS SHOULD BE STABILIZED WITH CONCRETE ENCASEMENTS IF PULLING TENSIONS DICTATE.
3. LARGER RADIUS SWEEPS, WHICH MAY BE REQUIRED BY PULLING TENSIONS, MAY INCREASE BURIAL DEPTHS.

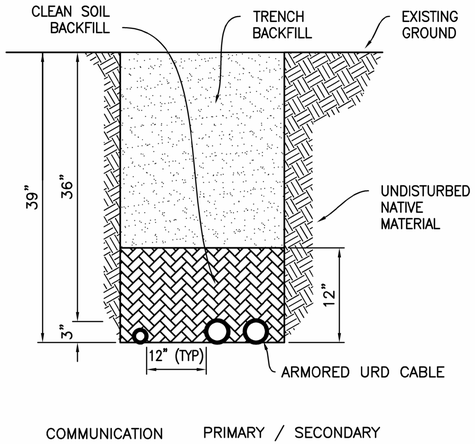
3 008E509 ONE–PHASE, RADIAL FEED, PADMOUNTED TRANSFORMER

NOT TO SCALE

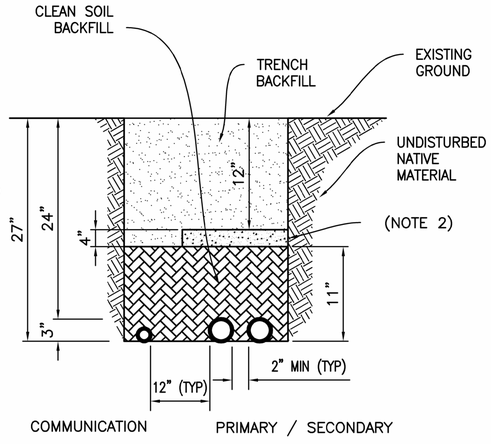
REGISTERED PROFESSIONAL ENGINEER OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE CADD N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E509	TITLE OF SHEET GENERAL ELECTRICAL DETAILS – TRANSFORMERS AND JUNCTION ENCLOSURE NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 84 OF 99

4/21/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\GEN\008-E509-E510.dwg



36" DEPTH TRENCH
N.T.S.

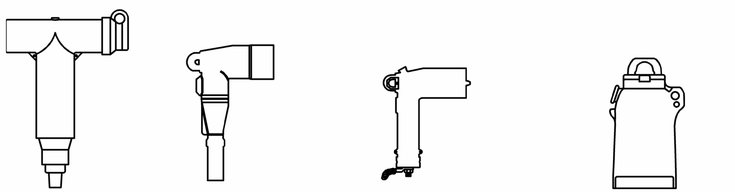


24" DEPTH TRENCH
N.T.S.

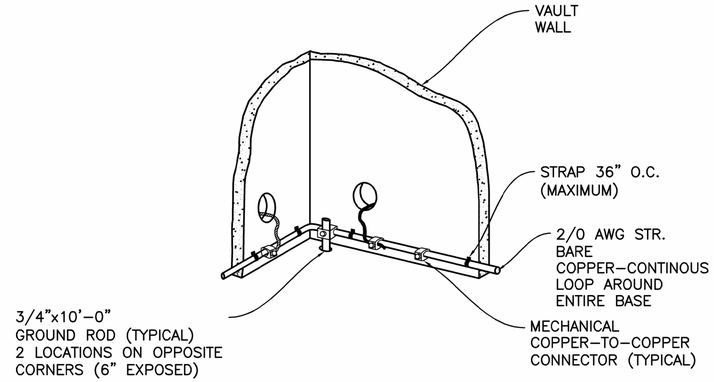
NOTES:

1. UTILIZE 36" BURY DEPTH FOR POWER CABLE TRENCH INSTALLATION.
2. IF 36" DEPTH CANNOT BE ACHIEVED, 24" DEPTH CAN BE USED WITH A CONCRETE CAP.
3. MINIMUM TRENCH WIDTH IS 1–INCH PLUS THE SUM OF THE HORIZONTALLY SPACED CONDUIT OUTSIDE DIAMETERS PLUS 1/2–INCH FOR EACH HORIZONTALLY SPACED CONDUITS (OVER ONE)
4. A MINIMUM 4–INCH CONCRETE CAP ON THE TOP AND 2 INCHES ON THE SIDES IS REQUIRED WHERE CABLES ARE INSTALLED IN REDUCED–DEPTH PLASTIC CONDUIT.
5. INSTALL UNDERGROUND PRIMARY CABLE ROUTER MARKERS AS NEEDED.
6. DISSIMILAR CONDUITS TO HAVE MINIMUM OF 12" OF SEPARATION IN ANY DIRECTION TO NON MV OR LV UTILITIES UNLESS OTHERWISE NOTED. SIMILAR UTILITY CONDUITS TO MAINTAIN 2" MINIMUM OF SEPARATION.
7. COMMUNICATION CONDUIT TO BE PLACED 12" FROM POWER CABLE FOR ALL NEW INSTALLATIONS.
8. SEE VOLUME 1 FOR ADDITIONAL UTILITY TRENCHING DETAILS.
9. TRENCH BACKFILL SHALL BE FREE FROM ROCK, GRAVEL, OR OTHER MATERIAL WHICH MIGHT DAMAGE THE CABLE JACKET. CLEANED SOIL BACKFILL WHEN USED SHALL CONTAIN NO SOLID MATERIAL LARGER THAN 1 INCH. THIS SOIL LAYER SHALL BE CAREFULLY COMPACTED SO THAT THE CABLE WILL NOT BE DAMAGE.
10. THIS DETAIL USED FOR SITE SPECIFIC LOCATIONS IN VOLUMES 5, 6, 7, AND 9. ALL OTHER TRENCHES FOR FO ROUTING WILL BE AS SHOWN IN VOLUME 1, 10, 11.

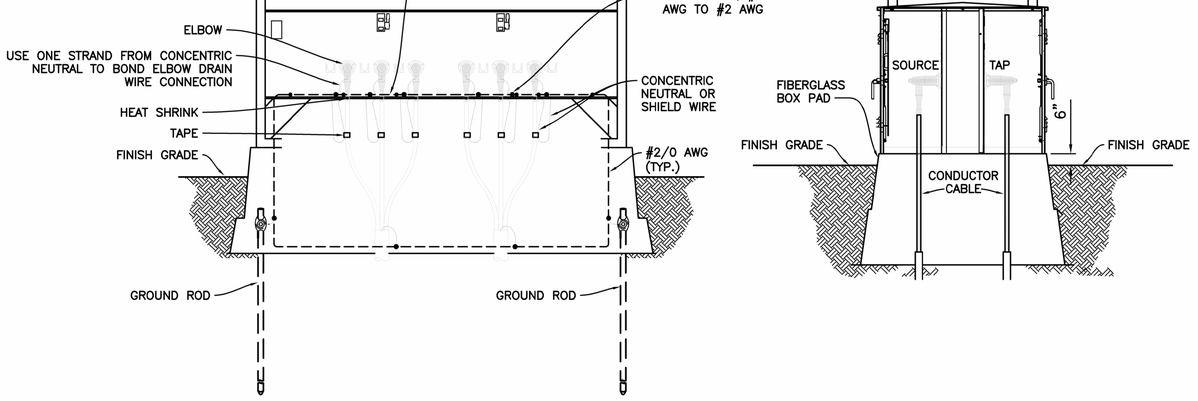
1 / 008E510 TYPICAL JOINT–USE, DIRECT–BURIED, POWER TRENCH DETAILS
NOT TO SCALE



2 / 008E510 ELBOWS AND CONNECTORS
NOT TO SCALE



3 / 008E510 TYPICAL EQUIPMENT VAULT GROUNDING DETAIL
NOT TO SCALE



4 / 008E510 TYPICAL THREE–PHASE SWITCHGEAR ENCLOSURE
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: N. LYNCH TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 008 E510	TITLE OF SHEET GENERAL ELECTRICAL DETAILS – GENERAL NORTH RIM – GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	85 OF 99

4/21/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\GEN\008-E509-E510.dwg

LINE TYPES

- MAIN PROCESS LINE
- SECONDARY PROCESS LINE
- PROCESS AIR
- AUXILIARY PROCESS LINE
- CHANNEL
- DIRECTION OF FLOW
- ANALOG SIGNAL
- ANALOG AND HARD SIGNAL
- FIELDBUS SIGNAL
- PNEUMATIC SIGNAL
- ELECTRICAL SIGNAL
- HYDRAULIC SIGNAL
- SOFTWARE OR ETHERNET
- SIGNAL CONNECTION
- CROSSOVER - NO CONNECTION
- CAPILLARY
- DOUBLE CONTAINMENT PIPING
- INTERIM PROCESS LINES

VALVE SYMBOLOGY

- BALL VALVE
- BUTTERFLY VALVE
- CHECK VALVE
- CONE VALVE
- DOUBLE CHECK VALVE
- DOUBLE-DISK CHECK VALVE
- BALL CHECK VALVE
- DIAPHRAGM VALVE
- GATE VALVE
- GLOBE VALVE
- KNIFE GATE VALVE
- MUD VALVE
- NEEDLE VALVE
- PINCH VALVE
- ENERGY DISSIPATION VALVE
- PLUG VALVE
- FOOT VALVE
- EMERGENCY STATION EYE WASH
- COMBINED BACK-PRESSURE WITH ANTI SIPHON
- PRESSURE-REDUCING VALVE
- PRESSURE-REGULATING VALVE
- THREE-WAY CONTROL VALVE
- OR — PRESSURE-RELIEF (PRESSURE-SAFETY) VALVE (PSV)
- AIR-RELEASE VACUUM VALVE
  A = AIR RELEASE
  VAC = VACUUM RELEASE
  CAV = COMBINED AIR AND VACUUM RELEASE
- SOLENOID VALVE

CROSS REFERENCE SYMBOLOGY

- B | 01Y105 — CONTINUATION ON SHEET 01Y105
- 01Y103 | B — CONTINUATION ON SHEET 01Y103

INSTRUMENT SYMBOLOGY

- XXX — LOCALLY MOUNTED FIELD INSTRUMENTATION (SUBSCRIPT INDICATES INSTRUMENTATION ABBREVIATION)
- MOUNTED ON PANEL FRONT
- MOUNTED INSIDE PANEL
- XXX — FRONT PANEL MOUNTED ON AUXILIARY PANEL (SUBSCRIPT INDICATES PANEL)
- MOUNTED INSIDE AUXILIARY PANEL
- PILOT LIGHT
- INSTRUMENT FUNCTIONS SHARING COMMON HOUSING
- I — COMPLEX INTERLOCK AS DEFINED IN CONTROL DIAGRAM OR IN SPECIFICATIONS
- SHARED DISPLAY, SHARED CONTROL, FIELD MOUNTED
- SHARED DISPLAY, SHARED CONTROL, PRIMARY LOCATION - NORMALLY ACCESSIBLE TO OPERATOR
- PROGRAMMABLE LOGIC CONTROL, PRIMARY LOCATION - NORMALLY ACCESSIBLE TO OPERATOR
- PROGRAMMABLE LOGIC CONTROL, FIELD MOUNTED
- DISCRETE INPUT
- DISCRETE OUTPUT
- ANALOG INPUT
- ANALOG OUTPUT
- I/I — SIGNAL ISOLATOR
- ISR — INTRINSICALLY SAFE RELAY
- ISB — INTRINSICALLY SAFE BARRIER

ACTUATOR SYMBOLOGY

- X — OPERATOR ABBREVIATIONS:
  M = MOTOR
  P = PNEUMATIC
  S = SOLENOID
- FLOAT OPERATOR
- SPRING-OPPOSED SINGLE-ACTING PNEUMATIC CYLINDER
- DOUBLE-ACTING PNEUMATIC CYLINDER
- PNEUMATIC DIAPHRAGM
- PNEUMATIC DIAPHRAGM WITH POSITIONER

TYPES OF POWER SUPPLY

A	PLANT COMPRESSED AIR
IA	INSTRUMENTATION AIR
ES	ELECTRIC SUPPLY
NG	NATURAL GAS
HYD	HYDRAULIC

INSTRUMENT IDENTIFICATION LETTERS

	FIRST LETTER		SUCCEEDING LETTERS		
	MEASURED OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	ANALYSIS		ALARM		AUTO
B	BURNER, COMBUSTION		USER'S CHOICE	USER'S CHOICE	USER'S CHOICE
C	USERS CHOICE			CONTROL	CLOSED
D	USERS CHOICE	DIFFERENTIAL			
E	VOLTAGE		SENSOR (PRIMARY ELEMENT)		
F	FLOW RATE	RATIO (FRACTION)			FORWARD
G	USER'S CHOICE		GLASS, VIEWING DEVICE		
H	HAND				HIGH
I	CURRENT (ELECTRICAL)		INDICATE		
J	POWER	SCAN			
K	TIME, TIME SCHEDULE	TIME; RATE OF CHANGE		CONTROL STATION	
L	LEVEL		LIGHT		LOW
M	MOISTURE	MOMENTARY			MIDDLE, INTERMEDIATE
N	STATUS		USER'S CHOICE	USER'S CHOICE	USER'S CHOICE
O	USER'S CHOICE		ORIFICE, RESTRICTION		OPEN
P	PRESSURE, VACUUM		POINT (TEST) CONNECTION		
Q	QUANTITY	INTEGRATE, TOTALIZE	PULSE		
R	RADIATION		RECORD	RESET	REMOTE, REVERSE
S	SPEED, FREQUENCY	SAFETY		SWITCH	STOP
T	TEMPERATURE			TRANSMIT	
U	MULTIVARIABLE		MULTIFUNCTION	MULTIFUNCTION	MULTIFUNCTION
V	VIBRATION, MECH. ANALYSIS			VALVE, DAMPER, LOUVER	
W	WEIGHT, FORCE		WELL		
X	COMMAND	X AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
Y	EVENT, STATE OR PRESENCE	Y AXIS		RELAY, COMPUTE, CONVERT	
Z	POSITION, DIMENSION	Z AXIS		DRIVER, ACTUATOR, UNCLASSIFIED FINAL CONTROL ELEMENT	

MISCELLANEOUS INSTRUMENTATION ABBREVIATIONS

AI	ANALOG INPUT
AO	ANALOG OUTPUT
CL2	CHLORINE (ANALYZER MODIFIER)
CO	CARBON MONOXIDE (ANALYZER MODIFIER)
CO2	CARBON DIOXIDE (ANALYZER MODIFIER)
COMB	COMBUSTIBLES (ANALYZER MODIFIER)
COND	CONDUCTIVITY (ANALYZER MODIFIER)
DEN	DENSITY (ANALYZER MODIFIER)
DI	DIGITAL INPUT
DO	DIGITAL OUTPUT
DO	DISSOLVED OXYGEN (ANALYZER MODIFIER)
E/P	VOLTAGE TO PNEUMATIC
FO	FIBER OPTIC
H2S	HYDROGEN SULFIDE (ANALYZER MODIFIER)
HCL	HYDROGEN CHLORIDE (ANALYZER MODIFIER)
I/O	INPUT/OUTPUT
I/P	CURRENT TO PNEUMATIC
I/P	CURRENT TO PNEUMATIC
LEL	LOWER EXPLOSIVE LIMIT
OI	OPERATOR INTERFACE
O2	OXYGEN (ANALYZER MODIFIER)
P&ID	PROCESS AND INSTRUMENTATION DIAGRAM
SC	STREAMING CURRENT (ANALYZER MODIFIER)
SO2	SULFER DIOXIDE
SS	SUSPENDED SOLIDS (ANALYZER MODIFIER)
TURB	TURBIDITY (ANALYZER MODIFIER)
WAN	WIDE AREA NETWORK

CONTROL SWITCH NOTATION ABBREVIATIONS

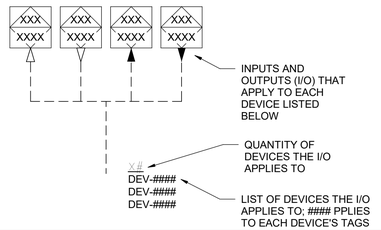
XXX / XXX

ACK	ACKNOWLEDGE
ESTOP	EMERGENCY STOP
FAIL	FAILURE
FOR	FORWARD-OFF-REVERSE
FR	FORWARD-REVERSE
FS	FAST-SLOW
HA	HAND-AUTO
HOA	HAND-OFF-AUTO
HOR	HAND-OFF-REMOTE
LL	LEAD-LAG
LLS	LEAD-LAG-STANDBY
LOR	LOCAL-OFF-REMOTE
LR	LOCAL-REMOTE
LS	LEAD-STANDBY
MA	MANUAL-AUTO
OAC	OPEN-AUTO-CLOSE
OC	OPEN-CLOSE
OO	ON-OFF
OSC	OPEN-STOP-CLOSE
PB	PUSH-BUTTON
RJ	RUN-JOG
RJR	RUN-JOG-REVERSE
SIL	SILENCE
SS	START-STOP

PRIMARY ELEMENT SYMBOLOGY

- ULTRASONIC ELEMENT
- ORIFICE PLATE
- SONIC OR ULTRASONIC ELEMENT
- M — MAGNETIC FLOW ELEMENT
- V — VELOCITY FLOW ELEMENT
- PROPELLER OR TURBINE ELEMENT
- TD — MASS DISPERSION FLOW ELEMENT
- THERMAL MASS FLOW ELEMENT
- FLUME
- WEIR
- VENTURI TUBE
- FLOAT SWITCH
- M — MOTOR
- THERMOWELL
- DIAPHRAGM SEAL
- ANNULAR SEAL
- BUBBLER LEVEL TUBE
- ULTRASONIC LEVEL SENSOR
- RADAR LEVEL SENSOR
- CABLE SUSPENDED PRESSURE TRANSMITTER
- IN-LINE FLANGED DIAPHRAGM SEAL
- PITOT TUBE OR ANNUBAR
- FG — SIGHT FLOW GLASS
- MASS DISPERSION FLOWMETER
- FT
- TE — TEMPERATURE ELEMENT WITH THERMOWELL
- FI — ROTOMETER
- SIGNAL ISOLATOR
- STOP PLATE

DUPLICATE INPUTS/OUTPUTS NOTATION



ISA PREDEFINED ABBREVIATIONS

ISA PREFIX	FIRST LETTER	SECOND LETTER	THIRD LETTER
AE	ANALYTICAL	ELEMENT	
AIT	ANALYTICAL	INDICATING	TRANSMITTER
ES	EMERGENCY	STOP	
ETM	ELAPSED	TIME	METER
FE	FLOW	ELEMENT	
FIT	FLOW	INDICATING	TRANSMITTER
FT	FLOW		TRANSMITTER
HSR	HAND	SWITCH	REMOTE
LI	LEVEL	INDICATION	
LIT	LEVEL	INDICATING	TRANSMITTER
LSH	LEVEL	SWITCH	HIGH
LSHH	LEVEL	SWITCH	HIGH-HIGH
LSL	LEVEL	SWITCH	LOW
LSLL	LEVEL	SWITCH	LOW-LOW
LT	LEVEL		TRANSMITTER
MSH	MOISTURE	INDICATION	HIGH
NI	STATUS	SWITCH	
PI	PRESSURE	INDICATION	
PIT	PRESSURE	INDICATING	TRANSMITTER
PSH	PRESSURE	SWITCH	HIGH
PSHH	PRESSURE	SWITCH	HIGH-HIGH
PSL	PRESSURE	SWITCH	LOW
SC	SPEED	COMMAND	
SI	SPEED	INDICATION	
TI	TEMPERATURE	INDICATION	
TIT	TEMPERATURE	INDICATING	TRANSMITTER
TE	TEMPERATURE	ELEMENT	
UA	MULTIVARIABLE	ALARM	
UU	MULTIVARIABLE	INFORMATION	
VE	VIBRATION	ELEMENT	
XC	START/STOP	COMMAND	
XCF	START/STOP	COMMAND	FORWARD
XCR	START STOP	COMMAND	REVERSE
XO	COMMAND	OPEN	
XR	COMMAND	RESET	
XS	COMMAND	SWITCH	
XSM	COMMAND	SWITCH	MASTER
XSS	COMMAND	SWITCH	STANDBY
XVC	COMMAND	VALVE	CLOSE
XVO	COMMAND	VALVE	OPEN
XVS	COMMAND	VALVE	STOP
ZF	POSITION	FORWARD	
ZR	POSITION	REVERSE	
ZS	POSITION	SWITCH	
ZSC	POSITION	SWITCH	CLOSE
ZSO	POSTION	SWITCH	OPEN

GENERAL NOTES:

A. THIS IS A STANDARD INSTRUMENTATION SYMBOLOGY AND ABBREVIATIONS SHEET. LISTING OF SYMBOLS AND ABBREVIATIONS DOES NOT IMPLY ALL SYMBOLS AND ABBREVIATIONS HAVE BEEN USED ON THIS PROJECT.

B. SCREENING OR SHADING OF WORK IS USED TO INDICATE EXISTING COMPONENTS OR TO DE-EMPHASIZE PROPOSED IMPROVEMENTS TO HIGHLIGHT SELECTED TRADE WORK. REFER TO CONTEXT OF EACH SHEET FOR USAGE.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. BRAINARD	009 Y001	GENERAL INSTRUMENTATION AND CONTROLS LEGEND
	DRAWN: N. LYNCH		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: K CHANDRA		
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	86 OF 99

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MISC. (CONT.)

VERTICAL TURBINE OR PROPELLER PUMP

CENTRIFUGAL PUMP

SUBMERSIBLE PUMP

CHEMICAL FEED PUMP (ALL TYPES)

PROGRESSING CAVITY PUMP

BUBBLE TRAP

ROTARY LOBE PUMP

AIR COMPRESSOR WITH INTERGAL DRYER

STATIC MIXER

POSITIVE DISPLACEMENT BLOWER

MEMBRANE

DIFFUSERS

PARSHALL FLUME

MECHANICAL SCREEN

M

MIXER

SLIDE GATE

MIXER NOZZLE

CALIBRATIONCOLUMN

ACCUMULATOR

EJECTOR / EDUCTOR

INLINE FLOW CONDITIONER

Y-TYPE STRAINER

EXPANSION JOINT

RS

REMOVABLE SPOOL

PULSATION DAMPER

FLEXIBLE HOSE

ELECTRIC CORD WITH PLUG

ELECTRIC CORD WITH RECEPTACLE

AUTOFILTER

CARTRIDGE FILTER

INJECTION QUILL WITH CORPORATION STOP

INJECTION QUILL WITH INTEGRAL CHECK VALVE

S

SNUBBER

FINE SCREEN

WEIGHT SCALE

120V FIT M

INTEGRAL MOUNTED MAGNETIC FLOW METER AND TRANSMITTER

120V FIT FE M

REMOTE MOUNTED MAGNETIC FLOW METER AND TRANSMITTER

FLUSHING CONNECTION, CAM-LOK

ED

EQUIPMENT DRAIN

S

VENT SILENCER

S

INLINE SILENCER

DAMPER

EXHAUST HEAD

VENT COVER

CONE STRAINER

TEMPORARY STRAINER

T-TYPE STRAINER

DUPLEX STRAINER

BACKFLOW PREVENTER

AIR GAP

RUPTURE DISK

HEAT EXCHANGER

CIRCUIT SETTER

TRUE UNION

THERMOMETER

PIPE IN SECTION

D

DRAIN (SECTION OR SCHEMATIC)

FS

FLOOR SINK

FLEXIBLE HOSE OR TUBING

HEAT TRACE

LINE SIZE CHANGE (CONCENTRIC REDUCER)

LINE SIZE CHANGE (ECCENTRIC REDUCER)

LINE TURNING DOWN

LINE TURNING UP

BLIND FLANGE

PIPE BREAK

PIPE JOINT (SEE SPECS FOR REQUIREMENTS)

COMPRESSION SLEEVE TYPE COUPLING

FLANGED COUPLING ADAPTER (FCA)

FLEXIBLE CONNECTION

HARNESSED MECHANICAL COUPLING

TRAP

QUICK DISCONNECT CAM & GROOVE COUPLING

CAP OR PLUG

CO

INTERIOR CLEANOUT

HB-X

HOSE VALVE, HOSE BIBB, OR FLUSHING CONNECTION

HR-X

HOSE RACK

FD-X

FLOOR DRAIN

X = TYPE DESIGNATED IN SPECIFICATIONS

ATA

AIR TOOL ASSEMBLY

AVS

AUTOMATIC VALVE STATION

PRS

PRESSURE-REDUCING STATION

HB

HOSE BIBB

HR

HOSE REEL

AIR FILTER

AIR DRYER

TANK HEATER

AIR RECEIVER

UV UNIT

VENT

GENERAL NOTES:

A. THIS IS A STANDARD INSTRUMENTATION SYMBOLOGY AND ABBREVIATIONS SHEET. LISTING OF SYMBOLS AND ABBREVIATIONS DOES NOT IMPLY ALL SYMBOLS AND ABBREVIATIONS HAVE BEEN USED ON THIS PROJECT.

B. SEE PROCESS, MECHANICAL AND PLUMBING LEGEND SHEET FOR MISCELLANEOUS PIPING SYMBOLS.

C. SCREENING OR SHADING OF WORK IS USED TO INDICATE EXISTING COMPONENTS OR TO DE-EMPHASIZE PROPOSED IMPROVEMENTS TO HIGHLIGHT SELECTED TRADE WORK. REFER TO CONTEXT OF EACH SHEET FOR USAGE.

D. VALVE SYMBOLS SHOWN HERE ARE APPLICABLE ONLY TO INSTRUMENTATION DIAGRAMS. SEE PROCESS, MECHANICAL AND PLUMBING LEGEND SHEET FOR VALVE SYMBOLS USED ELSEWHERE ON THE SHEETS.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: N. LYNCH TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y002	TITLE OF SHEET GENERAL INSTRUMENTATION AND CONTROLS LEGEND CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 87 OF 99

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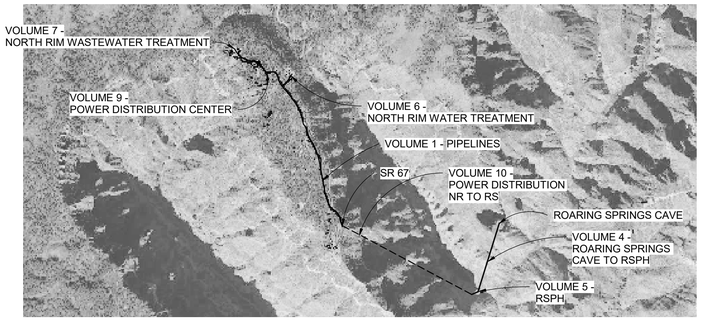


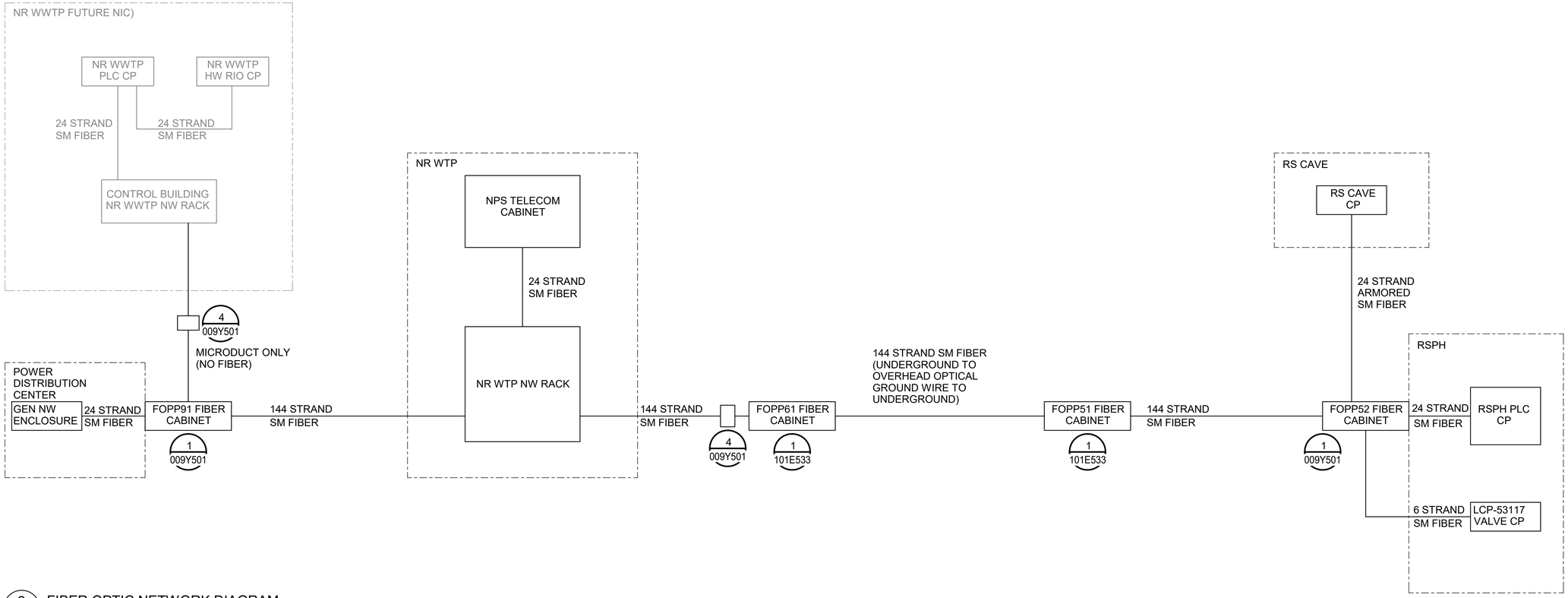
1 / 009Y101 OVERALL FIBER SITE MAP

NOT TO SCALE

LEGEND:

UNDERGROUND FIBER OPTIC

OVERHEAD OPTICAL GROUND WIRE FIBER OPTIC



2 / 009Y101 FIBER OPTIC NETWORK DIAGRAM

NOT TO SCALE

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